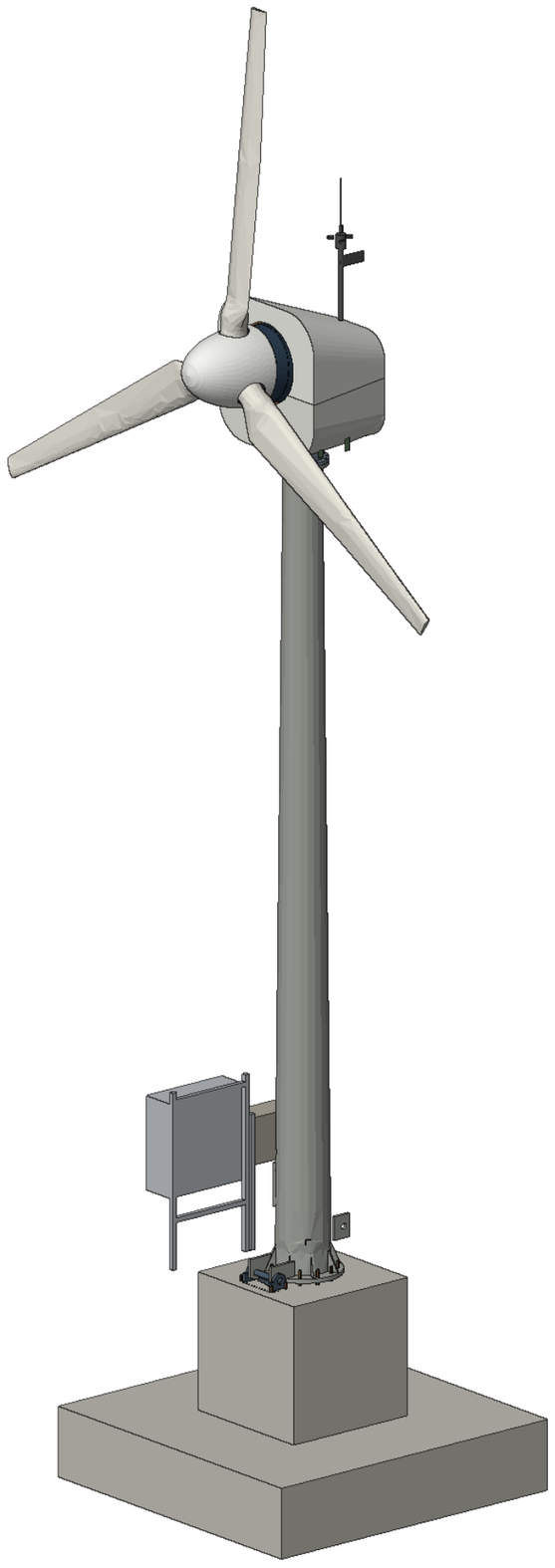
6 kW Wind Turbine S6

Build book · 8 m tower · revision B

Item	S6 on the 8 m tower
Rated power	6 kW electric at ≈ 9.7 m/s (hub height)
Rotor	Ø6.0 m, 3 glass/epoxy blades, TSR 5.5, fixed pitch −2°
Regulation	stall by speed (converter), storm ramp 25–30 m/s, yaw 90° out in storms
Generator	G6 = Rev C direct-drive PM (no fins/ribs), one 1.5 kNm spring brake + K1 braking resistor
Hub height	8.0 m; rotor tip 5.0 m above ground
Tower	one-piece tilt-up monopole Ø460 → Ø320, t 10/8, 661 kg, f1 3.5 Hz
Foundation	spread footing 2.8–3.1 m square, 5.5–6.6 m³; rock plinth 2 m³
Head mass	982 kg above the tower flange (CAD, every part)
Energy (vs 18 m tower)	sheltered 6.9 MWh (−42 %) · inland 12.6 MWh (−26 %) · coastal 18.3 MWh (−18 %) · highland 24.2 MWh (−15 %)



Contents of this book: design summary for the 8 m tower (§1); general arrangement and BOM (§2); part catalogue with an axonometric view of every part (§3); the S6 drawing set rev B (§4); the reused 12 kW drawings (§5); the assembly sequence in 13 steps (§6); erection, commissioning and inspection (§7); open items (§8).

Sources: docs 28–34, analysis/turbine_6kw (tower8.py, design6.py, sim6.py), cad/turbine_6kw (model6.py, drawings6.py, iso6.py, bom6.py). Every number in this book comes from those scripts; re-run them to update the book.

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How to use this book

- Every part has a card in §3 (axonometric view, part number, material, mass, drawing). The drawing follows in §4 (new S6 parts) or §5 (12 kW parts reused unchanged).
- Build in the order of §6. Each step lists the parts, the fasteners and torques (NS6-900) and the checks that must be signed before the next step.
- Site work (footing, erection, commissioning) is §7. Keep the signed checklists with the turbine file.
- All drawings are NOT CHECKED. Release for the prototype only after the review and the FE checks listed in §8.

1 Design summary: S6 on an 8 m tower

What changed with the 8 m tower

Tower. At 8 m a soft-stiff tower (first frequency between 1P at 41 rpm and 3P at 41 rpm, as the 18 m tower) is not possible: the head is 982 kg on a 6.9 m tube, and a tube slim enough to be that soft fails in strength. The tower is therefore stiff: f1 is kept 15 % above 1P at full speed (2.73 Hz × 1.15 = 3.14 Hz). The 3P excitation still crosses f1 at low speed, so the controller skips 60–77 rpm (3.4–4.4 m/s wind), in the same way it skipped the 1P zone of the 18 m tower.

	18 m tower (rev A)	8 m tower (rev B)
Concept	soft-stiff, 2 sections	stiff, one piece
Shell	Ø500 → Ø320, t 10/8	Ø460 → Ø320, t 10/8
Mass	1824 kg	661 kg
f1 rigid / soft soil + ice	0.87 / 0.81 Hz	3.5 / 3.28 Hz
Speed exclusion zone	47.5–57.4 rpm (1P)	59.6–77.0 rpm (3P)
Governing case, use	parked V_e50, 0.974	earthquake 0.5 g (ground D) + operation, 0.445
Base design moment	617.2 kNm	238.6 kNm
Base bolts	24 × M24 PCD 600	12 × M24 PCD 560
Footing firm / soft	3.8 / 4.2 m, 9.5 / 11.4 m³	2.8 / 3.1 m, 5.5 / 6.6 m³
Tilt-up	6 m gin pole, 68 kN	3 m gin pole, ≈ 38 kN, or a 3 t truck crane

Loads. The extreme winds stay at the class S hub-height values (IEC 61400-2 defines them at hub height), so no credit is taken for the lower wind near the ground. Earthquake (0.5 g, ground D) now governs the tower, because the stiff tower sits on the plateau of the response spectrum. Fatigue spectra were increased by the higher turbulence (× 1.234); tower fatigue damage 0.047 at the base.

Energy, turbulence and noise at 8 m

The wind is slower and more turbulent near the ground. With the log wind profile for each site roughness, the hub-height mean wind drops 10–19 % and the energy 15–42 %:

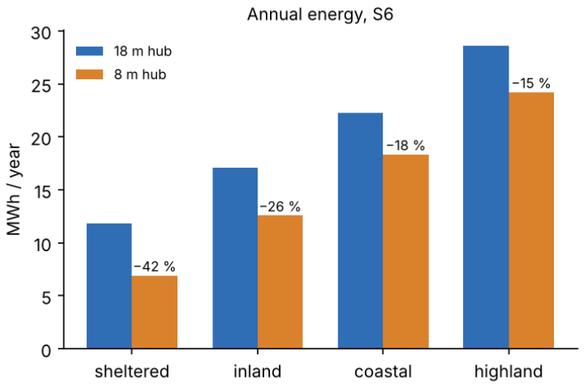
Site (roughness z0)	Mean wind 18 m → 8 m	AEP 18 m	AEP 8 m	Loss	Turbulence 18 → 8 m
Sheltered valley / farm (0.25 m)	5.0 → 4.05 m/s	11.8 MWh	6.9 MWh	−42 %	0.234 → 0.289 (+23 %)
Open inland lowland (0.05 m)	6.0 → 5.17 m/s	17.1 MWh	12.6 MWh	−26 %	0.170 → 0.197 (+16 %)
Coast / exposed lowland (0.01 m)	7.0 → 6.24 m/s	22.2 MWh	18.3 MWh	−18 %	0.133 → 0.150 (+12 %)
Highland / headland (0.03 m)	8.5 → 7.42 m/s	28.6 MWh	24.2 MWh	−15 %	0.156 → 0.179 (+14 %)

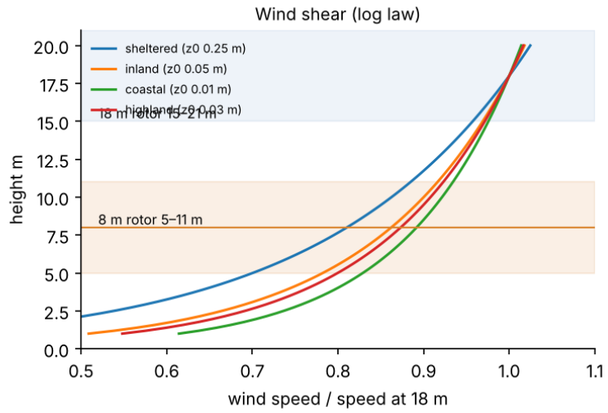
- Coast, highland and open land lose 15–26 %. A sheltered farm site loses 42 % and is not a good place for an 8 m tower.
- Turbulence intensity rises 12–23 %. Blade and tower fatigue were checked with the higher value; both stay far below their limits (doc 31 margins ≥ 3×).
- Noise: at the same wind at 10 m (8 m/s), the rotor turns slower at 8 m: sound power 85.0 vs 88.2 dB(A), 41.2 vs 44.1 dB(A) at 60 m. At rated speed the source is the same and the receiver is a little closer: 45.5 vs 45.2 dB(A) at 60 m. Less masking wind noise at the house is the practical difference.
- Safety: the lowest blade tip is 5.0 m above ground. Fence or mark a 4 m radius round the tower; keep the 21 m ice-fall zone when icing is possible.
- Siting: the rotor works in the wake of buildings and shelter belts; place it ≥ 20 obstacle heights upwind of them in the main wind directions.

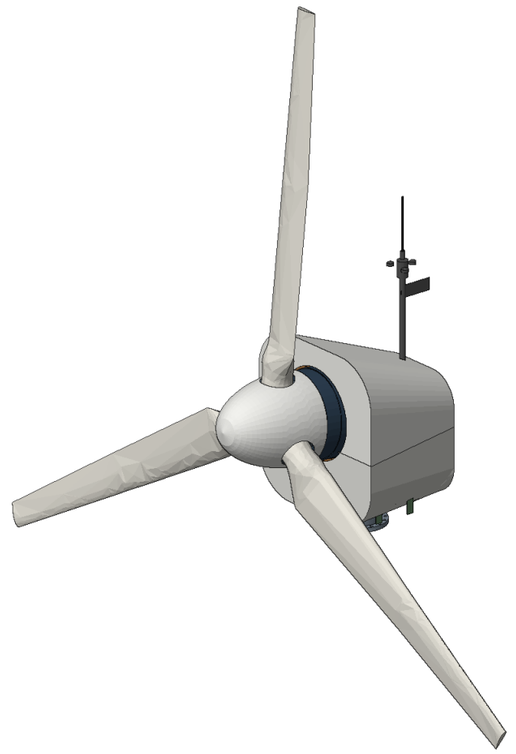
Design decisions taken for this book (design authority)

- Pitch setting by ±2.5° arc slots in the hub root flange (doc 30's wedge shims could not change pitch); gauge NS6-135.
- Hub shell lengthened to 295 mm so the arms close on it; hub 79 kg (CAD).
- GRP nacelle cover in two halves, drum outside (free-air cooling), met mast with air terminal on the rear rail.
- Tower one piece with base hinge, hand hole and gin-pole lugs; no climbing, service by tilting down.
- Spindle cooling insert NP1-110 left out (G6 runs without forced cooling, doc 29).
- Only the left caliper bracket NP1-310 fitted; braking resistor box and base cabinet on a stand beside the tower.

1 Design summary: figures and masses







Mass by group (assembled turbine)

Group	kg (CAD)
nacelle	168
electrical	134
yaw	266
drivetrain	378
rotor	162
tower	738
foundation	13344
Head above the tower flange	982

The head is 70 kg heavier than the doc 32 list (912 kg): the GRP cover and its arms (63 kg vs 20 kg allowed), the lock-pin holder and the junction box. The tower was sized with the CAD value.

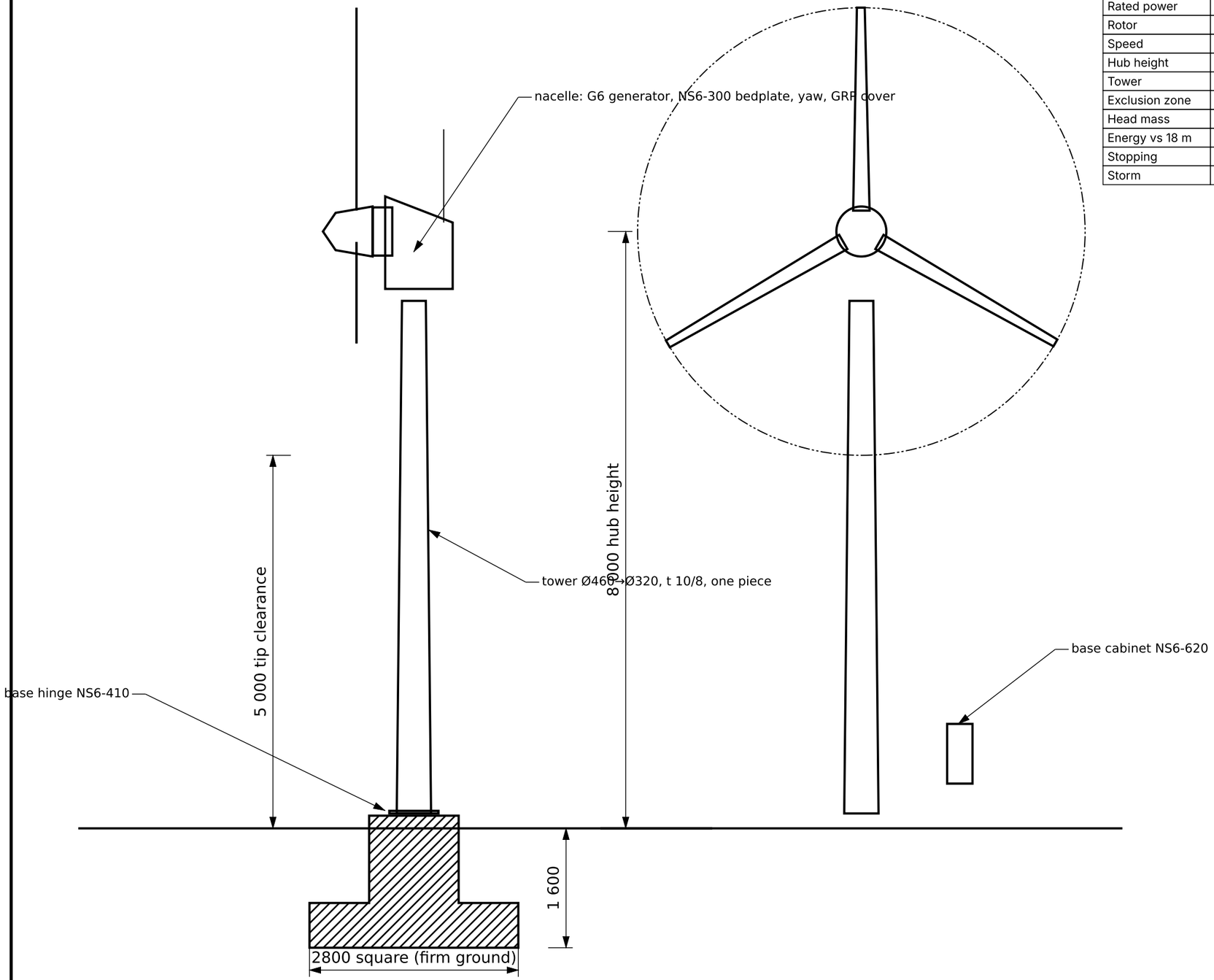
Checks carried over unchanged

- Blade, hub, drivetrain, bedplate and yaw: doc 30–32 checks (SLM, simulated loads) do not depend on tower height; the higher turbulence adds ≤ 20 % to fatigue ranges where the margins are ≥ 3×.
- Brake 1.5 kNm and the K1 resistor chain: doc 32 §3; unchanged.
- Yaw-out storm parking: doc 30 §5; the parked yaw moment is unchanged.
- Generator G6 thermal: doc 29; unchanged (lower mean wind only helps).

2 General arrangement and BOM

NS6-000 … NS6-020

SIDE VIEW (wind from the left)



VIEW FROM UPWIND

Item	Value
Rated power	6 kW electric at ≈ 9.7 m/s (hub)
Rotor	Ø6.0 m, 3 blades, TSR 5.5, pitch −2° fixed
Speed	41–164 rpm, trip 189 rpm
Hub height	8.0 m; lowest tip 5.0 m
Tower	661 kg, f1 3.5 Hz (3.28 soft+ice)
Exclusion zone	60–77 rpm (3P × f1)
Head mass	982 kg above the tower flange (CAD)
Energy vs 18 m	−18 % coast … −26 % inland
Stopping	K1 resistor 3 × 4 Ω + spring brake 1.5 kNm
Storm	stop, yaw 90° out, hold

AXONOMETRIC VIEW (not to scale)



NOTES

1. Docs 28–34. Rev B: hub height 8 m (was 18 m), tower one piece, footing re-sized; all new parts drawn (index NS6-001).
2. Keep people and vehicles out of the rotor disc: fence or marked zone of radius 4 m round the tower base; ice-fall zone 1.5 × (H + D) = 21 m when icing is possible.
3. Siting at 8 m: rotor bottom must be clear of the wake of buildings and shelter belts (≥ 20 × obstacle height upwind in the main wind directions).

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

GENERAL ARRANGEMENT · TURBINE ON 8 m TOWER

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-000	B · 2026-11-06	1:60
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see BOM NS6-002/003	≈ 982 kg head + 661 kg tower	Claude / not checked

Drawing	Title	Source
NS6-000	GENERAL ARRANGEMENT · TURBINE ON 8 m TOWER	S6 rev B
NS6-001	DRAWING INDEX	S6 rev B
NS6-002	BILL OF MATERIALS (1/2)	S6 rev B
NS6-003	BILL OF MATERIALS (2/2)	S6 rev B
NS6-010	GENERAL ARRANGEMENT · NACELLE SECTION	S6 rev B
NS6-020	ROTOR ASSEMBLY · HUB, BLADES, SPINNER	S6 rev B
NS6-100	BLADE · PLANFORM, TWIST AND ROOT	S6 rev B
NS6-101	BLADE · SECTIONS AND LAMINATE SCHEDULE	S6 rev B
NS6-105	BLADE ROOT INSERT M12 (BONDED)	S6 rev B
NS6-110	BLADE MOULD SET (PLUG, TWO HALVES, FRAME)	S6 rev B
NS6-115	ROOT INSERT FIXTURE AND WEB MOULD (TOOLS)	S6 rev B
NS6-130	PITCH SETTING · ROOT SLOTS, WASHER PLATES, GAUGE	S6 rev B
NS6-200	FIXED HUB WELDMENT	S6 rev B
NS6-210	SPINNER (GRP) AND BRACKETS NS6-211	S6 rev B
NS6-300	BEDPLATE WELDMENT 6 kW (LIGHT)	S6 rev B
NS6-310	CALIPER BRACKET NP1-310 AND YAW DRIVE BRACKET NP1-420	S6 rev B
NS6-400	TOWER 8 m · ONE-PIECE TILT-UP MONOPOLE	S6 rev B
NS6-410	BASE HINGE BRACKETS AND PIN	S6 rev B
NS6-420	GIN POLE AND TILT-UP RIGGING (ERECTION TOOL)	S6 rev B
NS6-500	FOUNDATION · SPREAD FOOTING AND ROCK PLINTH (8 m)	S6 rev B
NS6-510	ANCHOR CAGE AND TEMPLATE	S6 rev B
NS6-600	BRAKING RESISTOR BOX AND SAFETY CHAIN	S6 rev B
NS6-610	G6 PLAIN REAR COVER (NO FAN BLADES)	S6 rev B
NS6-620	TOWER BASE CABINET, STAND AND RESISTOR BOX POSITION	S6 rev B
NS6-700	NACELLE COVER · LOWER TRAY AND UPPER LID (GRP)	S6 rev B
NS6-710	COVER ARMS, REAR RAIL, MET MAST NS6-720, JUNCTION BOX NS6-730	S6 rev B
NS6-800	CABLING ROUTES, EARTHING AND LIGHTNING	S6 rev B
NS6-900	FASTENER AND TORQUE SCHEDULE	S6 rev B
NS6-910	BOUGHT PARTS SPECIFICATION (ENVELOPES AND INTERFACES)	S6 rev B
NP1-100	SPINDLE (STATIONARY MAIN SHAFT)	12 kW NP1 rev B
NP1-101	FRONT SPACER + REAR SEAL SLEEVE NP1-102	12 kW NP1 rev B
NP1-211	MAIN BEARING SEAL COVERS (FRONT / REAR)	12 kW NP1 rev B
NP1-230	BRAKE DISC AND ROTOR LOCK HOLES	12 kW NP1 rev B
NP1-320	ROTOR LOCK PIN AND HOLDER	12 kW NP1 rev B
NP1-T01	DRUM ASSEMBLY GUIDE (MAGNET SAFETY TOOL)	12 kW NP1 rev B
NP1-G00	GENERATOR REV C · ASSEMBLY SECTION (G6: no fins, ribs, fan blades)	Rev C (doc 23)
NP1-200C	STATOR CARRIER WITH 45 DOVETAIL GROOVES	Rev C
NP1-201C	STATOR TOOTH SEGMENT LAMINATION	Rev C
NP1-202	STATOR FRONT CLAMP RING	Rev C
NP1-203C	COIL ON SEGMENT (WOUND SEGMENT)	Rev C
NP1-210C	ROTOR DRUM WELDMENT (G6: omit fins and ribs)	Rev C
NP1-220E	MAGNET BLOCK 30 × 15 × 12 (PURCHASE)	Rev C
NP1-221C	STEPPED POLE COMB BAR	Rev C
NP1-222C	MAGNET END RINGS	Rev C
NP1-223C	ROTOR MAGNET PACK	Rev C
NY-000	YAW SYSTEM · GENERAL ARRANGEMENT	doc 25
NY-400	TOWER-TOP ADAPTER WITH FRICTION TRACK	doc 25
NY-410	FRICTION PAD UNIT	doc 25
NY-430	YAW LOCK (SERVICE)	doc 25
NY-440	YAW ENCODER AND CAM LIMIT SWITCH UNIT	doc 25
NY-460	CABLE HANGER AND TWIST LOOP	doc 25

NOTES

1. Rev B (2026-11-06): 8 m tower; every S6 part drawn with an axonometric view; BOM NS6-002/003; fasteners NS6-900; cabling NS6-800.
2. Reused 12 kW drawings are used unchanged except where the BOM says 'G6 variant'. Their PDFs follow in the build book in the order of this index.
3. All sheets NOT CHECKED: prototype release only after review, FE checks (hub arm welds, bedplate ring seat) and the stall-torque test.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

DRAWING INDEX (S6 SET REV B + REUSED 12 kW DRAWINGS)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-001	B · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
–	–	Claude / not checked

Pos	Part no.	Description	Qty	Make/buy	Material / spec	kg each
		A ROTOR				
1	NS6-100	Blade, glass/epoxy, glass UD caps (B1–B3)	3	make	E-glass/epoxy, PVC core	20.6
2	NS6-105	Root insert M12 (10 per blade, bonded)	30	make	C45E+QT zinc-flake coated	0.3
3	NS6-130	Pitch washer plates 28 × 4 hardened (30)	1	make	C45 hardened 45 HRC, zinc-flake	0.5
4	NS6-200	Fixed hub weldment	1	make	S355NL, welded, stress relieved, machined	79.2
5	NS6-210	Spinner (GRP, 4 mm)	1	make	glass/polyester, gelcoat RAL 9010	8.7
6	NS6-211	Spinner brackets (3)	1	make	S235 galv.	2.0
		B DRIVETRAIN AND GENERATOR G6				
7	NP1-100	Spindle	1	make	42CrMo4+QT forged	56.5
8	NP1-101	Front spacer / seal counterface Ø180 (Rev B)	1	make	42CrMo4+QT, induction hardened	1.4
9	NP1-102	Rear seal sleeve Ø195 on the Ø155 shoulder (Rev B)	1	make	42CrMo4+QT, induction hardened	1.3
10	NP1-200C	Stator carrier with 45 dovetail grooves	1	make		39.9
11	NP1-201C	Stator: 45 tooth-segment lamination stacks	1	make		50.5
12	NP1-202	Stator front clamp ring	1	make	1.4404 (non-magnetic)	2.2
13	NP1-203C	Coils on segments, 45 × 28 turns (end windings shown)	1	make	Cu enamelled Ø1.55 class 200, NKN 0.30	20.0
14	NP1-210C	Rotor drum weldment (G6: no fins, no ribs)	1	make	S355J2+N	91.2
15	NP1-211-F	Front bearing seal cover	1	make	S355J2	1.5
16	NP1-211-R	Rear bearing seal cover, bore Ø198 (Rev B)	1	make	S355J2	1.4
17	NP1-220E	Magnets 200 × N45SH 30×15×12 + comb + sleeve	1	make	NdFeB N45SH	10.7
18	NP1-230	Brake disc Ø880/Ø652 with 12 lock holes (Rev B)	1	make	C45E+N 16 mm	33.6
19	NP1-310	Rotor brake caliper bracket (left, Rev B)	1	make	S355J2+N 25 plate	18.1
20	NP1-320	Rotor lock pin Ø20 (engaged, top)	1	make	42CrMo4+QT	0.5
21	NP1-321	Lock pin holder with 2 overspeed sensor seats	1	make	S355J2	24.5
22	NS6-610	G6 plain rear cover (no fan blades)	1	make	EN AW-5083-H111	7.2
23	NS6-913	Tapered roller bearing 32028 X (front + rear)	2	buy	bought 32028 X	4.0
24	NS6-914	Overspeed sensors M18 inductive (2, bought)	1	buy	bought M18 PNP	0.4
25	NS6-915	Spring-applied brake caliper, 1.5 kNm pack (bought)	1	buy	bought, spring-applied	9.0
		C NACELLE				
26	NS6-300	Bedplate weldment 6 kW (light)	1	make	S355J2+N plates, stress relieved, machined	122.1
27	NS6-700-1	Nacelle cover, lower tray (GRP 5 mm)	1	make	glass/polyester, gelcoat RAL 9010	15.7
28	NS6-700-2	Nacelle cover, upper lid (GRP 5 mm)	1	make	glass/polyester, gelcoat RAL 9010	17.5
29	NS6-710	Cover support arms + rear rail	1	make	S235 galv.	8.3
30	NS6-720	Met mast: anemometer, vane, air terminal	1	make	1.4404 tube 42.4 × 2 + bought sensors	4.5
		D YAW SYSTEM				
31	NP1-420	Yaw drive bracket	2	make	S355J2+N	7.7
32	NS6-911	Yaw slewing ring Dpw 360, z150 m4	1	buy	bought: 42CrMo4, induction hardened teeth	55.0
33	NS6-912	Yaw gearmotor 0.18 kW 1:1680, motor brake 2 Nm	2	buy	bought	19.0
34	NY-400	Tower-top adapter with friction track	1	make	S355J2+N weldment	106.7
35	NY-410	Friction pad unit	2	make	S355 / DIN 2093 springs / organic pads	14.0
36	NY-430	Yaw lock (screw block)	1	make	42CrMo4 / S355	6.5
37	NY-440	Yaw encoder + cam limit switch unit	1	make	S355 + absolute encoder + cam switch	6.0
38	NY-460	Cable hanger tube + loop cleat	1	make	S355 tube 168.3 × 4.5	10.0
		E ELECTRICAL				
39	NS6-600	Braking resistor box 3 × 4 Ω (vented)	1	make	1.4301 box + steel-grid resistors	18.0
40	NS6-620	Tower base cabinet IP55 800 × 600 × 300 on stand	1	make	stand S235 galv.; bought enclosure	95.0
41	NS6-730	Nacelle junction box IP66 (400 × 340 × 150)	1	make	bought: 1.4301 enclosure	9.0
42	NS6-800	Drop cable + base duct (cabling route)	1	make	H07RN-F 4G10 + 2 × LiYCY signal; PE duct Ø11	12.0
		F TOWER				
43	NS6-400	Tower 8 m, Ø460 → Ø320, t 10/8, one piece	1	make	S355J2+N, hot-dip galvanised	697.5
44	NS6-401	Hand-hole cover 200 × 350 × 6 with gasket	1	make	S355 galv.	3.3
45	NS6-410	Base hinge brackets + hinge pin Ø50	1	make	S355J2+N galv.; pin 42CrMo4+QT	37.4
		G FOUNDATION				

NOTES

1. Masses from the CAD model (cad/turbine_6kw/model6.py) for made parts; supplier data or estimates for bought parts. Full BOM with sources: bom6.csv.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

BILL OF MATERIALS (1/2)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-002	B · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see lines	see lines	Claude / not checked

Pos	Part no.	Description	Qty	Make/buy	Material / spec	kg each
46	NS6-500	Spread footing (firm ground) with pedestal	1	make	C30/37 XF4 air-entrained, B500C	13200.0
47	NS6-510	Anchor cage 12 × M24 + 4 hinge anchors	1	make	M24 8.8 HDG threaded rod, S355 template ring	143.5
		H TOOLS AND MOULDS				
48	NS6-110-1	Blade mould, lower (pressure side) half on frame	1	make	GRP tooling laminate 12 mm, vinylester tooli	75.4
49	NS6-110-2	Blade mould, upper (suction side) half	1	make	GRP tooling laminate 12 mm	75.4
50	NS6-110-3	Mould frame (RHS 80 × 80 × 4)	1	make	S235 RHS 80×80×4	155.4
51	NS6-115	Root insert fixture (tool)	1	make	S355, machined	9.2
52	NS6-135	Pitch setting gauge (tool)	1	make	Al 6082 + digital inclinometer	1.6
53	NS6-420	Gin pole 3.0 m (erection tool)	1	make	S355 tube 168.3 × 6.3	94.0
		J FASTENERS, CABLES, SITE MATERIALS				
54	NS6-900-01	Stud M12 × 75 10.9 zinc-flake + nut + washer (blade roots)	30	buy	10.9 zinc-flake	0.12
55	NS6-900-02	Bolt M16 × 70 10.9 (hub to drum)	12	buy	10.9 zinc-flake	0.17
56	NS6-900-03	Bolt M20 × 90 10.9 (spindle to bedplate)	12	buy	10.9 zinc-flake	0.32
57	NS6-900-04	HV set M16 × 110 10.9 (yaw inner ring)	16	buy	EN 14399-4	0.26
58	NS6-900-05	Bolt M16 × 70 10.9 (yaw outer ring, pads, caliper)	28	buy	10.9 zinc-flake	0.17
59	NS6-900-06	HV set M20 × 90 10.9 (adapter to tower)	16	buy	EN 14399-4	0.40
60	NS6-900-07	Nut M24 8.8 HDG + plate washer (base, hinge)	44	buy	8.8 HDG	0.15
61	NS6-900-08	Small fasteners kit M8–M12 A4 (cover, spinner, mast, cabinet)	1	buy	A4-70	2.50
62	NS6-800-W1	Cable H07RN-F 4G10 (generator → box)	3	buy	m	0.60
63	NS6-800-W2	Cable H07RN-F 4G10 (box → loop → hand hole)	12	buy	m	0.60
64	NS6-800-W3	Cable NYY-J 4 × 10 (hand hole → cabinet)	4	buy	m	0.50
65	NS6-800-W4	Cable ÖLFLEX 4G1.5 shielded (brake)	16	buy	m	0.10
66	NS6-800-W5	Cable LiYCY 12 × 0.5 (sensors)	16	buy	m	0.12
67	NS6-800-W6	Cable ÖLFLEX 7G1.5 (yaw motors)	16	buy	m	0.15
68	NS6-800-W7	Earthing conductor Cu 50 mm² + ring earth + 2 rods 3 m	1	buy	Cu	18.00
69	NS6-500-M1	Concrete C30/37 XF4 air-entrained (firm ground)	5.5 m³	buy	ready-mix	
70	NS6-500-M2	Rebar B500C (mesh Ø16/150 top + bottom, pedestal cage)	1	buy	≈ 420 kg	420.00
71	NS6-500-M3	Non-shrink grout ≥ 40 MPa, 30 mm under the base flange	60 l	buy		
72	NS6-600-K1	Contactor K1 3-pole NC (mirror contacts) + safety relay PL d	1	buy		2.00
73	NS6-620-PE	Converter, dump chopper, AC module, 48 V module, controller (d	1	make	in-house electronics	45.00

NOTES

1. Masses from the CAD model (cad/turbine_6kw/model6.py) for made parts; supplier data or estimates for bought parts. Full BOM with sources: bom6.csv.

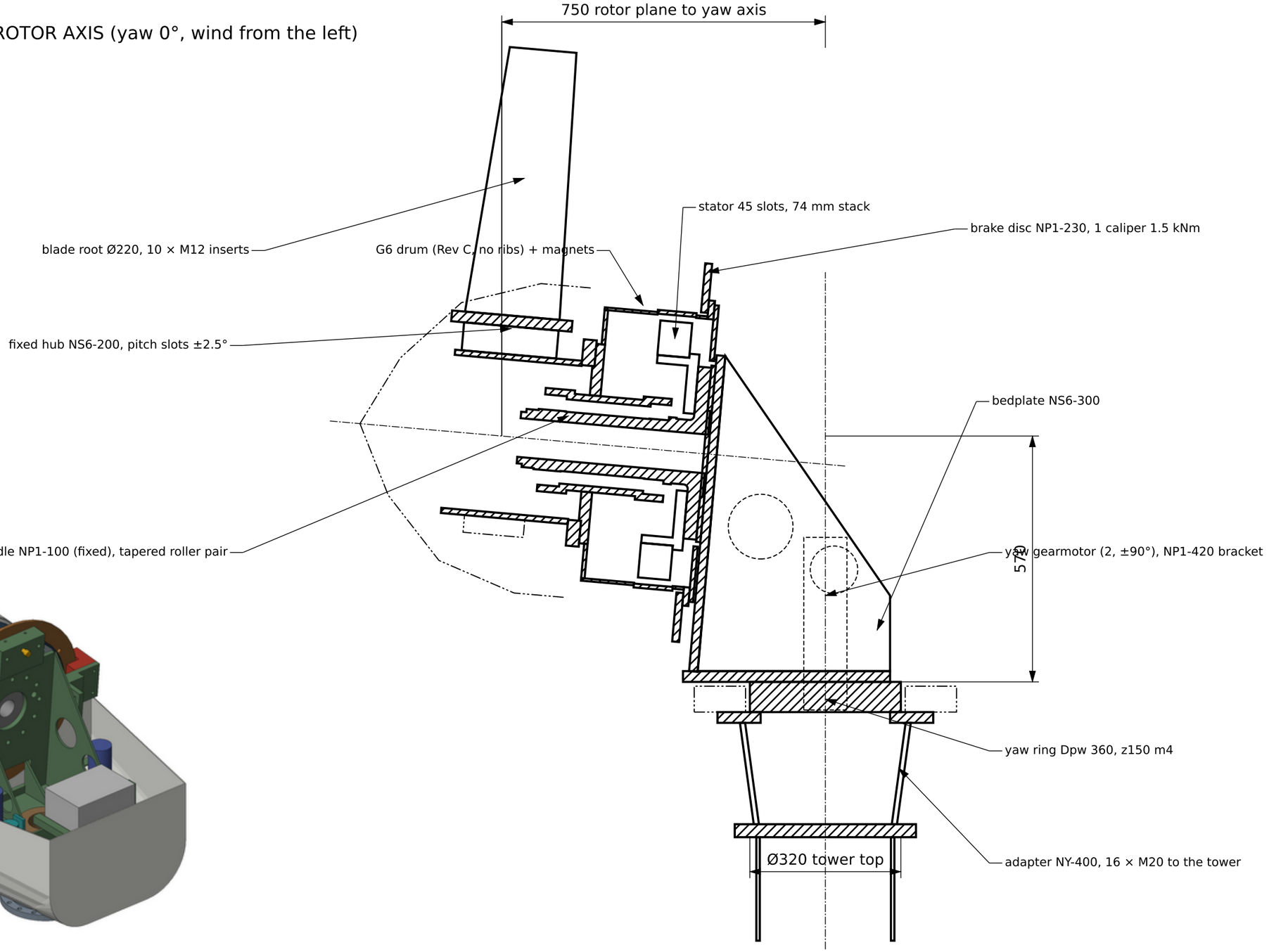
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

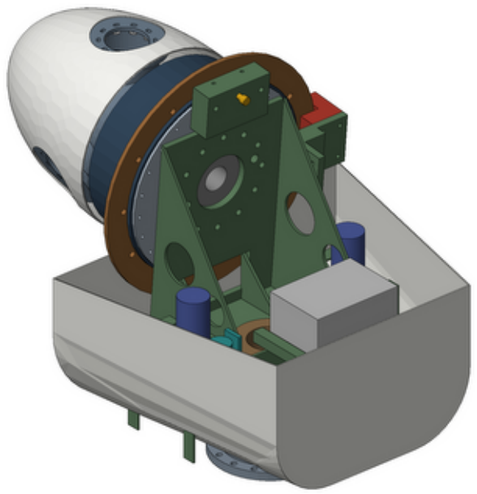
BILL OF MATERIALS (2/2)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-003	B · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see lines	see lines	Claude / not checked

SECTION ON THE ROTOR AXIS (yaw 0°, wind from the left)



AXONOMETRIC VIEW (not to scale)



Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

NOTES

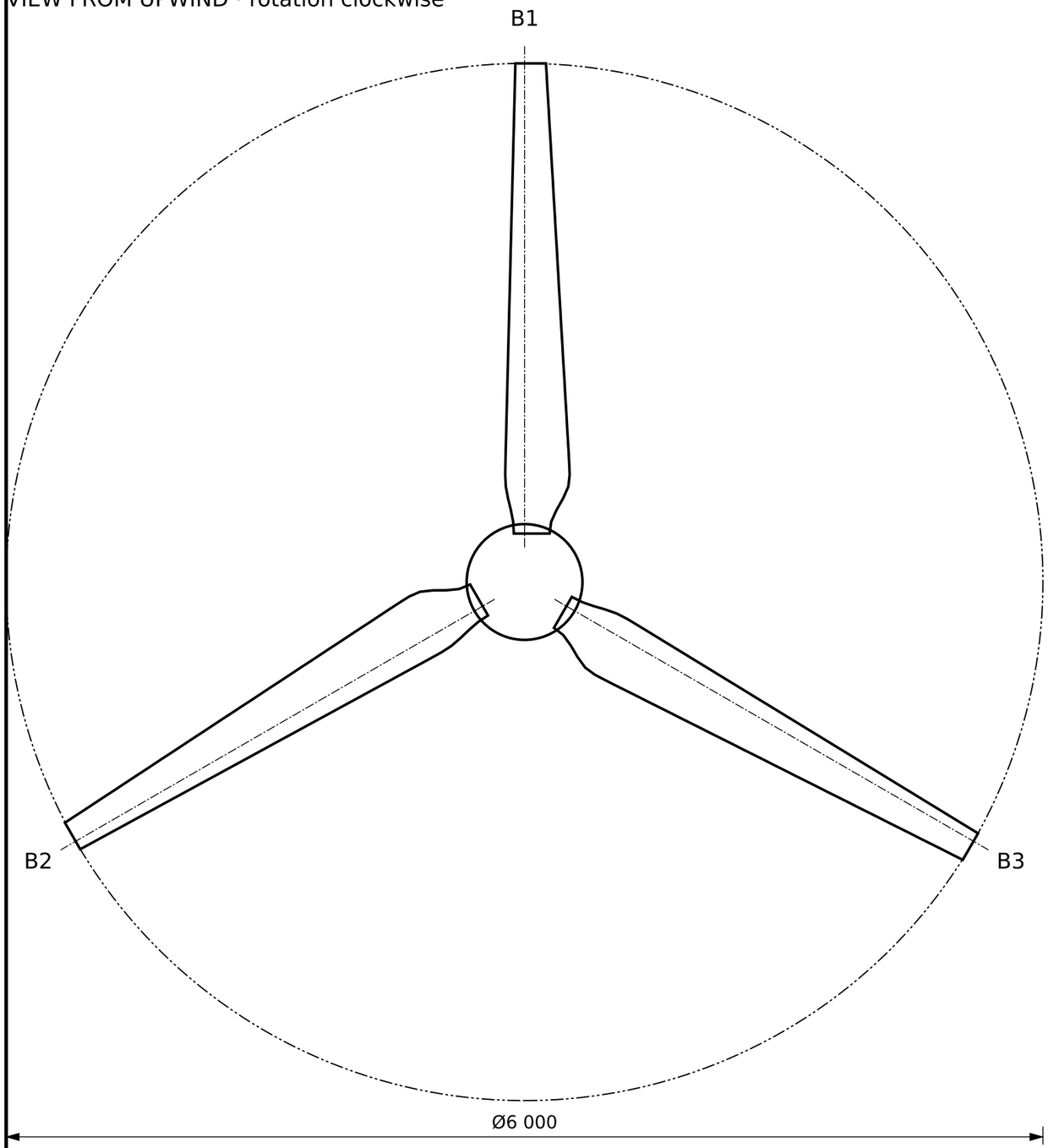
1. Drivetrain geometry from cad/nacelle_p1/geometry.py (Rev B nacelle, Rev C generator, doc 23); fins, ribs and cover blades not fitted on G6 (doc 29).
2. Rotor turns with the drum on two tapered roller bearings on the stationary spindle; the hub bolts to the drum front face (12 × M16 PCD 400).
3. Section shows the hub arm of the upper blade; the other two arms are at ±120°. Spinner (phantom) is GRP, NS6-210. GRP nacelle cover NS6-700 not shown (see NS6-700).
4. Braking resistor box NS6-600 is mounted on the tower base cabinet wall, not in the nacelle (heat). Phase leads run in the hanger tube.

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

GENERAL ARRANGEMENT · NACELLE SECTION

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-010	B · 2026-11-06	1:10
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
see NS6-001	≈ 982 kg above the tower flange	Claude / not checked

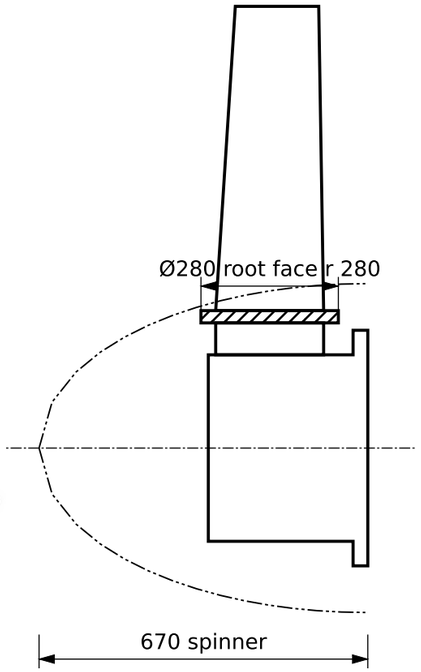
VIEW FROM UPWIND · rotation clockwise



AXONOMETRIC VIEW (not to scale)



SIDE VIEW, upper blade



Item	Value
Blade setting	−2.0° ± 0.1° at r = 2 000 (gauge NS6-135)
Blade set	static moments equal within 0.5 %
Track	tip track within ± 15 mm (string / laser)
Rotor balance	residual ≤ 0.2 kg·m (doc 09 R6.6)
Root bolts	30 × M12 10.9, 95 Nm + 90° (NS6-900)
Hub bolts	12 × M16 10.9 to the drum, 250 Nm

NOTES

1. Blades are numbered B1–B3 at the factory with their static moment; mount B1 on the arm marked 1 (lock-pin side).
2. Set pitch with the blade on the hub, bolts finger tight, gauge on the r = 2 000 station; tighten in a star pattern; re-check after 50 h of operation.
3. Balance the rotor on the turbine with the doc 09 procedure (one trial mass on the spinner brackets).

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

ROTOR ASSEMBLY · HUB, BLADES, SPINNER

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-020	B · 2026-11-06	1:30
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see BOM	≈ 162 kg (rotor + hub + spinner)	Claude / not checked

3 Part catalogue

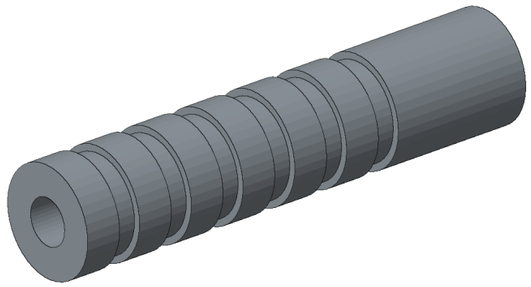
Axonometric view, part number, material and mass of every part

Axonometric view, part number, material and mass of every part

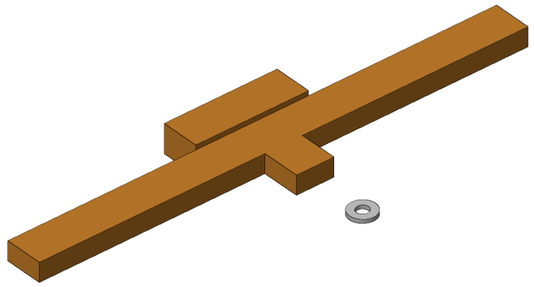
3 Part catalogue (axonometric views)



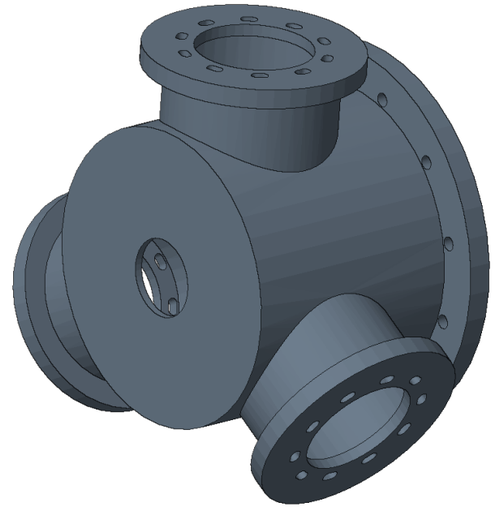
NS6-100 ×3
Blade, glass/epoxy, glass UD caps (B1–B3)
E-glass/epoxy, PVC core
make · 20.6 kg each · drawing NS6-100



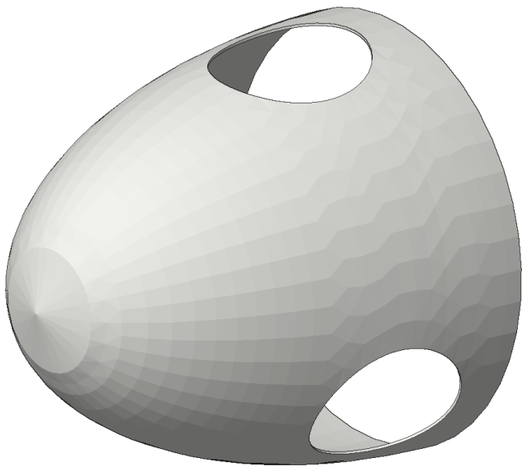
NS6-105 ×30
Root insert M12 (10 per blade, bonded)
C45E+QT zinc-flake coated
make · 0.3 kg each · drawing NS6-105



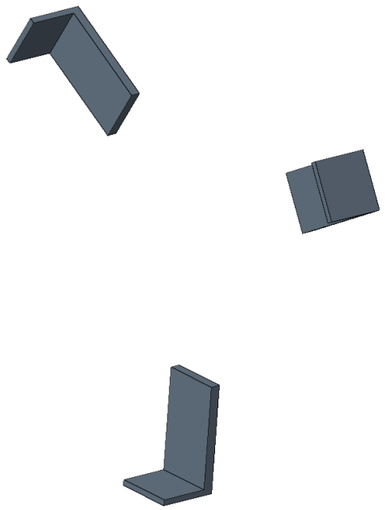
NS6-130 ×1
Pitch washer plates 28 × 4 hardened (30)
C45 hardened 45 HRC, zinc-flake
make · 0.5 kg each · drawing NS6-130



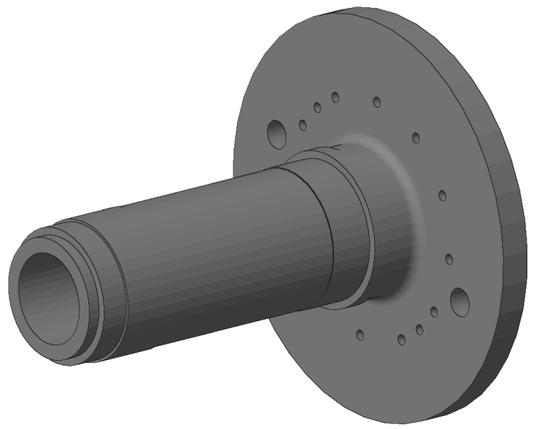
NS6-200 ×1
Fixed hub weldment
S355NL, welded, stress relieved, machined
make · 79.2 kg each · drawing NS6-200



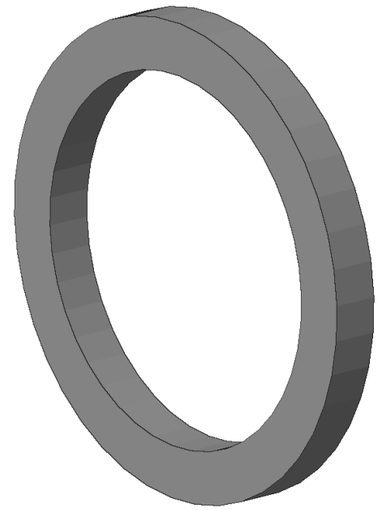
NS6-210 ×1
Spinner (GRP, 4 mm)
glass/polyester, gelcoat RAL 9010
make · 8.7 kg each · drawing NS6-210



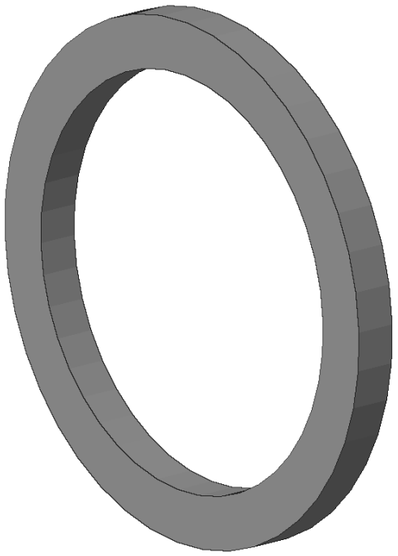
NS6-211 ×1
Spinner brackets (3)
S235 galv.
make · 2.0 kg each · drawing NS6-210



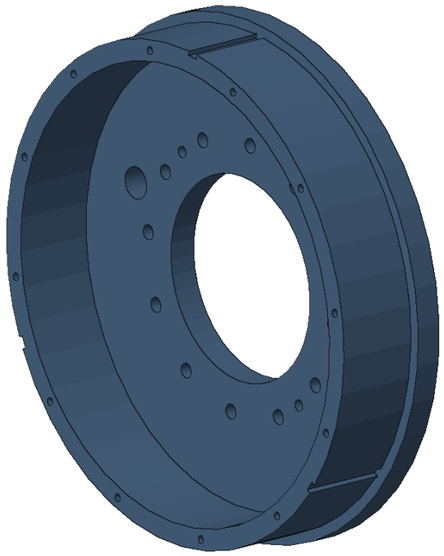
NP1-100 ×1
Spindle
42CrMo4+QT forged
make · 56.5 kg each · drawing NP1-100



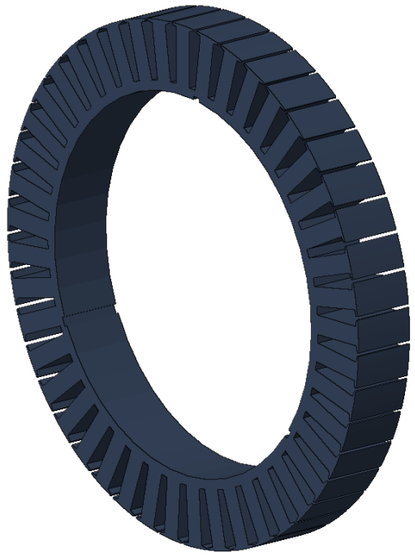
NP1-101 ×1
Front spacer / seal counterface Ø180 (Rev B)
42CrMo4+QT, induction hardened
make · 1.4 kg each · drawing NP1-101



NP1-102 ×1
Rear seal sleeve Ø195 on the Ø155 shoulder (Rev B)
42CrMo4+QT, induction hardened
make · 1.3 kg each · drawing NP1-102



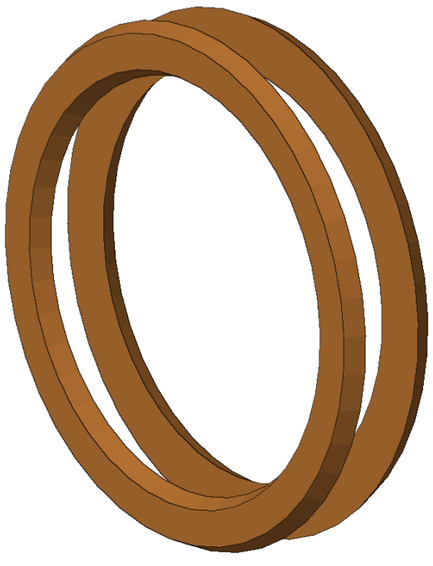
NP1-200C ×1
Stator carrier with 45 dovetail grooves
–
make · 39.9 kg each · drawing NP1-200C



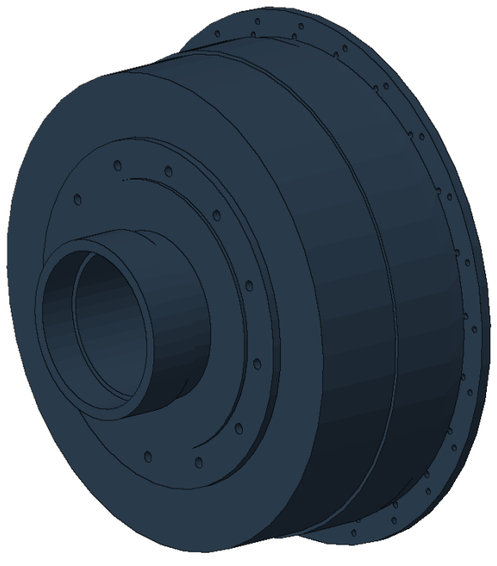
NP1-201C ×1
Stator: 45 tooth-segment lamination stacks
–
make · 50.5 kg each · drawing NP1-201C



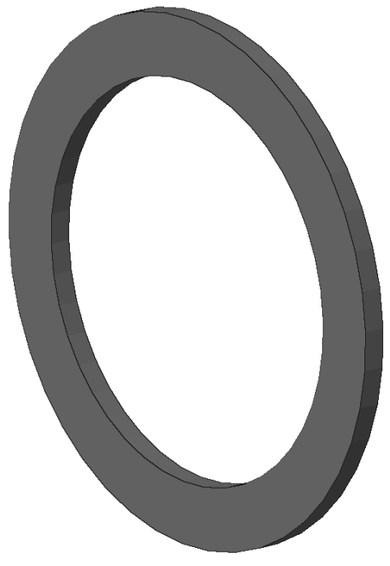
NP1-202 ×1
Stator front clamp ring
1.4404 (non-magnetic)
make · 2.2 kg each · drawing NP1-202



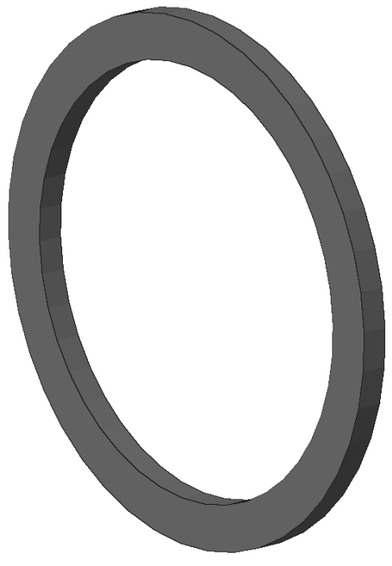
NP1-203C ×1
Coils on segments, 45 × 28 turns (end windings shown)
Cu enamelled Ø1.55 class 200, NKN 0.30
make · 20.0 kg each · drawing NP1-203C



NP1-210C ×1
Rotor drum weldment (G6: no fins, no ribs)
S355J2+N
make · 91.2 kg each · drawing NP1-210C



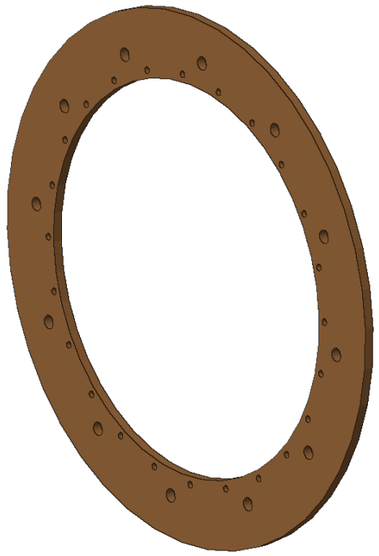
NP1-211-F ×1
Front bearing seal cover
S355J2
make · 1.5 kg each · drawing NP1-211



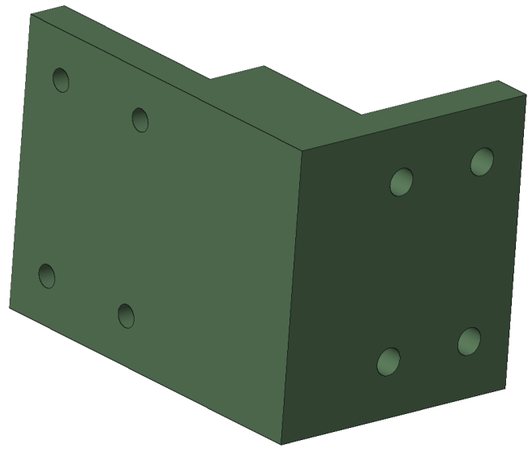
NP1-211-R ×1
Rear bearing seal cover, bore Ø198 (Rev B)
S355J2
make · 1.4 kg each · drawing NP1-211



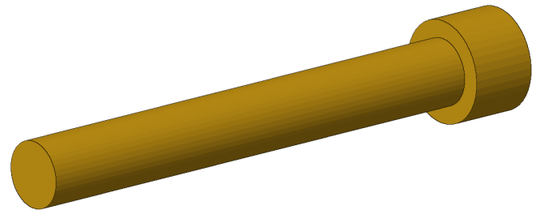
NP1-220E ×1
Magnets 200 × N45SH 30×15×12 + comb + sleeve
NdFeB N45SH
make · 10.7 kg each · drawing NP1-220E/221C/222C/223C



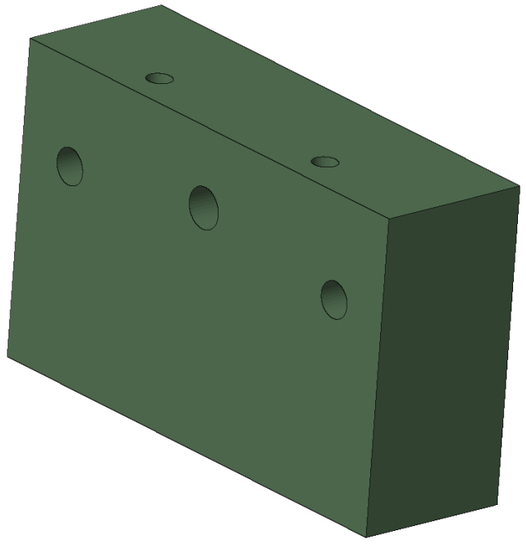
NP1-230 ×1
Brake disc Ø880/Ø652 with 12 lock holes (Rev B)
C45E+N 16 mm
make · 33.6 kg each · drawing NP1-230



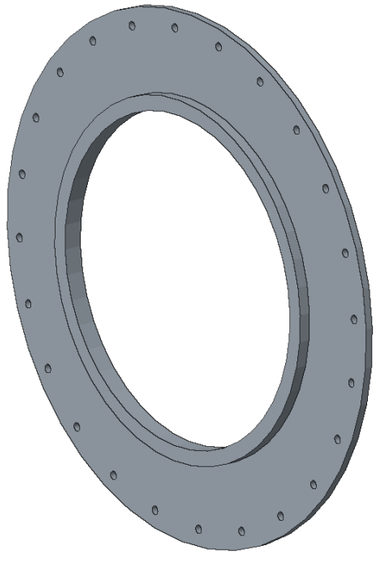
NP1-310 ×1
Rotor brake caliper bracket (left, Rev B)
S355J2+N 25 plate
make · 18.1 kg each · drawing NP1-310



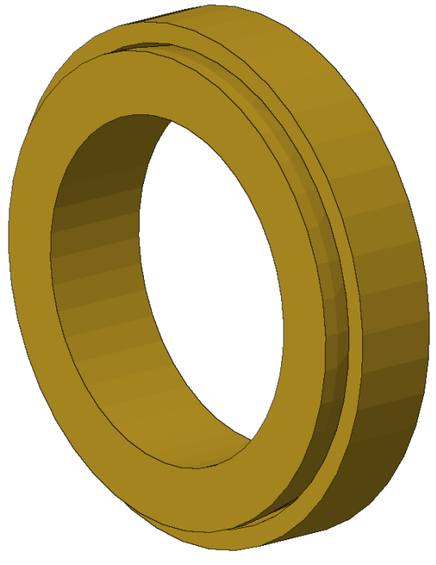
NP1-320 ×1
Rotor lock pin Ø20 (engaged, top)
42CrMo4+QT
make · 0.5 kg each · drawing NP1-320



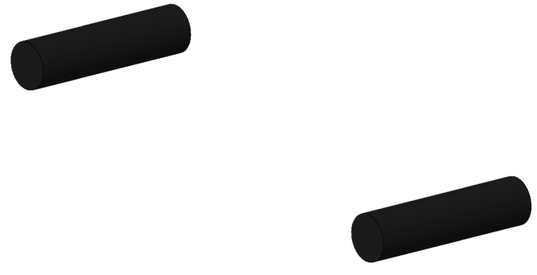
NP1-321 ×1
Lock pin holder with 2 overspeed sensor seats
S355J2
make · 24.5 kg each · drawing NP1-321



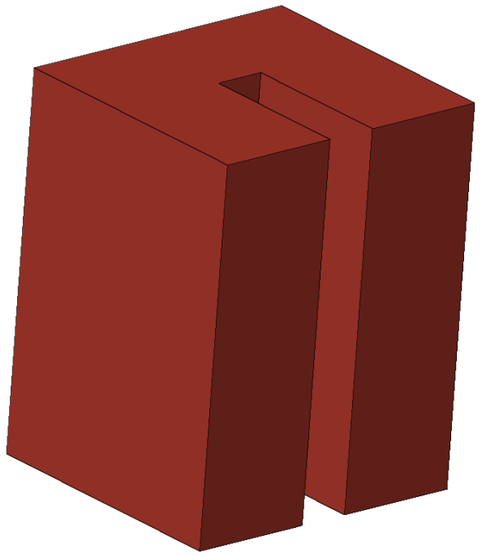
NS6-610 ×1
G6 plain rear cover (no fan blades)
EN AW-5083-H111
make · 7.2 kg each · drawing NS6-610



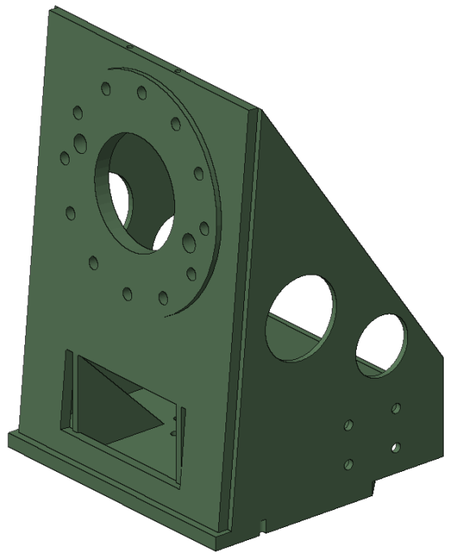
NS6-913 ×2
Tapered roller bearing 32028 X (front + rear)
bought 32028 X
buy · 4.0 kg each · drawing NS6-910



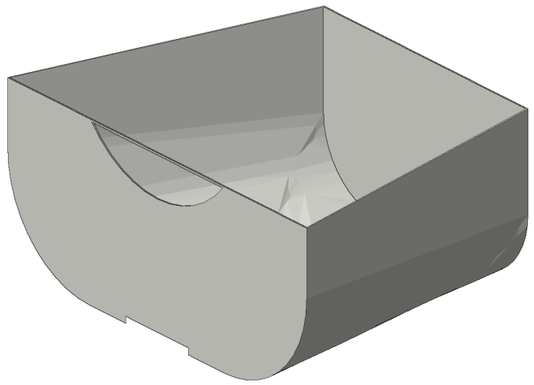
NS6-914 ×1
Overspeed sensors M18 inductive (2, bought)
bought M18 PNP
buy · 0.4 kg each · drawing NS6-910



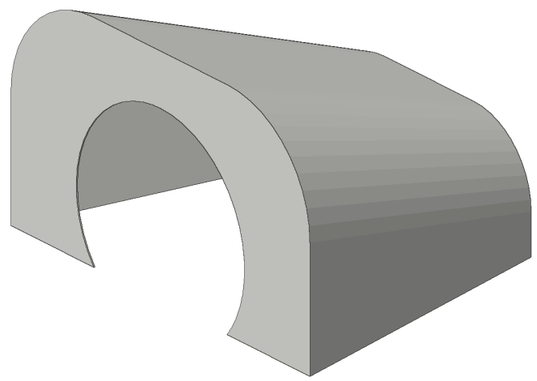
NS6-915 ×1
Spring-applied brake caliper, 1.5 kNm pack (bought)
bought, spring-applied
buy · 9.0 kg each · drawing NS6-910



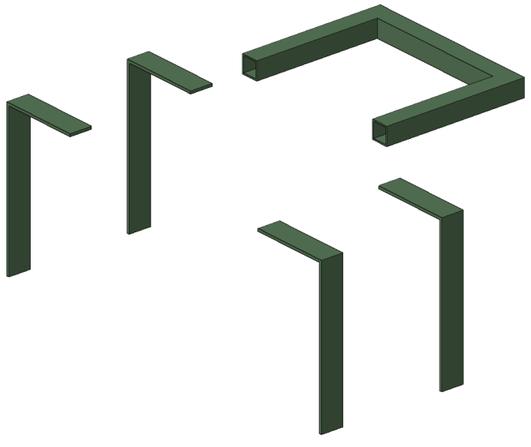
NS6-300 ×1
Bedplate weldment 6 kW (light)
S355J2+N plates, stress relieved, machined
make · 122.1 kg each · drawing NS6-300



NS6-700-1 ×1
Nacelle cover, lower tray (GRP 5 mm)
glass/polyester, gelcoat RAL 9010
make · 15.7 kg each · drawing NS6-700



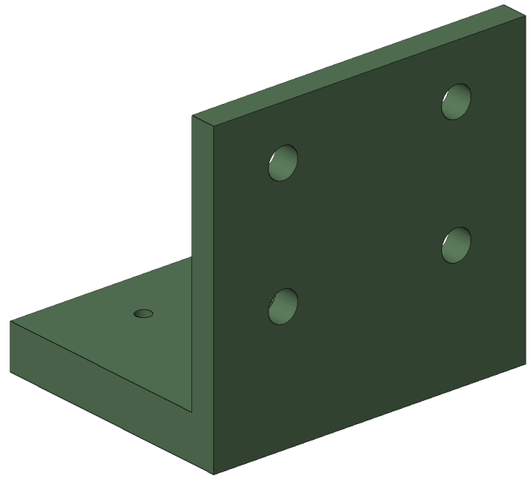
NS6-700-2 ×1
Nacelle cover, upper lid (GRP 5 mm)
glass/polyester, gelcoat RAL 9010
make · 17.5 kg each · drawing NS6-700



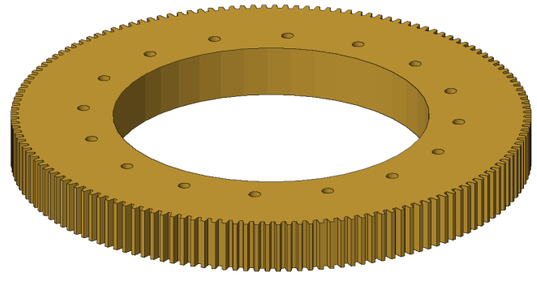
NS6-710 ×1
Cover support arms + rear rail
S235 galv.
make · 8.3 kg each · drawing NS6-710



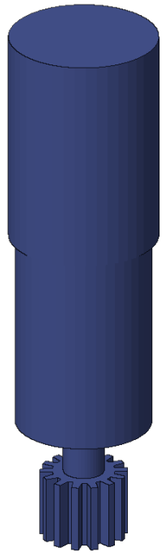
NS6-720 ×1
Met mast: anemometer, vane, air terminal
1.4404 tube 42.4 × 2 + bought sensors
make · 4.5 kg each · drawing NS6-720



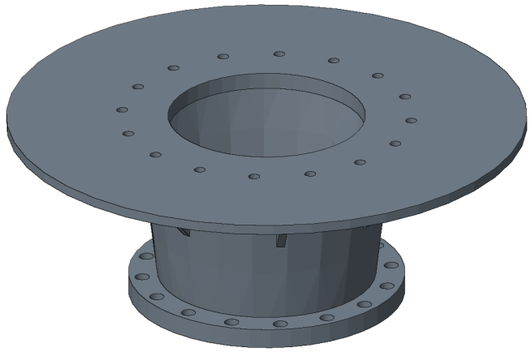
NP1-420 ×2
Yaw drive bracket
S355J2+N
make · 7.7 kg each · drawing NP1-420



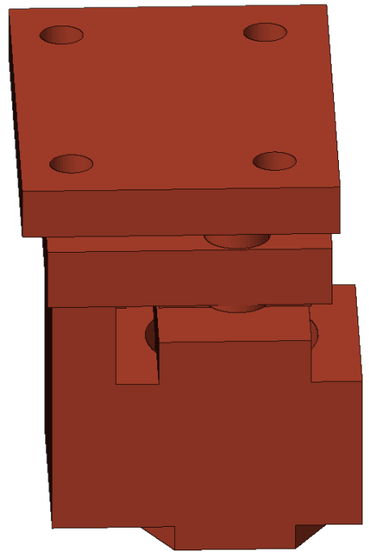
NS6-911 ×1
Yaw slewing ring Dpw 360, z150 m4
bought: 42CrMo4, induction hardened teeth
buy · 55.0 kg each · drawing NS6-911



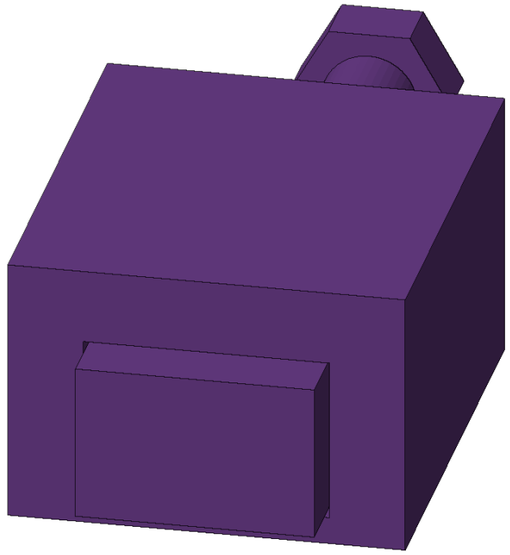
NS6-912 ×2
Yaw gearmotor 0.18 kW 1:1680, motor brake 2 Nm
bought
buy · 19.0 kg each · drawing NS6-912



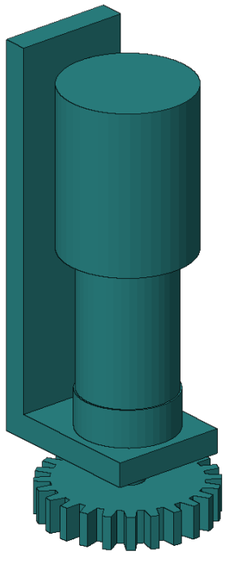
NY-400 ×1
Tower-top adapter with friction track
S355J2+N weldment
make · 106.7 kg each · drawing NY-400



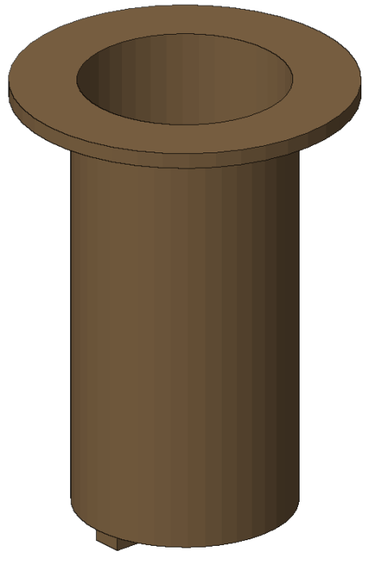
NY-410 ×2
Friction pad unit
S355 / DIN 2093 springs / organic pads
make · 14.0 kg each · drawing NY-410



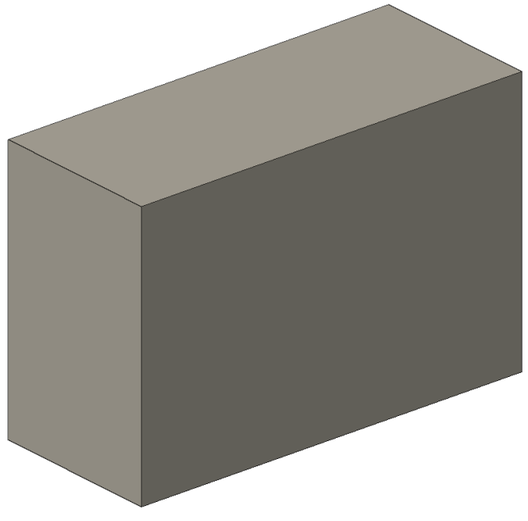
NY-430 ×1
Yaw lock (screw block)
42CrMo4 / S355
make · 6.5 kg each · drawing NY-430



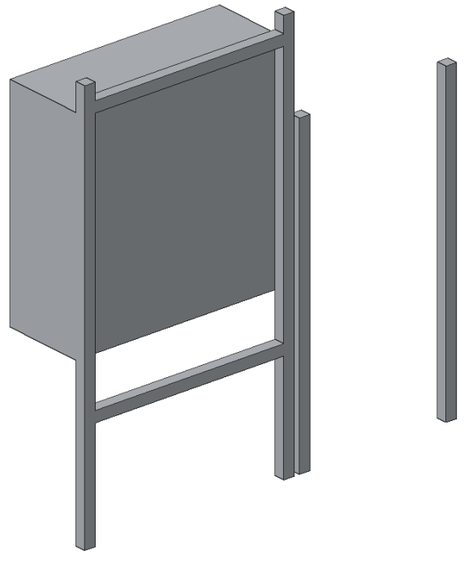
NY-440 ×1
Yaw encoder + cam limit switch unit
S355 + absolute encoder + cam switch
make · 6.0 kg each · drawing NY-440



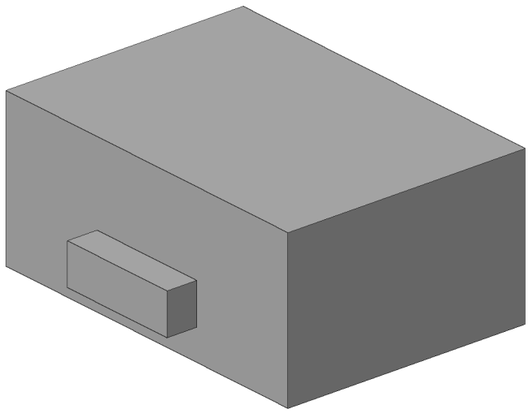
NY-460 ×1
Cable hanger tube + loop cleat
S355 tube 168.3 × 4.5
make · 10.0 kg each · drawing NY-460



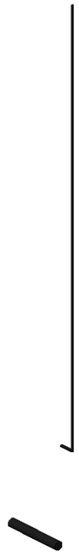
NS6-600 ×1
Braking resistor box 3 × 4 Ω (vented)
1.4301 box + steel-grid resistors
make · 18.0 kg each · drawing NS6-600



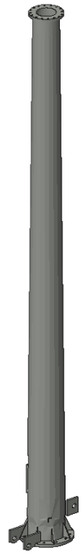
NS6-620 ×1
Tower base cabinet IP55 800 × 600 × 300 on stand
stand S235 galv.; bought enclosure
make · 95.0 kg each · drawing NS6-620



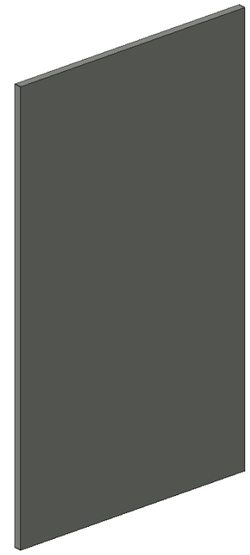
NS6-730 ×1
Nacelle junction box IP66 (400 × 340 × 150)
bought: 1.4301 enclosure
make · 9.0 kg each · drawing NS6-800



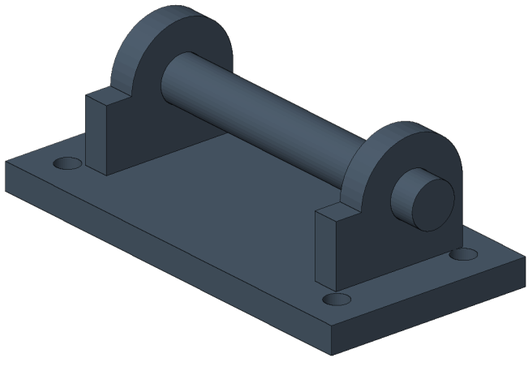
NS6-800 ×1
Drop cable + base duct (cabling route)
H07RN-F 4G10 + 2 × LiYCY signal; PE duct Ø110
make · 12.0 kg each · drawing NS6-800



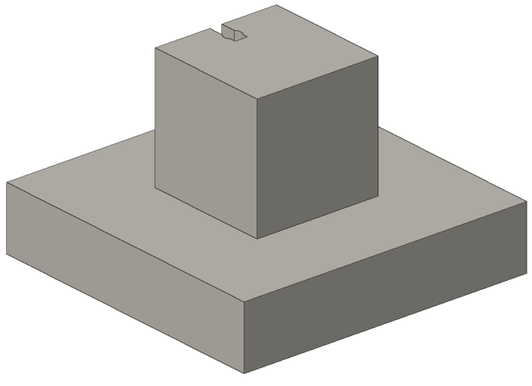
NS6-400 ×1
Tower 8 m, Ø460 → Ø320, t 10/8, one piece
S355J2+N, hot-dip galvanised
make · 697.5 kg each · drawing NS6-400



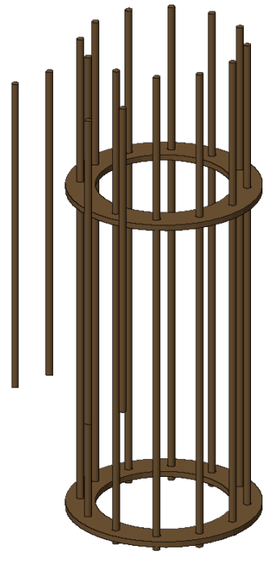
NS6-401 ×1
Hand-hole cover 200 × 350 × 6 with gasket
S355 galv.
make · 3.3 kg each · drawing NS6-400



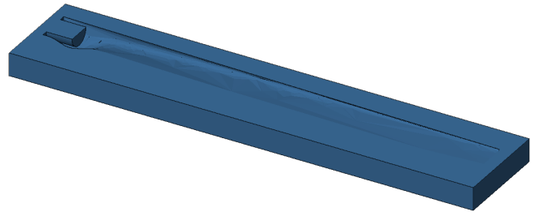
NS6-410 ×1
Base hinge brackets + hinge pin Ø50
S355J2+N galv.; pin 42CrMo4+QT
make · 37.4 kg each · drawing NS6-410



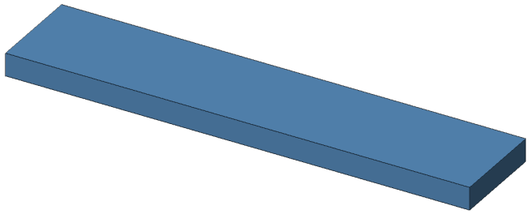
NS6-500 ×1
Spread footing (firm ground) with pedestal
C30/37 XF4 air-entrained, B500C
make · 13200.0 kg each · drawing NS6-500



NS6-510 ×1
Anchor cage 12 × M24 + 4 hinge anchors
M24 8.8 HDG threaded rod, S355 template rings
make · 143.5 kg each · drawing NS6-510



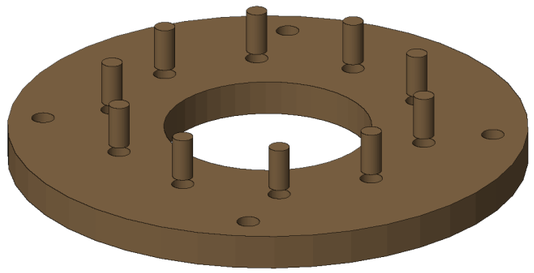
NS6-110-1 ×1
Blade mould, lower (pressure side) half on frame
GRP tooling laminate 12 mm, vinylester tooling gelcoat
make · 75.4 kg each · drawing NS6-110



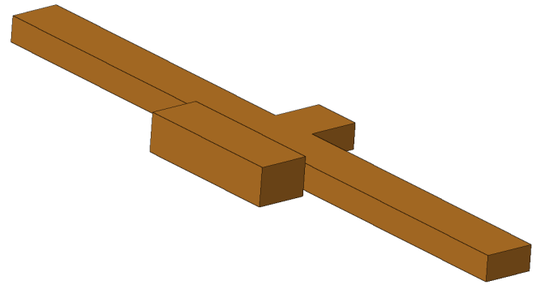
NS6-110-2 ×1
Blade mould, upper (suction side) half
GRP tooling laminate 12 mm
make · 75.4 kg each · drawing NS6-110



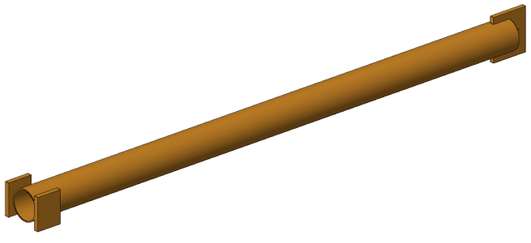
NS6-110-3 ×1
Mould frame (RHS 80 × 80 × 4)
S235 RHS 80×80×4
make · 155.4 kg each · drawing NS6-110



NS6-115 ×1
Root insert fixture (tool)
S355, machined
make · 9.2 kg each · drawing NS6-115



NS6-135 ×1
Pitch setting gauge (tool)
Al 6082 + digital inclinometer
make · 1.6 kg each · drawing NS6-130

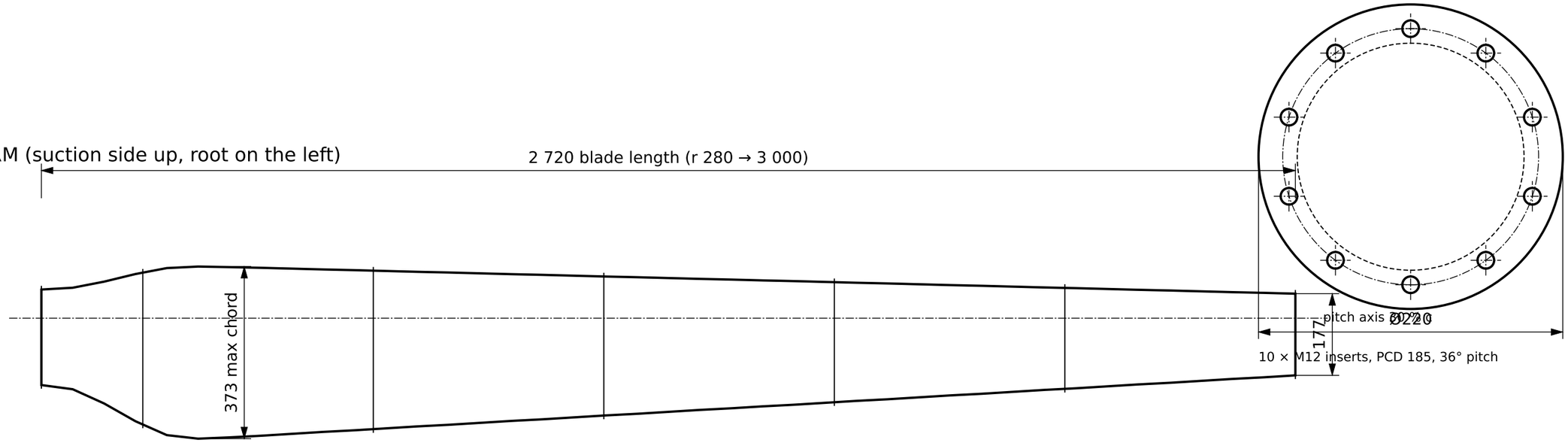


NS6-420 ×1
Gin pole 3.0 m (erection tool)
S355 tube 168.3 × 6.3
make · 94.0 kg each · drawing NS6-420

4 S6 drawing set, revision B

New and changed S6 parts, each with its axonometric view

PLANFORM (suction side up, root on the left)



ROOT END (from the hub)

r mm	chord	twist °	t/c	cap mm
280	207	20.0	1.00	4.25
400	254	20.0	0.71	4.25
600	370	20.0	0.32	4.25
800	360	17.1	0.28	4.25
1000	343	13.4	0.26	4.10
1400	310	8.5	0.24	3.38
1800	277	5.5	0.21	2.41
2200	244	3.6	0.19	1.44
2600	210	2.2	0.18	1.00
3000	177	1.2	0.18	1.00

AXONOMETRIC VIEW (not to scale)



NOTES

1. Aerofoil family and twist from analysis/rotor/bem.py (TSR 5.5, cl 1.0, R 3.0 m, root r 0.28 m); twist given at the pitch axis, positive nose up towards feather; blade set at −2° in the arc slots of the hub root flange (NS6-130).
2. Structure doc 30 §2 / doc 10 model: two glass UD caps (table), ±45° glass skins, PVC foam core, single shear web 1.5 mm glass; root wall builds up 12 mm over the first 0.7 m. Cap buckling at minimum thickness governs (use 1.00); strength 0.43 (SLM H), fatigue 0.10 with simulated loads (doc 31).
3. Root: 10 bonded steel inserts M12 NS6-105, PCD 185, pull-out ≥ 60 kN each (test). Bolt force 43 kN at SLM H. Sections and laminate: NS6-101; mould NS6-110.
4. Mass 20.6 kg ± 0.3 kg; static moment matched within 0.5 % per set of three (balance class doc 09). Erosion tape on the outer 1 m of the leading edge.
5. First flap 15.2 Hz, first edge 38.3 Hz at standstill. Tip deflection 0.06 m, tower clearance 0.83 m.

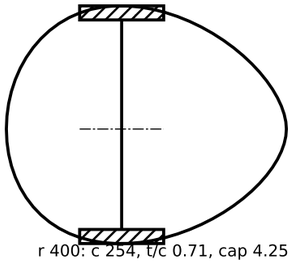
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

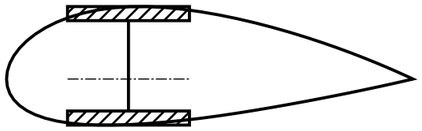
BLADE · PLANFORM, TWIST AND ROOT

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-100	B · 2026-11-06	1:10
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
E-glass/epoxy infused; glass UD caps; PVC core shear web	20.6 kg	Claude / not checked

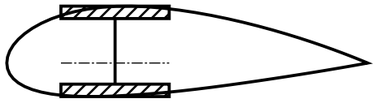
SECTIONS (caps hatched ×3 thickness for clarity, web at 30 % c; LE to the left)



r 400: c 254, t/c 0.71, cap 4.25



r 700: c 368, t/c 0.29, cap 4.25



r 1200: c 327, t/c 0.25, cap 3.75



r 2000: c 260, t/c 0.20, cap 1.93



r 2800: c 194, t/c 0.18, cap 1.00

Zone (r mm)	Shell skin	Core	Spar cap (each side)	Web
280–700 root	2 × 600 g ±45° + 0–12 mm UD/biax build-up	–	merges into root	–
700–1 400	2 × 450 g ±45°	PVC H80 8 mm	UD 1 200 g: 4.25→3.75 mm	2 × 450 g ±45° on 10 mm core
1 400–2 200	2 × 450 g ±45°	PVC H80 6 mm	UD: 3.50→2.25 mm	same
2 200–3 000 tip	2 × 300 g ±45°	PVC H80 5 mm	UD: 2.00→1.00 mm (min)	ends at r 2 900
LE r 2 000–3 000	erosion tape 0.3 mm PU			

NOTES

1. Vacuum infusion, epoxy (Tg ≥ 80 °C, cold-curing then post-cure 8 h at 70 °C), fibre volume ≥ 50 %. Each blade: two shells + web, bonded with structural paste (1–3 mm gaps), leading-edge and trailing-edge flanges 30 mm.
2. Cap thickness follows design6.json t_cap_mm (doc 30 §2); minimum cap 1.0 mm governed by buckling: do not reduce. UD glass E-glass 1 200 g/m² stitched, 0° ± 2° to the pitch axis.
3. Root: laminate builds up from r 700 to a 28 mm wall at the root face around the 10 bonded inserts NS6-105 (fixture NS6-115). Machine the root face square to the pitch axis within 0.2 mm after cure.
4. Gelcoat: polyurethane, white RAL 9010, 0.4 mm. Drain hole Ø4 at the tip trailing edge. Lightning: none on 3 m blades (LPL I, doc 02: tower and nacelle carry it; blade receptor option).
5. Quality: weigh and record mass + static moment; tap test the bonds; one blade per 20 static-tested to SLM H × 1.35 (doc 30).

AXONOMETRIC VIEW (not to scale)

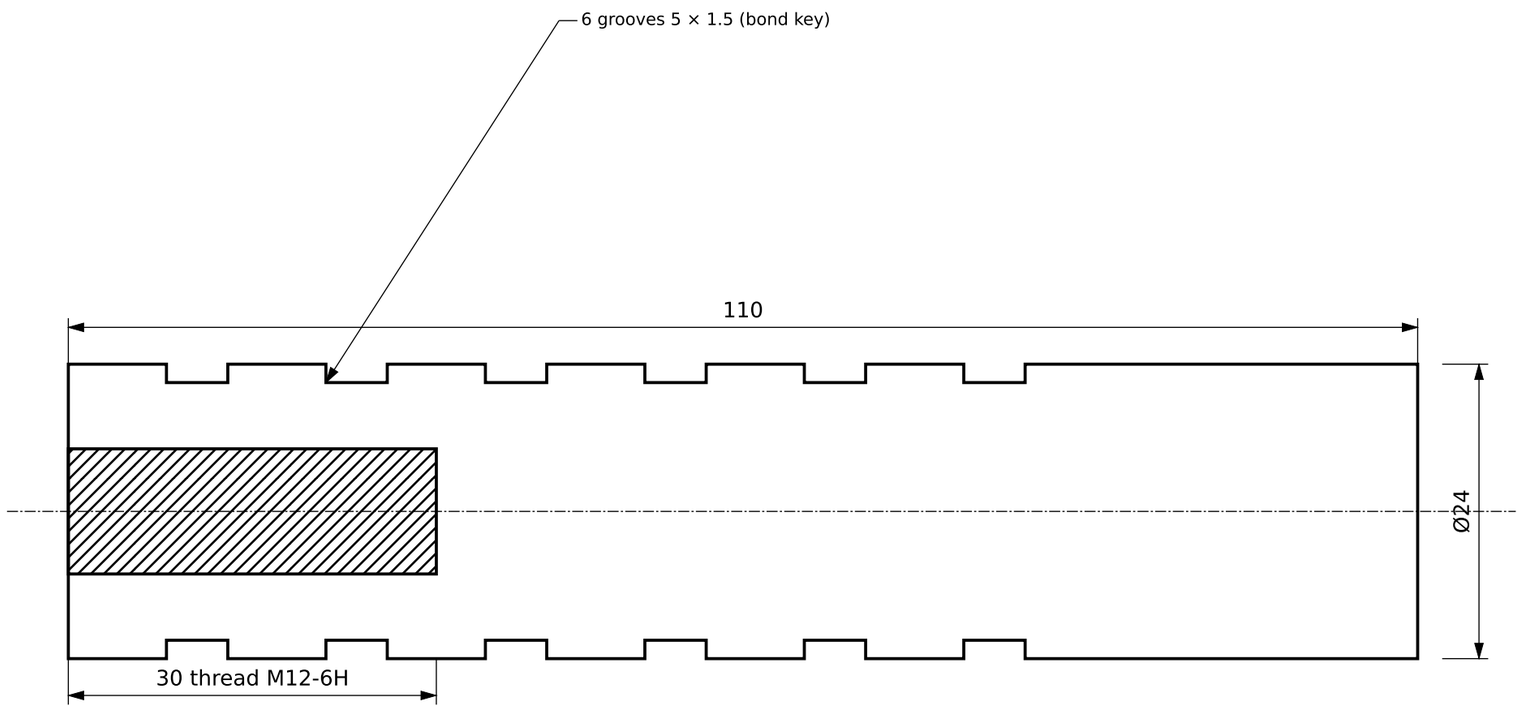


Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

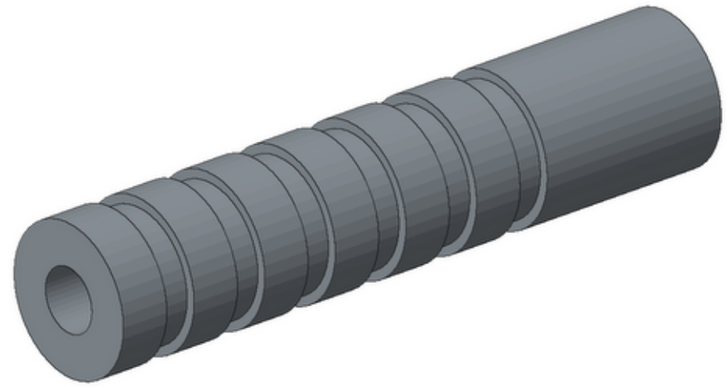
6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

BLADE · SECTIONS AND LAMINATE SCHEDULE

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-101	B · 2026-11-06	1:2 / 1:4
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
E-glass/epoxy (infusion), PVC H80 core	20.6 kg	Claude / not checked



AXONOMETRIC VIEW (not to scale)



NOTES

1. Bonded into the blade root laminate during lay-up with the root fixture NS6-115: insert axis parallel to the pitch axis within 0.3°, PCD 185 ± 0.2, face flush ± 0.2 after root facing.
2. Pull-out test: 3 inserts per batch in a root coupon, ≥ 60 kN each (design 43 kN at SLM H, doc 30 §2).
3. Threads protected with plastic plugs until assembly; studs M12 10.9 × 75 (NS6-900) screwed in 30 mm with Loctite 270, 40 Nm.

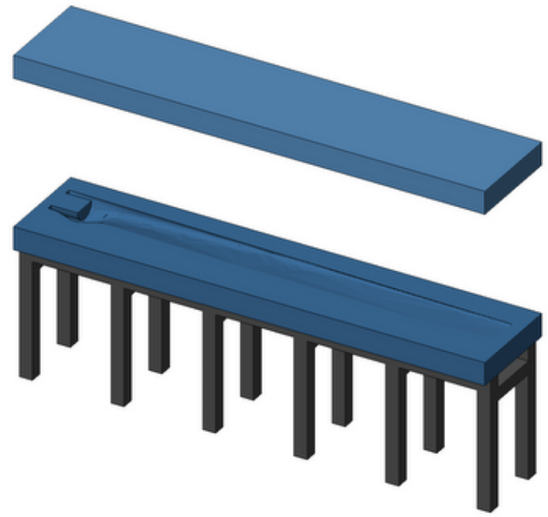
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

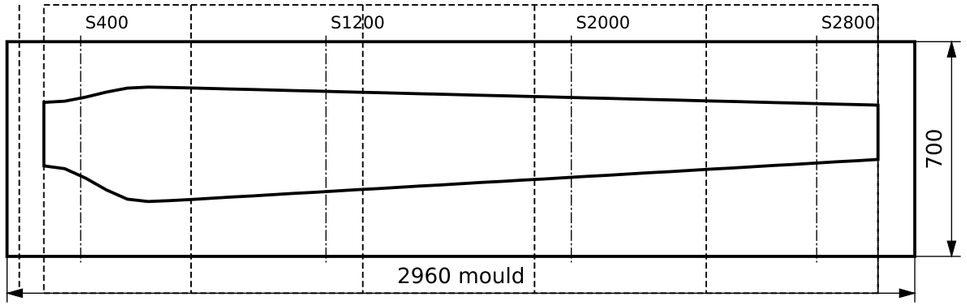
BLADE ROOT INSERT M12 (BONDED)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-105	B · 2026-11-06	2:1
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
C45E+QT, zinc-flake coated (Geomet 500)	0.33 kg	Claude / not checked

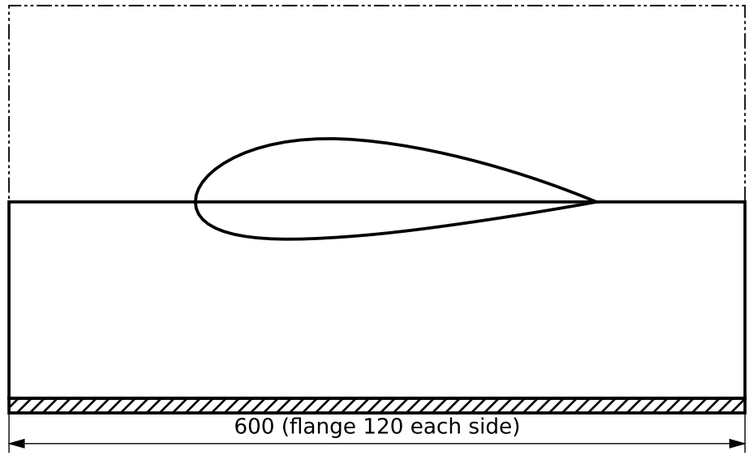
AXONOMETRIC VIEW (not to scale)



PLAN, LOWER HALF (pressure side), frame RHS 80 × 80 × 4 hidden



SECTION S1200: lower half solid, upper half phantom; parting line on LE/TE



Step	Process
1 Plug	CNC-machined tooling board (polyurethane 0.7 g/cm³) on an MDF/steel core; ruled parting surface along LE and TE; sand P600, seal, release
2 Halves	Tooling gelcoat 1 mm, skin 2 × 300 g CSM, 10 mm vinylester/glass build-up, egg-crate ribs 12 mm GRP every 400 mm
3 Frame	RHS 80 × 80 × 4 cradle, 6 legs, levelling feet, pins Ø20 at 4 places on the flanges for closing
4 Heating	silicone heating mats 300 W/m² under the lower half, 70 °C post-cure; thermocouples ×4
5 Web	flat-plate web mould: tapered aluminium plate 2 900 × 220 on the bench (NS6-115)

NOTES

1. One mould set makes all blades (≥ 300 parts before refurbishment). Mould flange 120 mm with a 25 mm vacuum sealing land and a 6 mm resin trap groove.
2. Tolerances: profile ± 0.5 mm, twist ± 0.2°, length ± 2 mm. Check each new mould against the plug with the five section templates S400 … S2800.
3. Root end: 20 mm steel end plate with the bolt pattern of the root fixture NS6-115 (4 × M12), square to the pitch axis.

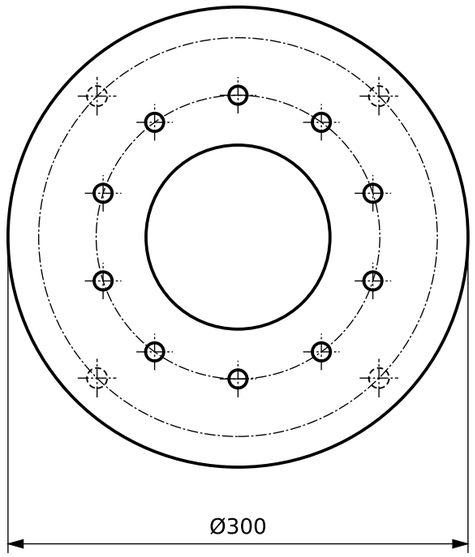
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

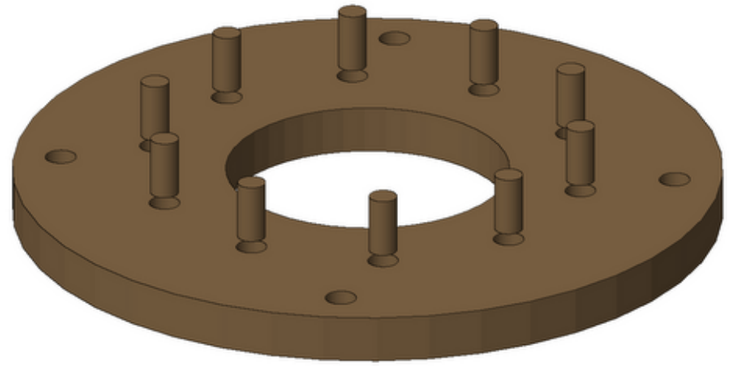
BLADE MOULD SET (PLUG, TWO HALVES, FRAME)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-110	B · 2026-11-06	1:20
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
GRP tooling laminate on S235 RHS frame	≈ 2 × 75 kg + frame 155 kg	Claude / not checked

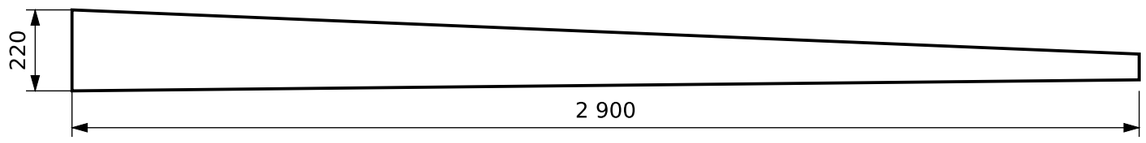
ROOT FIXTURE, view on the spigots: 10 × Ø11.8 g6 spigots × 30 on PCD 185



AXONOMETRIC VIEW (not to scale)



WEB MOULD PLATE (tapered, 10 mm Al)



NOTES

1. Fixture bolts to the root end plate of the lower mould half (4 × M12); inserts NS6-105 slide over the spigots and are screwed on with M12 bolts so their faces sit on the fixture face.
2. After demoulding, remove the fixture, face the root on the root-facing jig (lathe or portable facer) to the root length, check PCD with the hub-flange template.
3. Web mould: flat plate with 25 mm edge flanges; web is infused flat (core 10 mm, 2 × 450 g ±45° each face) and cut to the template lines.

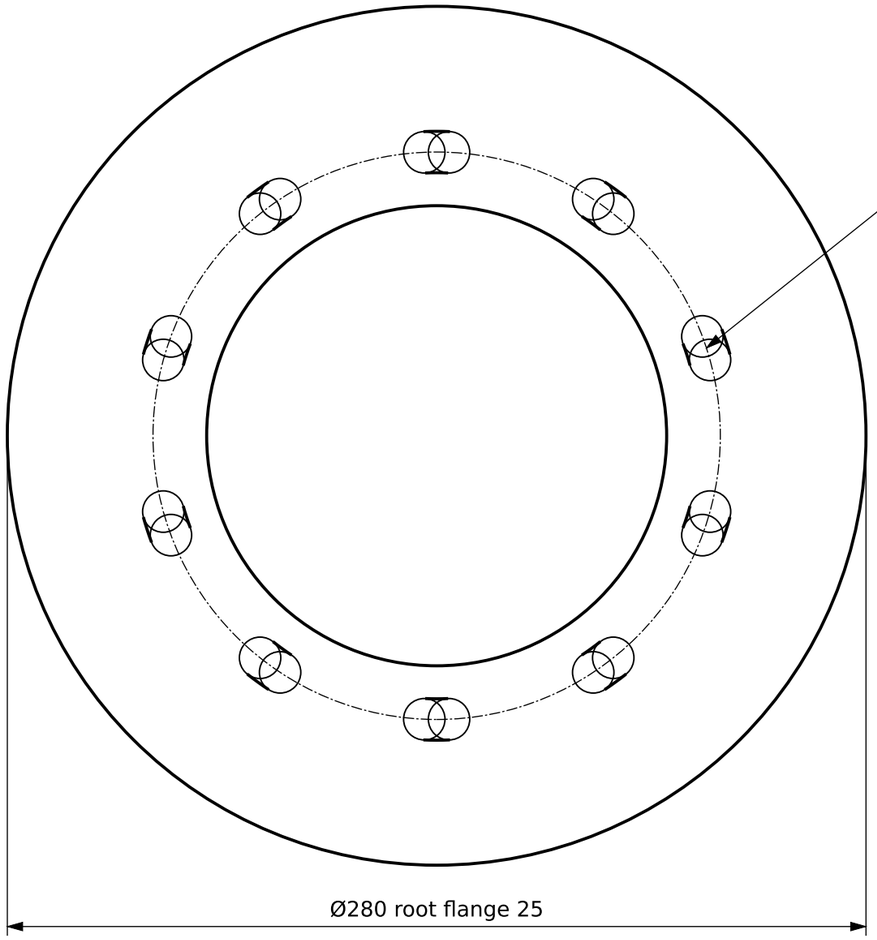
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

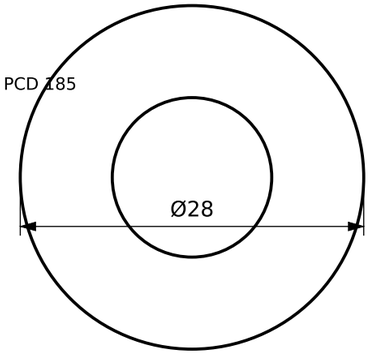
ROOT INSERT FIXTURE AND WEB MOULD (TOOLS)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-115	B · 2026-11-06	1:4
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355 machined; Al 5083 plate	9 kg (fixture)	Claude / not checked

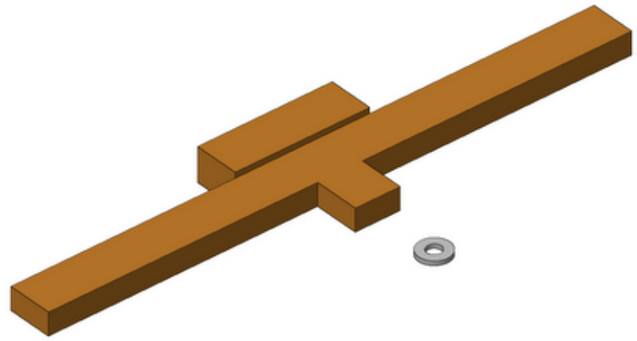
HUB ROOT FLANGE (part of NS6-200), from the blade



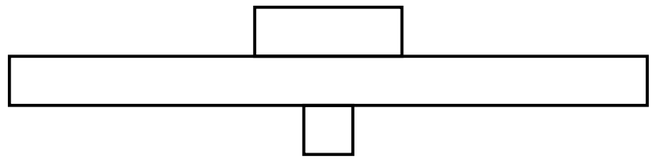
WASHER PLATE 28 × 4, 45 HRC



AXONOMETRIC VIEW (not to scale)



GAUGE NS6-135: clamps on r = 2 000 (chord line pads), inclinometer seat



NOTES

1. Doc 30 called for wedge shims; a wedge tilts the blade axis (cone), it cannot change pitch. Rev B sets pitch by rotating the blade about its axis in arc slots of the hub root flange: ±2.5° around the −2.0° standard (−4.5° … +0.5°).
2. Setting: rotor locked with B1 horizontal, gauge on the blade at r = 2 000, rotor plane reference from a level on the hub front face; set −2.0° ± 0.1°, torque the studs, re-check.
3. Joint: 10 × M12 10.9 studs + hardened washer plates over the slots; friction carries the blade pitching moment (≤ 0.05 kNm) with μ 0.2 ≥ 10× margin.

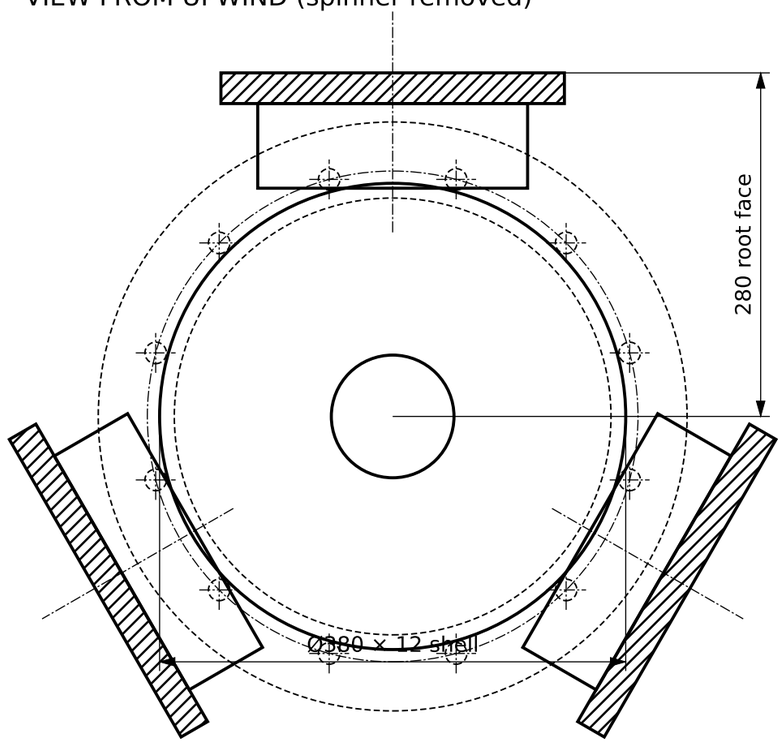
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

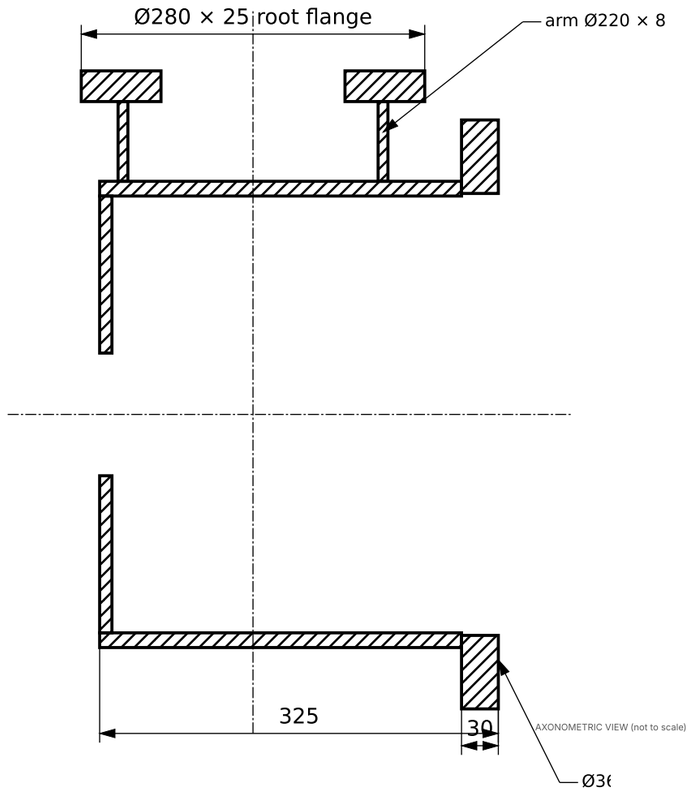
PITCH SETTING · ROOT SLOTS, WASHER PLATES, GAUGE

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-130	B · 2026-11-06	1:2 / 1:5
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
C45 hardened 45 HRC; gauge Al 6082	30 × 15 g; gauge 1.6 kg	Claude / not checked

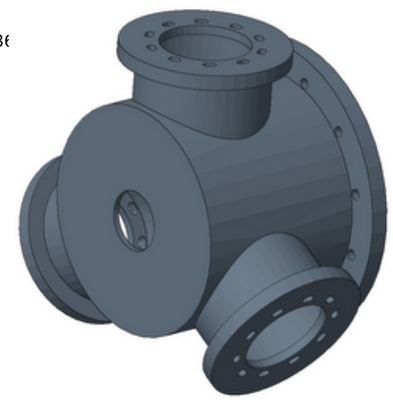
VIEW FROM UPWIND (spinner removed)



SECTION THROUGH THE UPPER ARM (x = 0 rotor plane)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Arms Ø220 × 8: parked SLM stress 68 MPa (allow 323); fatigue at the flange weld 5.7 MPa with simulated loads (allow 30.8, detail 71).
2. Drum joint: Ø360 H7 bore on the drum spigot, 12 × M16 10.9 PCD 400, 250 Nm; friction carries the 5.7 kNm design torque.
3. Root flanges machined after stress relief: face square to the arm axis within 0.05, 10 arc slots ±2.5° on PCD 185 centred on −2.0° pitch (NS6-130); arm 1 marked at the lock-pin azimuth.
4. Shell lengthened to 295 (rev A 240) so the arm intersections close on the shell; front plate 10 with Ø100 access hole. Welds full penetration, ISO 5817 B, 100 % MT, UT on flange welds. Zinc spray + C5 paint.
5. Spinner brackets NS6-211 weld on the front plate edge at 60° between the arms. FE check of the arm-to-shell weld before release (doc 30 §11).

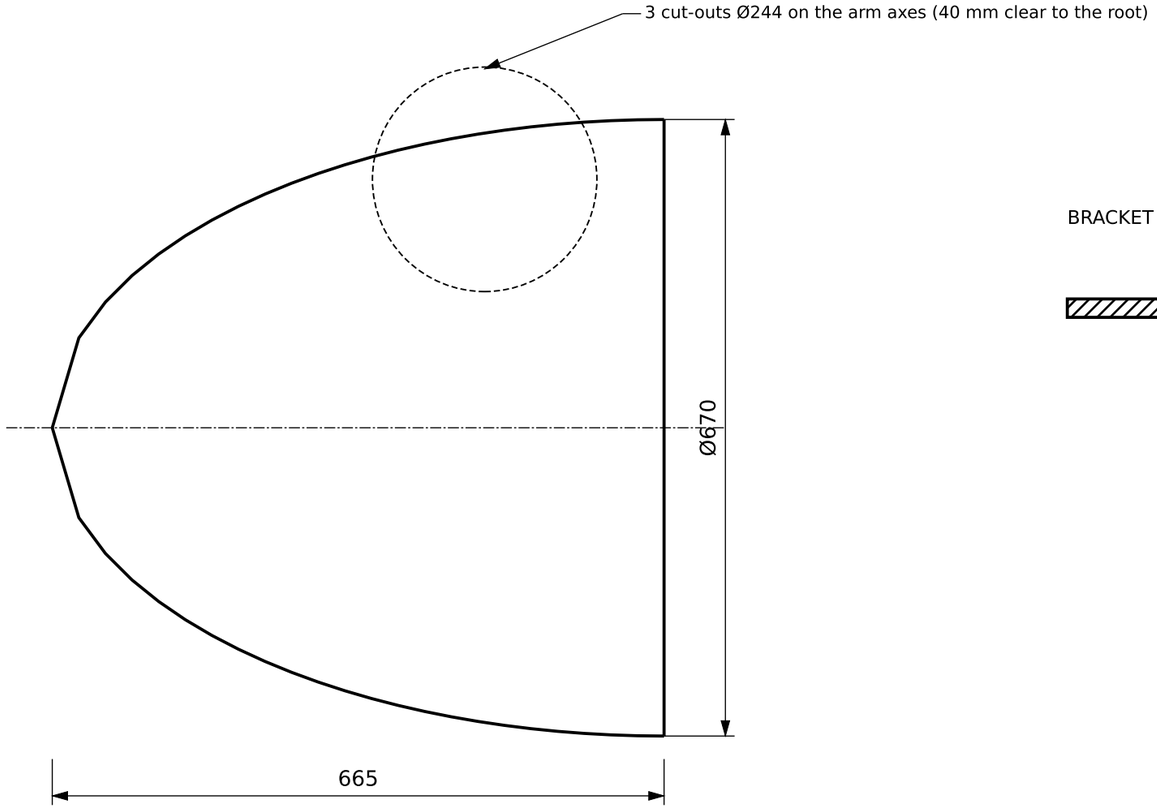
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

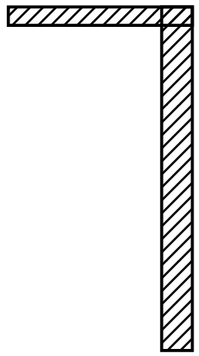
FIXED HUB WELDMENT

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-200	B · 2026-11-06	1:5
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355NL, welded, stress relieved	79 kg (CAD)	Claude / not checked

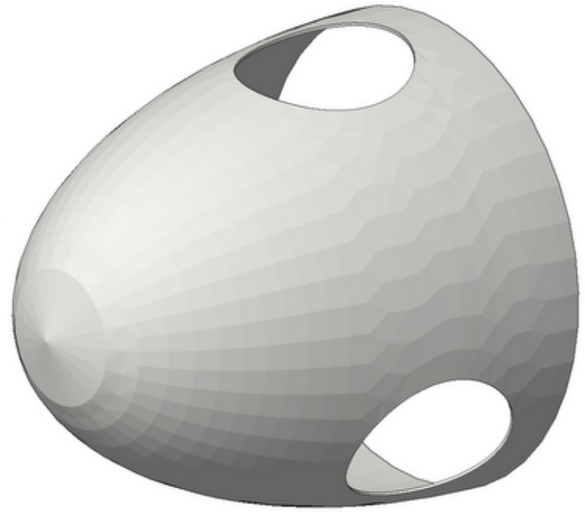
SIDE VIEW



BRACKET NS6-211: 60 × 10 flat, 3 per hub, M8 rivnut in the s



AXONOMETRIC VIEW (not to scale)



NOTES

1. Hand lay-up or RTM in a two-part female mould from the same plug supplier as the blades; 4 mm: gelcoat 0.5 + 3 × 450 g CSM + 1 × 300 g woven; flange-less rear edge with a 20 mm stiffening bead.
2. Rear edge runs 5 mm from the drum front web; the spinner does not seal (drain gaps at the cut-outs).
3. Fix with 3 × M8 1.4404 screws + rubber washers into the brackets; brackets weld to the hub front plate edge at 60° between the arms.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

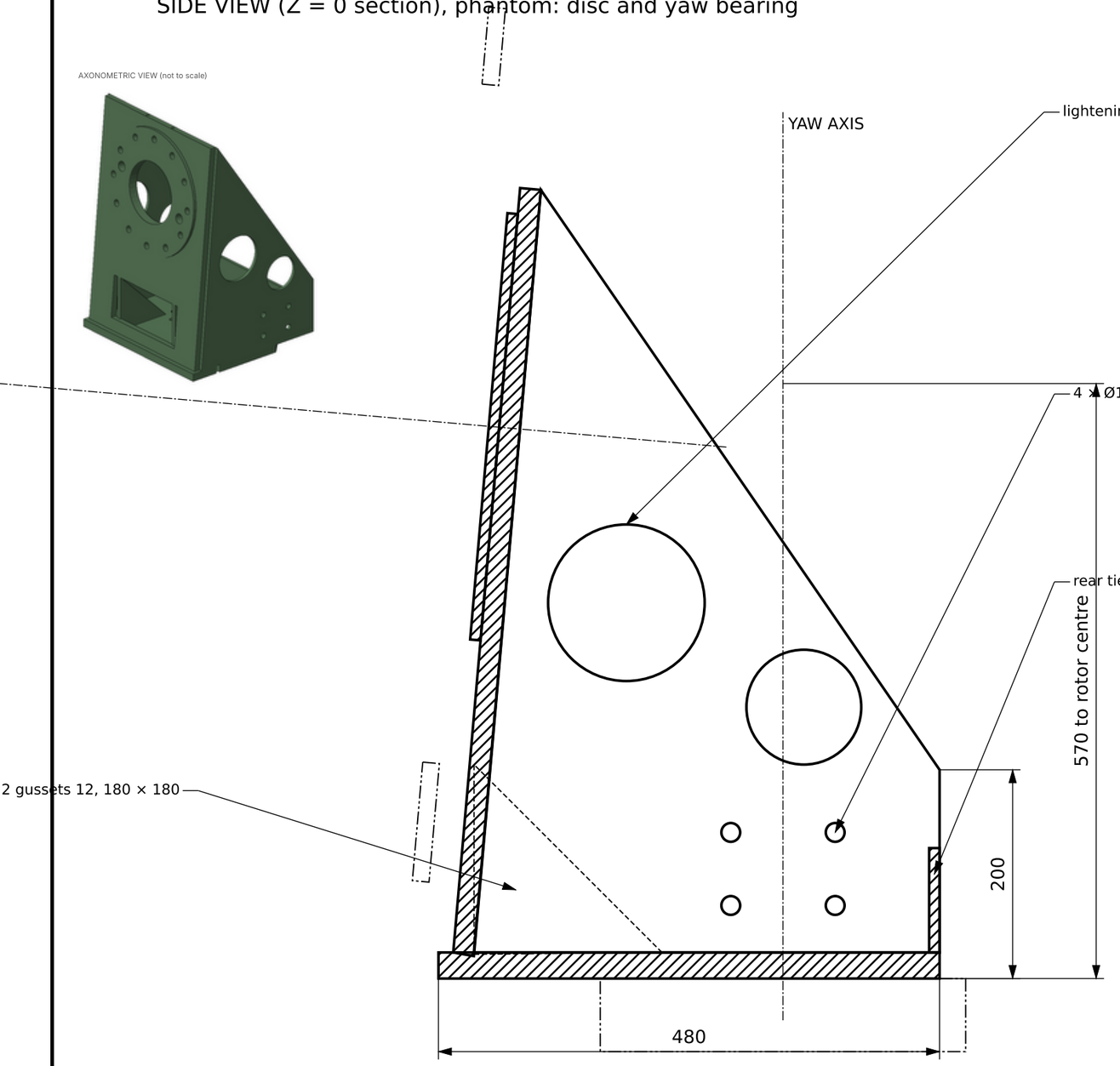
6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

SPINNER (GRP) AND BRACKETS NS6-211

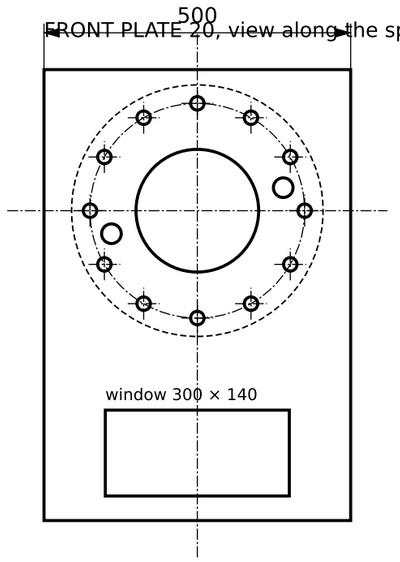
DRAWING NO.	REV / DATE	SCALE (A3)
NS6-210	B · 2026-11-06	1:5
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
glass/polyester 4 mm, gelcoat RAL 9010; brackets S275 galv.	8.7 kg + 3.6 kg	Claude / not checked

SIDE VIEW (Z = 0 section), phantom: disc and yaw bearing

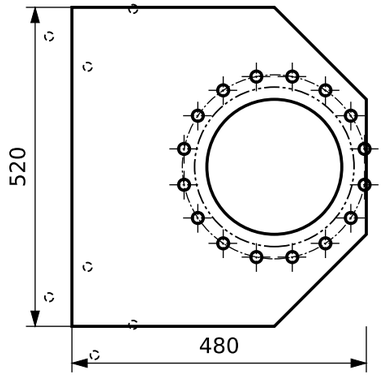
AXONOMETRIC VIEW (not to scale)



FRONT PLATE 20, view along the spindle axis



BOTTOM PLATE (yaw flange), from below



NOTES

1. Replaces NP1-300 (12 kW, 333 kg as drawn) on S6 (doc 32). Same interfaces: Ø410 × 10 pad at x 470, Ø200 H7 spigot, 12 × Ø22 PCD 350 (M20 10.9 from inside, 540 Nm), 2 × Ø32 cable holes PCD 290; bottom Ø220 bore, 16 × Ø18 PCD 300 (yaw inner ring, M16 10.9 with nuts).
2. Plates: front 20, bottom 25, sides 10 (outer faces Y ±260 for NP1-420), rear tie 10, gussets 12. Rear corners of the bottom plate chamfered 150 × 150.
3. Pad units NY-410 bolt under the front corners (4 × M12 each, tapped, hidden); hanger tube flange NY-460 on Ø260 at the bore; caliper bracket NP1-310 on the left side plate front edge; lock pin holder NP1-321 on 2 × M12 in the front plate top edge.
4. Design checks (SLM γ 3, doc 32): root section 21 MPa, front plate 51 MPa, bottom plate 144 MPa (allow 323); fatigue 0.07. Thicknesses are set by machining and slewing ring flatness, not strength: FE stiffness check of the ring seat before release.
5. Welds ISO 5817 B, 100 % MT; stress relieve; machine front and bottom plates in one set-up on a 5° fixture: front face square to the yaw flange within 0.05/400 at 5.00° ± 0.05°; ring seat flatness 0.05.

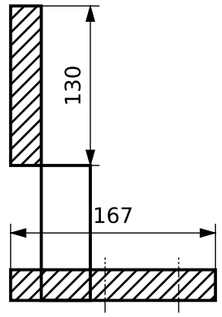
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

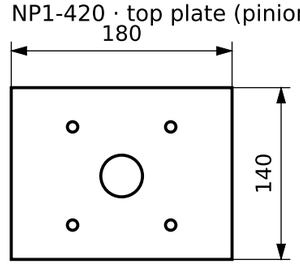
BEDPLATE WELDMENT 6 kW (LIGHT)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-300	B · 2026-11-06	1:5
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N plates, stress relieved, machined	127 kg	Claude / not checked

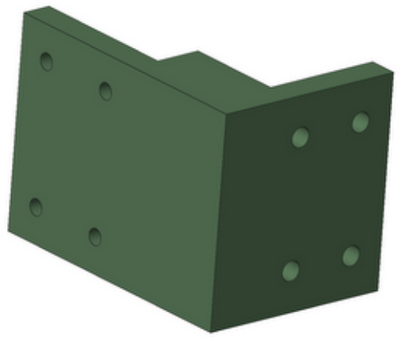
NP1-310 · plan (x along the spindle, y lateral)



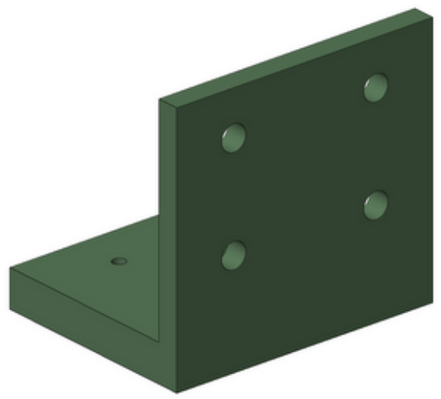
NP1-420 · top plate (pinion bore Ø34, 4 × M10)



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



NOTES

1. NP1-310 (12 kW Rev B geometry, cad/nacelle_fitcheck): only the LEFT bracket is fitted on S6 (one caliper). Mounting plate 25 with 4 × M16 tapped to suit the chosen caliper; web 25 to the bedplate side plate, 4 × M16 10.9 through, 210 Nm.
2. NP1-420 (doc 25): top plate 25 + web 25; web bolts to the side plate outer face with 4 × M16 at x ±50, z 70/140; gearbox flange 4 × M10 8.8, 49 Nm. Centre distance pinion–ring 328.0 ± 0.1 (shim the web).
3. Both: welds a = 8 fillet all round, ISO 5817 C; hot-dip galvanise after welding, then machine the caliper face.

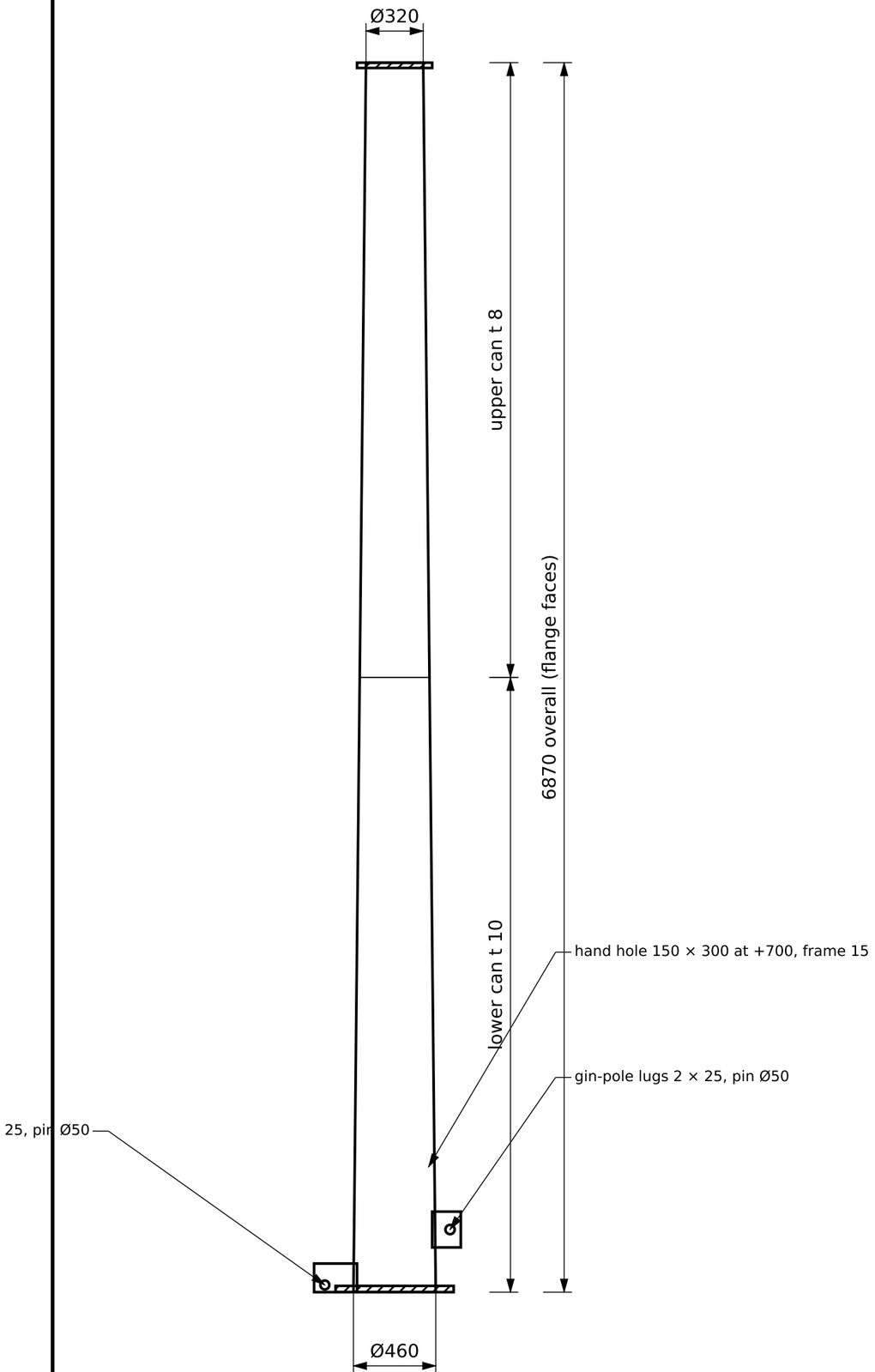
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

CALIPER BRACKET NP1-310 AND YAW DRIVE BRACKET NP1-420

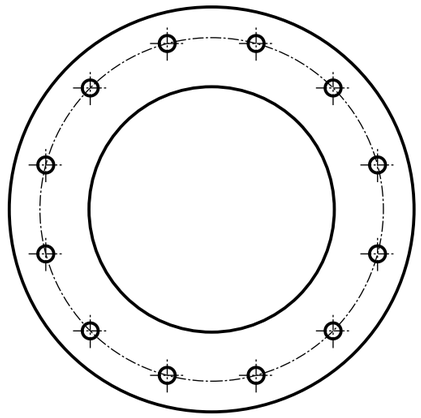
DRAWING NO.	REV / DATE	SCALE (A3)
NS6-310	B · 2026-11-06	1:5
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N plate 25, welded, galv.	18.1 kg; 2 × 7.7 kg	Claude / not checked

ELEVATION

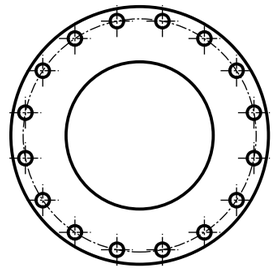


Item	Value
Shell	Ø460 × 10 → Ø320 × 8, linear taper, D/t ≤ 48.3
Cans	2 × 3.44 m rolled, one circumferential butt weld
f1 (rigid / soft+ice)	3.5 / 3.28 Hz; f2 24.04 Hz
Window	f1 ≥ 1.15 × 1P max (3.14 Hz); 3P crosses at 60–77 rpm
Governing	earthquake 0.5 g (ground D) + operation, use 0.445
Base moment (design)	238.6 kNm; SLM sensitivity use 0.601
Base bolts	12 × M24 10.9 anchors: 184.6 / 254.2 kN
Fatigue (base / mid)	0.047 / 0.0245 (turbulence × 1.234)
Tilt-up	base moment 114.7 kNm, use 0.214

FLANGES (plan)

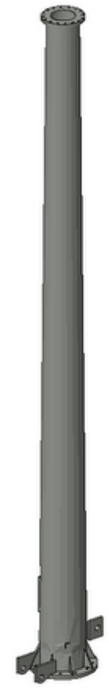


BASE FLANGE 35: 12 × Ø26 PCD 560



TOP FLANGE 30: 16 × Ø22 PCD 380 (to NY-400)

AXONOMETRIC VIEW (not to scale)



NOTES

1. Rev B replaces the 18 m two-section tower. At 8 m a soft tower (f1 below 3P at 41 rpm) is impossible with the 983 kg head, so the tower is stiff: f1 above 1P at full speed; the controller skips the 3P crossing zone like the old 1P zone.
2. Seam and can welds full penetration, ISO 5817 B, 100 % UT on flange and can welds; flanges machined flat after welding (0.2 mm), top flange square to the axis within 0.5 mm/m.
3. No climbing: service by tilting the tower down (NS6-420). Earthing bosses 2 × M12 at +300. Hand-hole cover NS6-401 with EPDM gasket and 6 × M8.

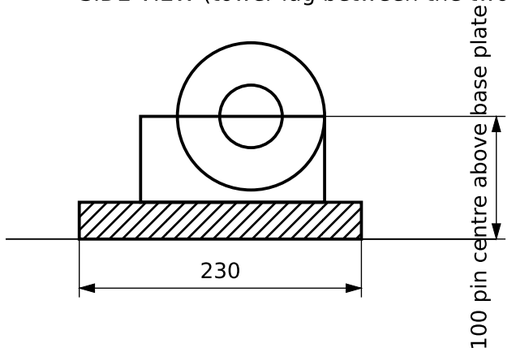
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

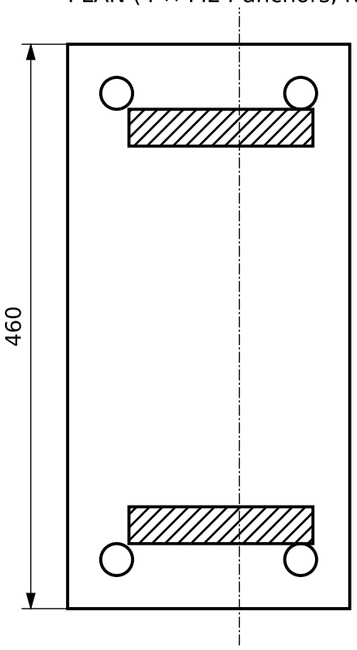
TOWER 8 m · ONE-PIECE TILT-UP MONOPOLE

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-400	B · 2026-11-06	1:25 / 1:10
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N, rolled and seam welded; hot-dip galvanized	681 kg (CAD 698)	Claude / not checked

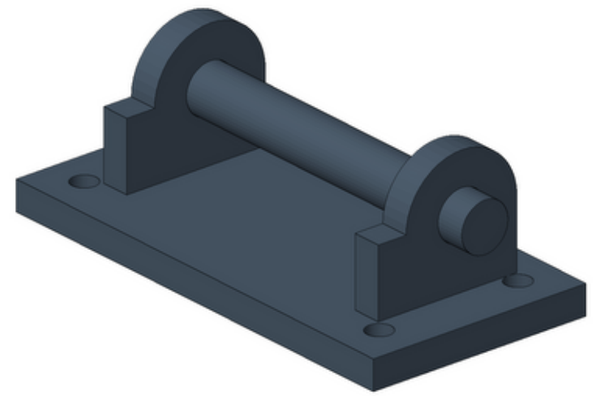
SIDE VIEW (tower lug between the two bracket plates)



PLAN (4 × M24 anchors, NS6-510)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Hinge axis 390 mm upwind of the tower axis (−X side), 40 mm above the base-flange underside: the tower lands on the levelling nuts as it reaches vertical.
2. Pin Ø50 h9 in Ø51 holes, secured with R-clips; pin load at tilt-up 2 × 55 kN shear (double shear, use 0.15).
3. Base plate 30 sits on the pedestal under the grout line; anchors 4 × M24 8.8, 350 Nm after the grout has cured.

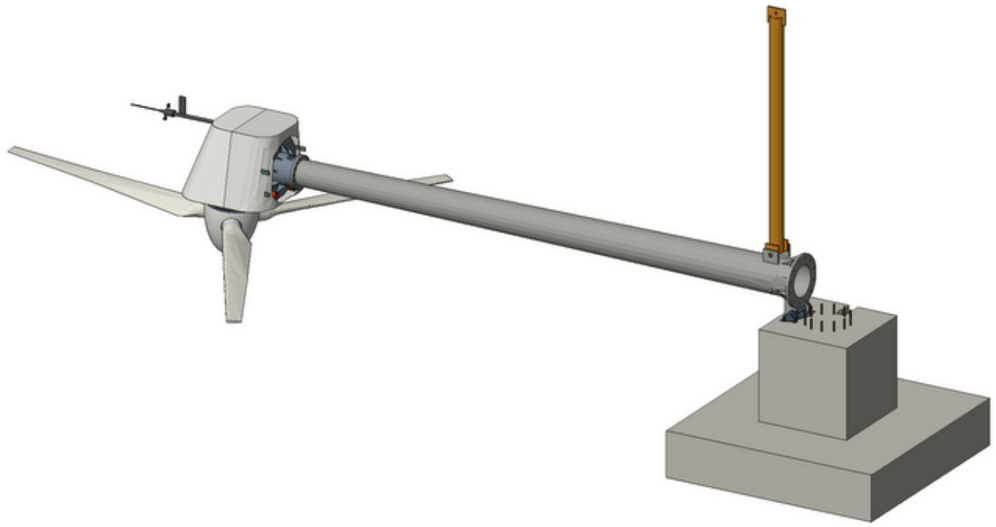
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

BASE HINGE BRACKETS AND PIN

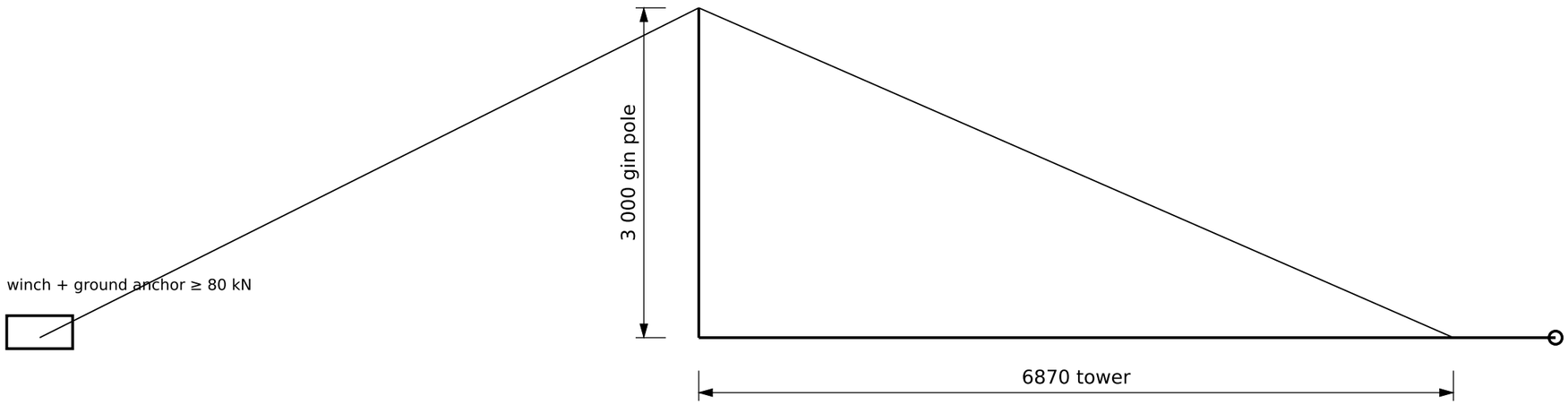
DRAWING NO.	REV / DATE	SCALE (A3)
NS6-410	B · 2026-11-06	1:5
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N 30 mm plate, galv.; pin 42CrMo4+QT Ø50 h9	37 kg	Claude / not checked

AXONOMETRIC VIEW (not to scale)



Item	Value
Gin pole	3.0 m, fixed square to the tower on its lugs
Peak line pull	≈ 38 kN (wind ≤ 10 m/s)
Bridle	2 × wire rope Ø16 6×36 IWRC to the top flange lugs
Pull line	Ø16 wire rope, winch ≥ 50 kN with 1:2 block
Ground anchor	screw anchors or concrete block, ≥ 80 kN at 6 m
Alternative	truck crane ≥ 3 t at 9 m reach (lift by the top flange)

TILT-UP (tower horizontal on the hinge, head fitted, rotor locked)



NOTES

1. Two people + winch operator; wind below 10 m/s; nobody inside the fall zone (1.5 × tower + rotor). Rotor locked with pin NP1-320, blade B1 pointing up away from the ground.
2. Lower the tower the same way for service; support the head on a trestle 1.0 m high so the lower blade clears the ground.
3. After erection: fit base nuts and torque (NS6-900), remove the gin pole, keep it with the site kit.

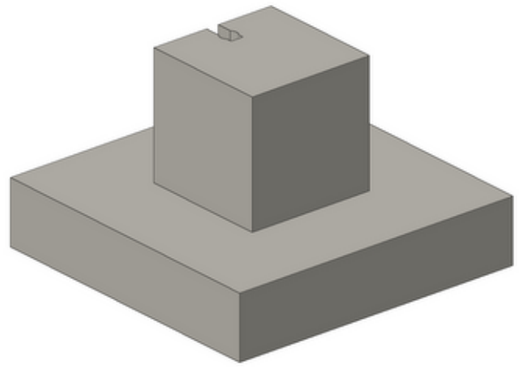
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

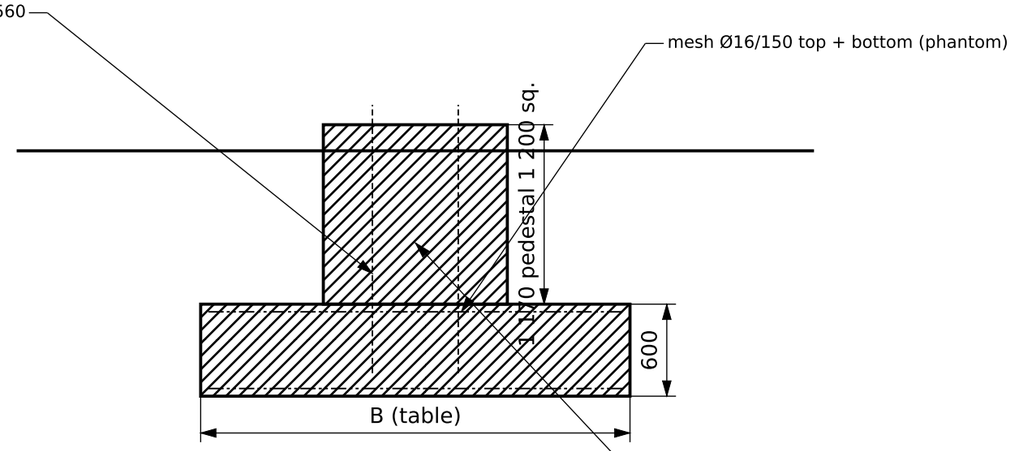
GIN POLE AND TILT-UP RIGGING (ERECTION TOOL)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-420	B · 2026-11-06	1:50
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355 tube 168.3 × 6.3, plates 25	94 kg	Claude / not checked

AXONOMETRIC VIEW (not to scale)

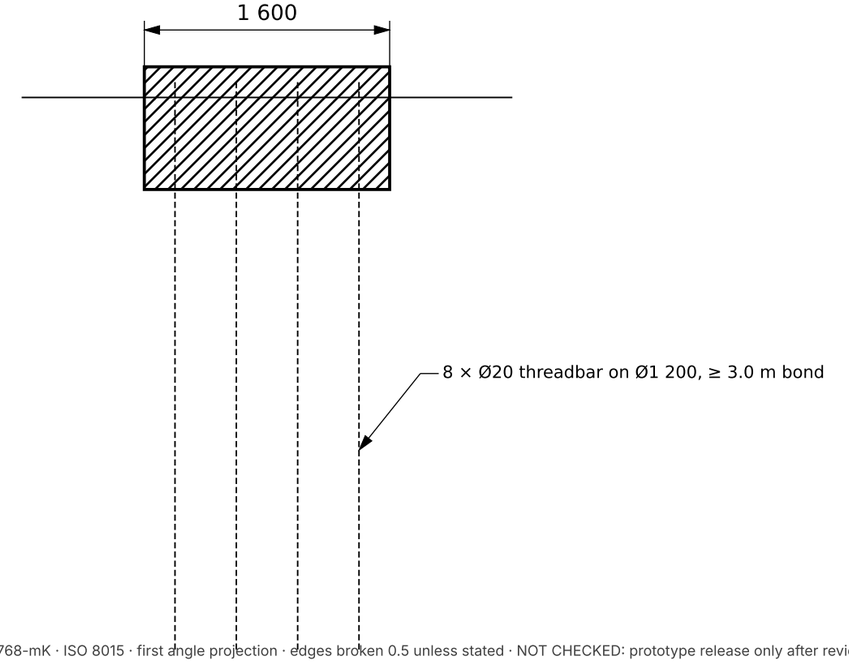


SPREAD FOOTING, SECTION



Ground	B m	Concrete m³	Bearing kPa	q_rd kPa
firm soil (dense gravel, moraine)	2.8	5.5	178.6	300.0
soft volcanic soil (loose ash, pumice sand)	3.1	6.6	99.0	100.0
frost-susceptible silt (thawed)	2.9	5.9	135.5	150.0
rock (anchored plinth)	1.6	2.0	–	anchors 92.1 kN

ROCK PLINTH (lava, basalt)



NOTES

1. Base 1.6 m below ground (frost depth); pedestal 1.2 × 1.2 m to +0.17, grout 30 to +0.20 under the base flange. Design base moment 238.6 kNm (earthquake 0.5 g + operation governs), parked V_e50 175.3 kNm.
2. Characteristic eccentricity ≤ B/4, operation ≤ B/6, bearing on the effective area ≤ q_rd (EN 1997 approach 2 style), as doc 09.
3. Concrete C30/37 XF4 air-entrained, cover 50; mesh Ø16/150 both ways top and bottom, pedestal 12 × Ø16 verticals + Ø10/150 links; earthing ring 50 mm² Cu round the footing, 2 rods 3 m.
4. Soil class to be confirmed by a site inspection; soft volcanic ash needs a plate-load test. Rock: drill Ø76, grout, proof-load each anchor to 1.25 × design.

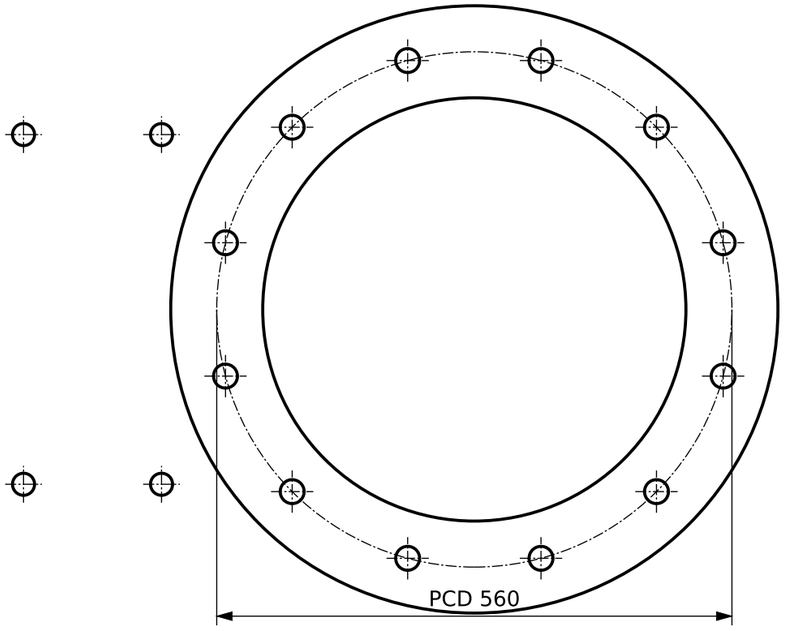
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

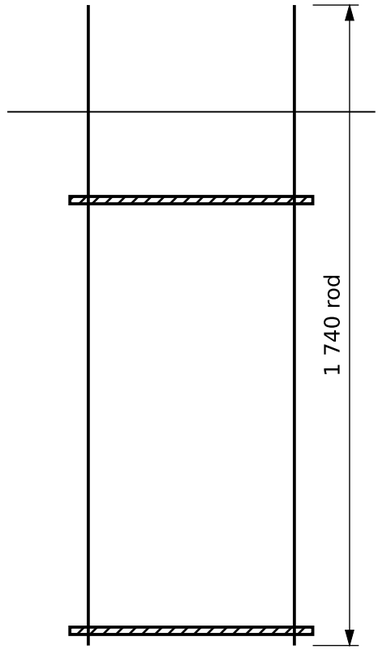
FOUNDATION · SPREAD FOOTING AND ROCK PLINTH (8 m)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-500	B · 2026-11-06	1:40
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
C30/37 XF4 air-entrained; B500C rebar	5.5–6.6 m³ (soil), 2.0 m³ (rock)	Claude / not checked

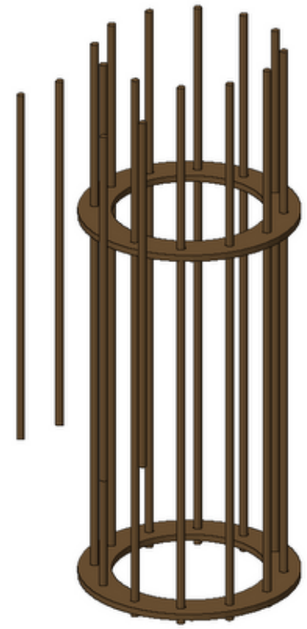
PLAN: template ring (top) on the anchor cage; 4 hinge anchors at −X



ELEVATION (ground at 0)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Anchor rods M24 8.8 HDG × 1 740, rolled thread both ends, 2 nuts + plate washer at the bottom ring; top 260 above the pedestal: levelling nut, base flange 35, washer, nut, lock nut.
2. Set the cage with the template ring (= base flange drilling) bolted on top; position ± 3 mm, level ± 1 mm; the hinge anchors from a second template bolted to the same ring.
3. Pretension after the grout (≥ 40 MPa) has cured 7 days: 450 Nm (≈ 110 kN) with a calibrated wrench, re-check after 1 month.

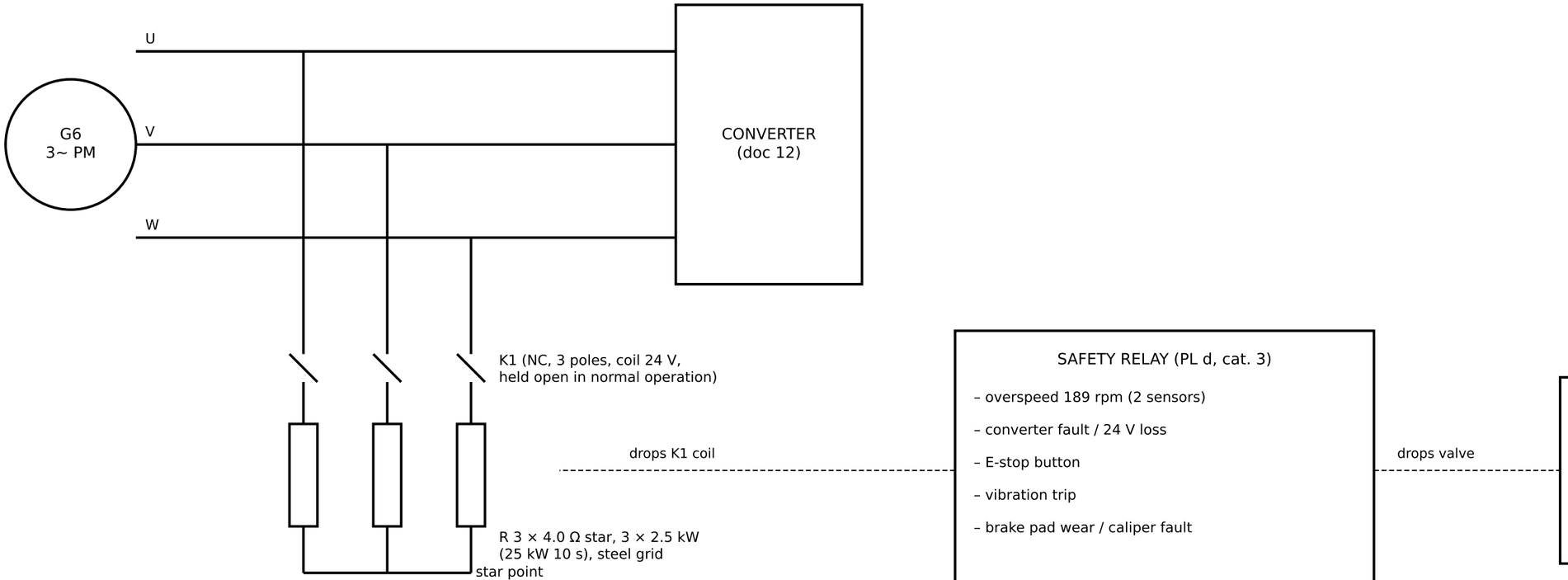
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

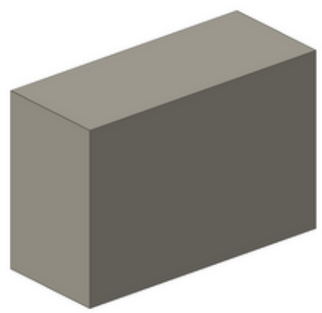
ANCHOR CAGE AND TEMPLATE

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-510	B · 2026-11-06	1:10
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
M24 8.8 HDG threaded rod; S355 rings 100 × 20	144 kg	Claude / not checked

SINGLE-LINE: BOTH STOPPING CHANNELS DROP TOGETHER ON A TRIP



AXONOMETRIC VIEW (not to scale)



NOTES

1. Doc 29 / 31: K1 is normally closed. While the safety relay is healthy it holds K1 open; any trip or loss of 24 V closes K1 and drops the brake at the same time. Either channel alone keeps the rotor below the trip speed (doc 31: brake alone stops in ≤ 2.4 s, peak 173 rpm; resistor alone slows to a 3–15 rpm crawl).
2. Resistor current 43 A peak at trip speed with hot magnets; energy of one stop ≈ 120 kJ. Grid resistors in a ventilated 1.4301 box (600 × 400 × 250) on the tower base cabinet wall, not in the nacelle.
3. K1: 3-pole contactor with mechanically linked mirror contact (EN 60947-4-1 Annex F) for monitoring; AC-3 rating ≥ 63 A at 690 V.
4. Brake fault alarm: with the brake failed the rotor only crawls, so a failed or worn brake keeps the turbine parked and yawed out until repaired (doc 31 §6).
5. Spring pack stays 1.5 kNm (Martin, 2026-11-06). Parked yaw-lost 50-year peak aero torque 1.45 kNm; if exceeded the rotor creeps slowly against the brake and the closed K1 (doc 32 §1). Use the lock pin NP1-320 for long unattended outages.

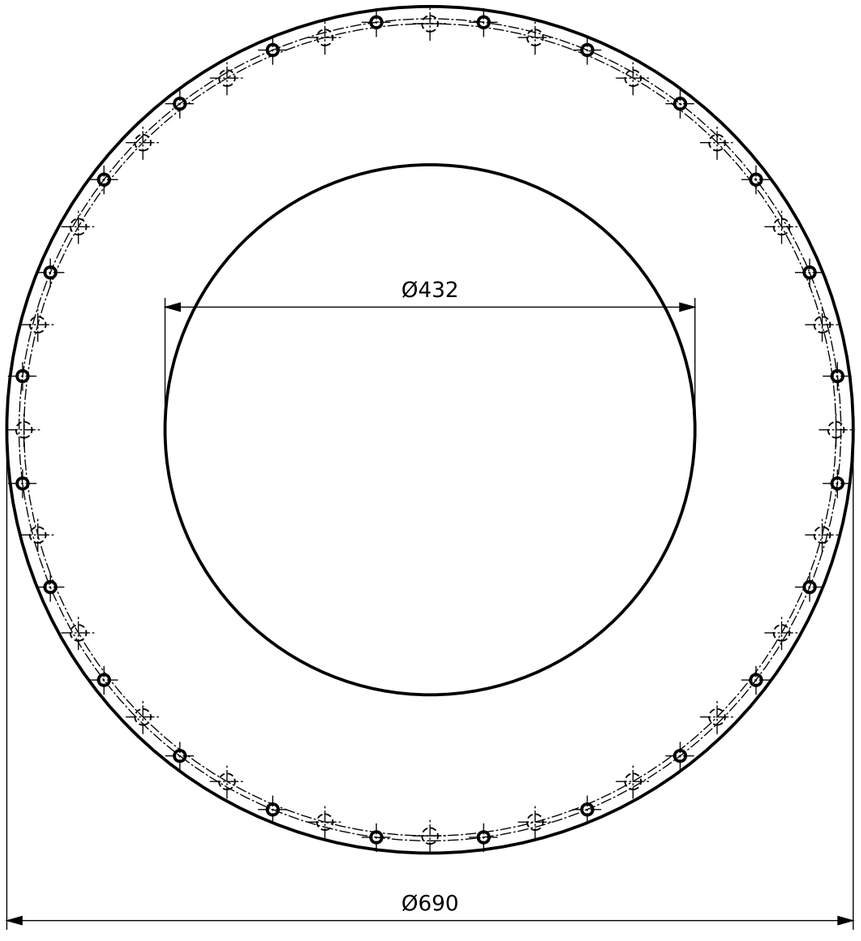
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

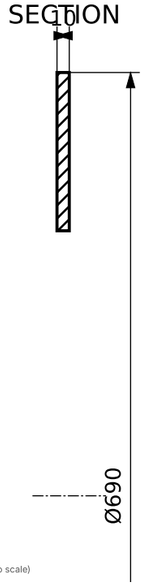
BRAKING RESISTOR BOX AND SAFETY CHAIN

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-600	B · 2026-11-06	schematic / 1:5
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
Box 1.4301, IP54; resistors steel-grid	≈ 18 kg	Claude / not checked

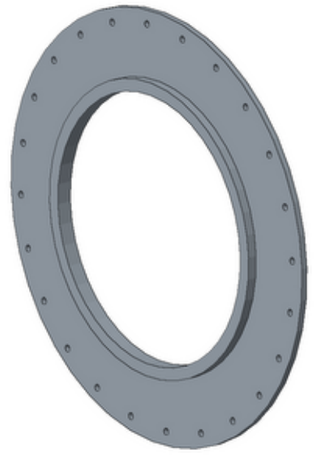
VIEW FROM THE REAR



SECTION



AXONOMETRIC VIEW (not to scale)



NOTES

1. Replaces the Rev C rear cover with 24 fan blades (doc 23/29): G6 runs at half load with no forced cooling; hot spot 86 °C at 1.2 × rated torque without fins, ribs or blades (doc 29).
2. 24 × M8 on PCD 670 to the drum rear ring; seal carrier as Rev C (NP1-212 inner ring).
3. Rear face runs 5 mm from the bedplate front plate outside the Ø410 pad (as Rev B).

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

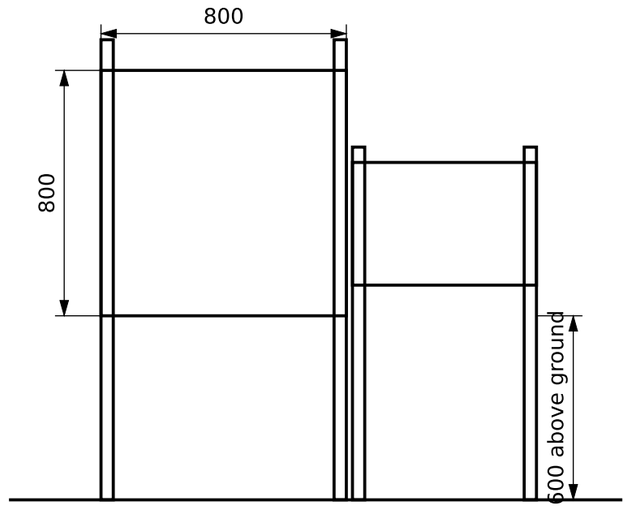
6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

G6 PLAIN REAR COVER (NO FAN BLADES)

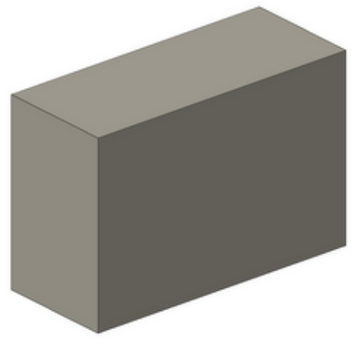
DRAWING NO.	REV / DATE	SCALE (A3)
NS6-610	B · 2026-11-06	1:5
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
EN AW-5083-H111, 10 mm plate	≈ 7 kg	Claude / not checked

Cabinet content (doc 12 modules)	Note
SiC active rectifier 1 200 V, 750 V DC link	6 kW firmware limits
Dump chopper 12 kW + heater feed	hardware trigger 820 V
AC module 6.5 kW (grid / heater)	single- or three-phase
48 V battery module + 200 Wh aux battery	yaw-out without grid
Safety relay PL d, K1 contactor	NS6-600 chain
Controller, modem, surge arresters T1+T2	earth bar to the tower lug

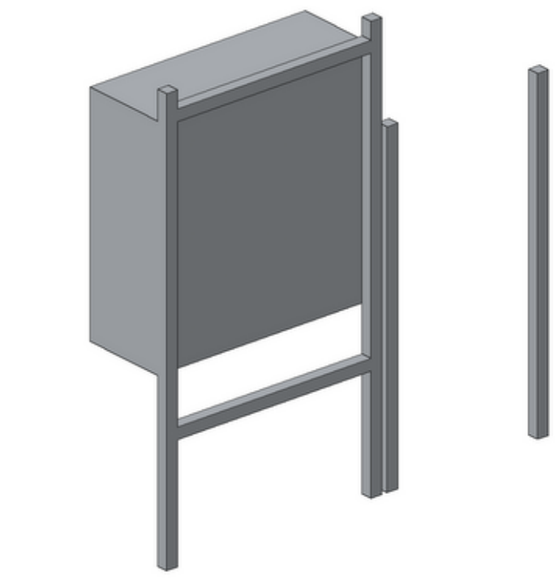
FRONT (from +Y): cabinet left, braking resistor box right (vented)



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Stand on two concrete pads 400 × 400 × 600 (or bolted to the pedestal side); cabinet door faces away from the tower; minimum 1.0 m from the tower axis, outside the rotor disc projection is not required (tip at 5 m).
2. Resistor box NS6-600 mounted 20 mm proud of the stand, louvres down, ≥ 300 mm from the cabinet (heat).
3. Cables enter from below through the two PE ducts in the footing (NS6-800).

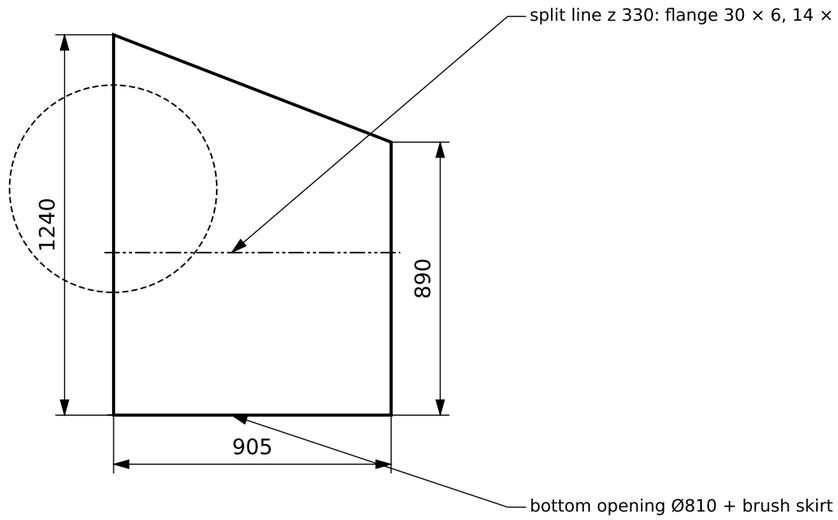
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

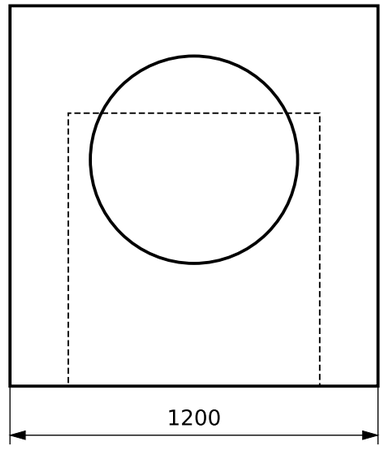
TOWER BASE CABINET, STAND AND RESISTOR BOX POSITION

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-620	B · 2026-11-06	1:20
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
enclosure 1.4301 IP55; stand S235 galv.	95 kg + 18 kg	Claude / not checked

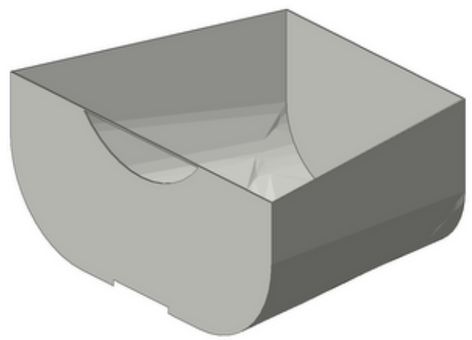
SIDE VIEW (yaw axis at x 0)



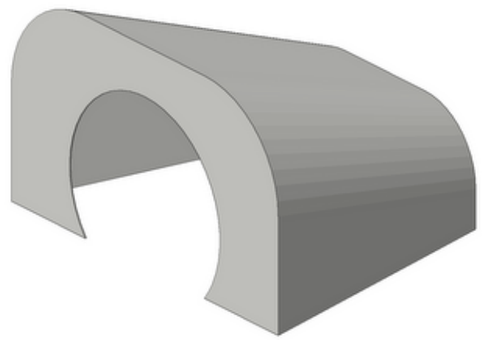
FRONT (opening Ø676 round the drum, 15 gap)



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Lofted rounded-rectangle shell, front 1 200 × 1 240 (R330) to rear 820 × 890 (R230); encloses the disc, caliper, bedplate, yaw drives and pad units; drum and spinner stay outside (drum cools in the free air).
2. Lay-up 4 mm: gelcoat 0.5 + 3 × 450 g CSM + 1 × 300 g woven; PU foam hat-section ribs every 300 mm; lid has 2 lifting points and a gas-strut-held service hatch 500 × 400 over the junction box.
3. Lower tray hangs on the NS6-710 arms (4 × M10 rubber mounts); the lid bolts to the tray flange. Air in at the bottom skirt, out through a labyrinth vent at the rear top (no forced cooling).

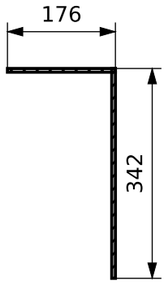
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

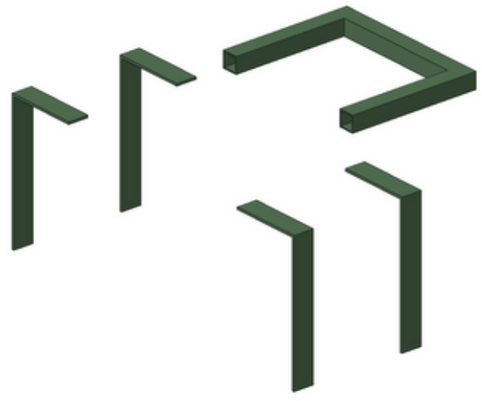
NACELLE COVER · LOWER TRAY AND UPPER LID (GRP)

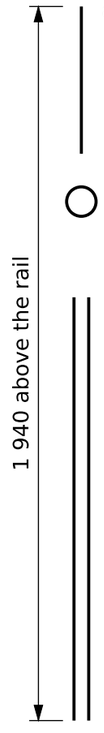
DRAWING NO.	REV / DATE	SCALE (A3)
NS6-700	B · 2026-11-06	1:20
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
glass/polyester 4 mm, gelcoat RAL 9010, PU foam ribs	≈ 22 kg	Claude / not checked

ARM (4 off): flat 50 × 6, bolted 2 × M10 to the side plate



AXONOMETRIC VIEW (not to scale)

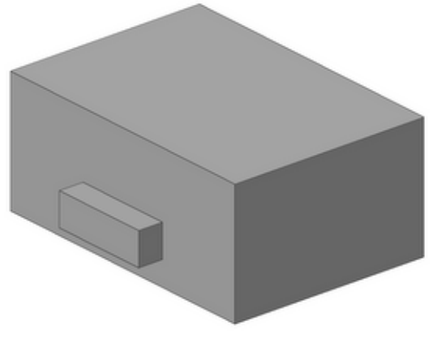




AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Rear rail RHS 40 × 40 × 3 on the bedplate rear tie carries the junction box NS6-730 (IP66, 400 × 340 × 150) and the met mast clamp.
2. Met mast: wind sensors 0.5 m above the cover top (doc 25), air terminal bonded with 50 mm² Cu to the bedplate; the bedplate–tower path goes through the yaw pad brush (NY-410).
3. Junction box: phase terminals 3 × 16 mm², brake/sensor terminals, surge arresters for the sensor lines, 24 V supply; cable glands at the bottom only.

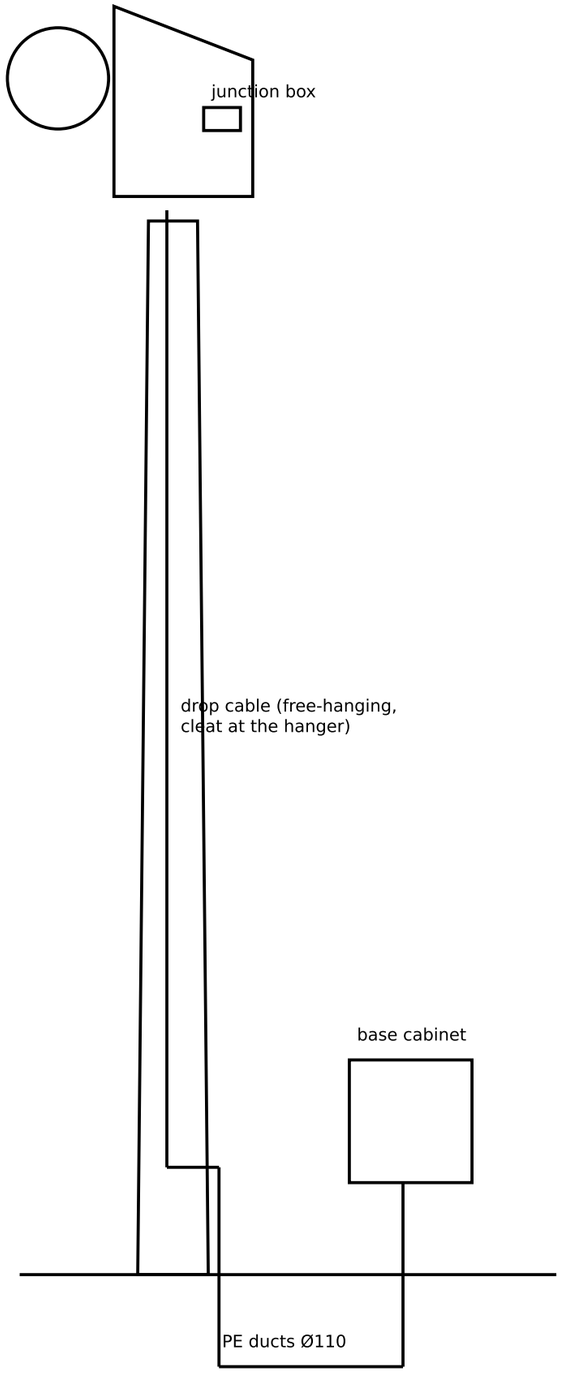
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

COVER ARMS, REAR RAIL, MET MAST NS6-720, JUNCTION BOX NS6-730

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-710	B · 2026-11-06	1:10
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S235 galv.; mast 1.4404 Ø42.4 × 2	22 kg; 4.5 kg; 9 kg	Claude / not checked

ROUTE (nacelle → loop → tower → cabinet)

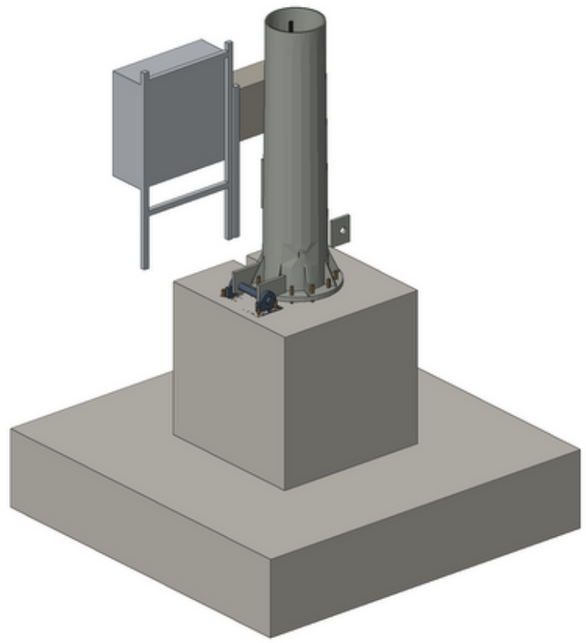


Cable	Type	From → to	Length m
W1 power	H07RN-F 4G10 (3 ph + PE)	generator terminals → junction box	3
W2 power	H07RN-F 4G10, 2.5 turns loop	junction box → hanger loop → hand hole	12
W3 power	NYY-J 4 × 10 in duct	hand hole → cabinet	4
W4 brake	ÖLFLEX 4G1.5 shielded	caliper → junction box → cabinet	16
W5 sensors	LiYCY 12 × 0.5	overspeed, encoder, cam, met, PT100 → cabinet	16
W6 yaw motors	ÖLFLEX 7G1.5	2 gearmotors + brakes → cabinet	16
W7 earth	Cu 50 mm² green/yellow	air terminal → bedplate; tower lug → earth ring	6
W8 data	CAT6 outdoor or fibre	cabinet → house / grid meter	site

NOTES

1. Cable loop NY-460: power and control cables hang free in the tower below the cleat with 2.5 turns of slack; cable-twist limit ±2.5 turns from the encoder/cam switch (doc 25), auto-untwist when idle.
2. Lightning (LPL I, doc 02): air terminal → bedplate → yaw pad brush → adapter → tower → base flange → 2 earthing lugs → ring earth (≤ 10 Ω). Generator winding star point not earthed (IT, insulation monitor in the cabinet).
3. Surge protection: T1+T2 arresters at the cabinet entry for W2/W3, data-line arresters for W5/W8; shields bonded at both ends via the junction box gland plate.
4. All cables UV and cold resistant to −40 °C; drip loops at the hand hole; glands IP68; label every core at both ends.

AXONOMETRIC VIEW (not to scale)



Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

CABLING ROUTES, EARTHING AND LIGHTNING

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-800	B · 2026-11-06	schematic
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see cable schedule	–	Claude / not checked

Joint	Fastener	Qty	Grade / finish	Tightening	Notes
Blade root → hub flange	stud M12 × 75 + nut + washer plate	30	10.9, zinc-flake	95 Nm + 90°	Loctite 270 in the insert, 40 Nm
Hub → drum web	M16 × 70 ISO 4017	12	10.9, zinc-flake	250 Nm (μ 0.12)	check after 50 h
Spinner → brackets	M8 × 20 + rubber washer	3	A4-70	12 Nm	
Spindle → bedplate front plate	M20 × 90 ISO 4014	12	10.9	540 Nm	from inside, NP1-100
Stator carrier → spindle	M12 × 45	12	10.9	110 Nm	NP1-200
Brake disc → drum	M12 × 40	12	10.9	110 Nm	NP1-230
Rear cover → drum	M8 × 20	24	A4-80	20 Nm	NS6-610
Caliper → NP1-310	M16 (caliper pattern)	4	10.9	250 Nm	per caliper supplier
NP1-310 → side plate	M16 × 70 + nut	4	10.9 HV	210 Nm	
Lock pin holder → front plate	M12 × 50	2	8.8	80 Nm	
Yaw inner ring → bedplate	M16 × 110 + nut	16	10.9 HV	250 Nm	doc 25
Yaw outer ring → NY-400	M16 × 70	16	10.9	250 Nm	doc 25
Pad unit → bedplate	M16 × 50	8	10.9	250 Nm	NY-410
Gearmotor bracket → side plate	M16 × 60 + nut	8	8.8	195 Nm	NP1-420
Gearmotor → bracket	M10 × 30	8	8.8	49 Nm	
NY-400 → tower top flange	M20 × 90 + nut	16	10.9 HV	450 Nm	through bolts
Cover arms → side plates	M10 × 25	8	A4-70	35 Nm	rubber mounts
Cover lid → tray	M8 × 25 + EPDM	14	A4-70	10 Nm	
Base flange → anchors	M24 nuts (levelling + top + lock)	12	8.8 HDG anchors	450 Nm	after grout cured 7 d
Hinge brackets → anchors	M24 nuts	4	8.8 HDG	350 Nm	
Hand-hole cover	M8 × 20	6	A4-70	12 Nm	EPDM gasket
Cabinet stand → pads	M12 anchors	4	A4-70	40 Nm	

NOTES

1. Torques for lightly oiled zinc-flake threads (μ ≈ 0.12) unless stated; HV sets per EN 14399-4 with the manufacturer's k-class.
2. Mark every torqued bolt with a torque stripe; record values in the installation report. Re-check all structural joints (blade, hub, yaw, tower) after 50 h and then yearly.
3. Never re-use M16 / M20 / M24 HV sets that have been fully tightened; replace studs that show thread damage.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

FASTENER AND TORQUE SCHEDULE

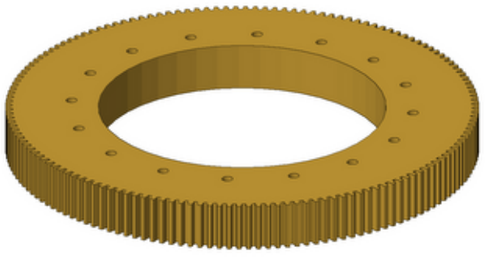
DRAWING NO.	REV / DATE	SCALE (A3)
NS6-900	B · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
ISO 4014/4017, EN 14399-4 (HV), 10.9 zinc-flake unless stated	–	Claude / not checked

Item	Specification	Interface	Candidates / note
NS6-911 yaw ring	4-point ball slewing ring Dpw 360, ext. gear z150 m4, ind. hardened teeth	inner 16 × M16 PCD 300; outer 16 × M16 PCD 440	Rollix / Kaydon / IMO; tilt 34.6 kNm s0 ≥ 1.44
NS6-912 yaw gearmotor (2)	0.18 kW, i ≈ 1 680, output ≥ 1 500 Nm static, brake 2 Nm	flange 4 × M10 on 80 square, pinion z14 m4	SEW / Bonfiglioli / NORD
NS6-913 main bearings (2)	tapered roller 32028 X (140 × 210 × 45)	spindle Ø140 f6 / j6	SKF / FAG / Timken, C3 not needed
NS6-914 overspeed sensors (2)	inductive M18 PNP, −40 °C, IP68	holder NP1-321	ifm / Pepperl+Fuchs
NS6-915 rotor brake caliper	spring-applied, power-released, 16 mm disc, 1.5 kNm at r 0.41 m	4 × M16 on NP1-310	Svendborg BSFI / Altra / Ringspann
Magnets	200 + 10 × N45SH 30 × 15 × 12, NiCuNi + epoxy	drawing NP1-220E	Neo / VAC (non-Asian, doc 23)
K1 contactor	3-pole AC-3 ≥ 63 A 690 V, mirror contacts	NS6-600	ABB / Siemens
Converter, chopper, modules	doc 12 SiC rectifier etc., 6 kW firmware	NS6-620	in-house (doc 12)
Met sensors	heated cup anemometer + vane, −40 °C	NS6-720	Thies / Lufft
Yaw encoder	absolute multi-turn SSI, IP67	NY-440	Kübler / Sick

NOTES

1. Envelopes in the CAD model are estimates; replace them with the supplier STEP files before the clash check of the prototype build.
2. All bought parts: operating −30 °C, survival −40 °C, salt fog C5 (doc 02); certificates 3.1 for structural parts (slewing ring, bearings, bolts).

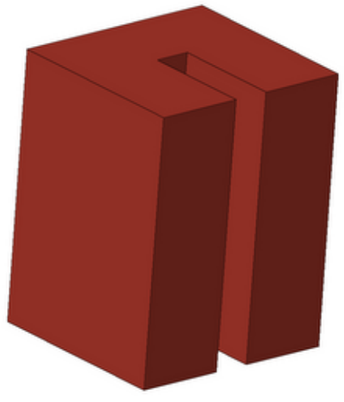
AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

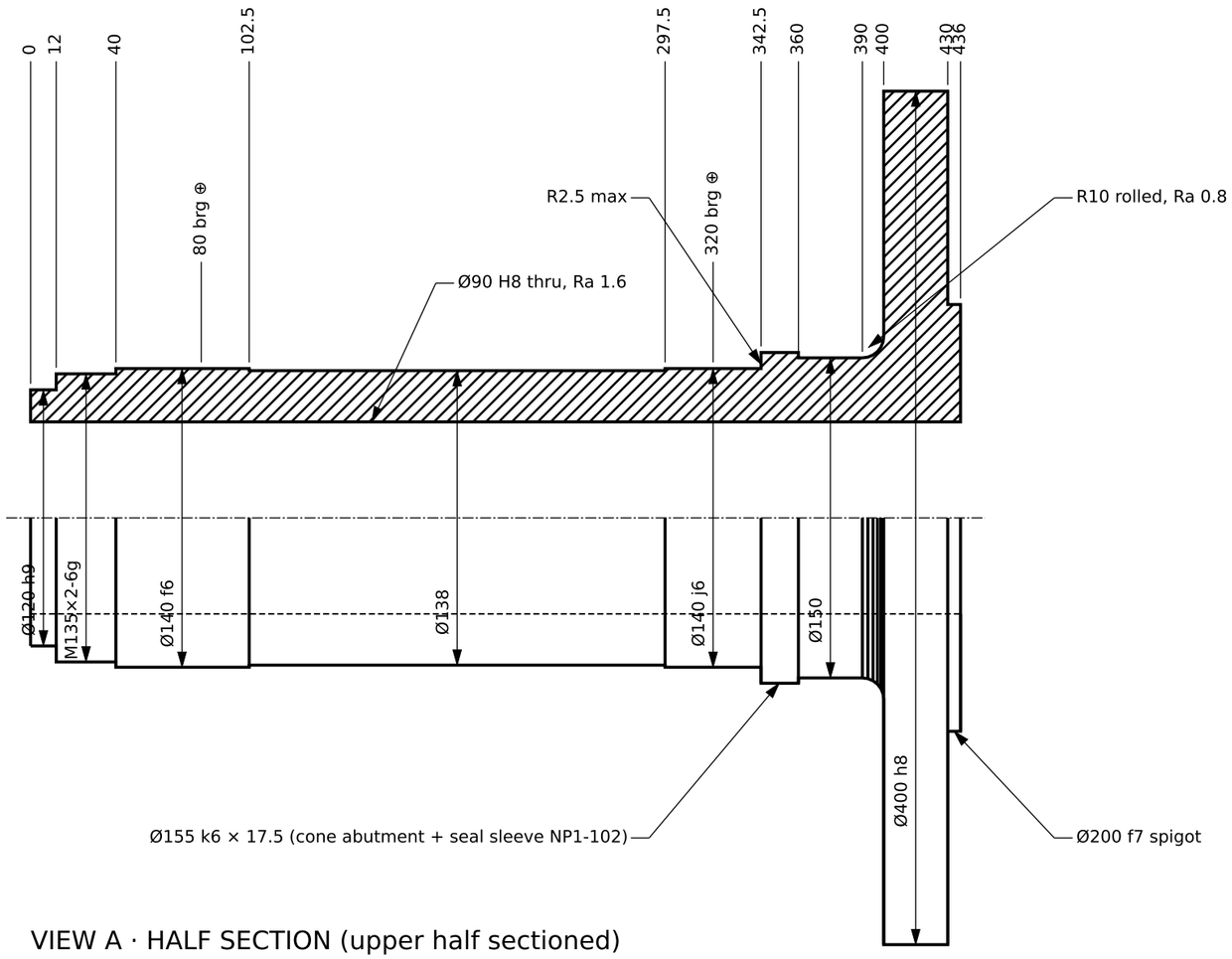
6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

BOUGHT PARTS SPECIFICATION (ENVELOPES AND INTERFACES)

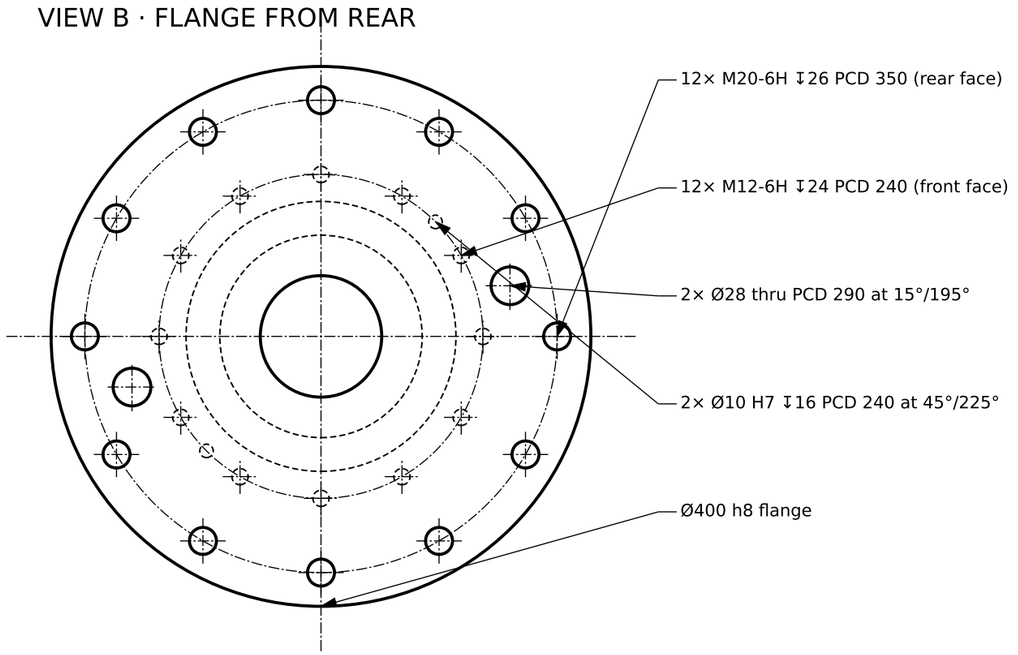
DRAWING NO.	REV / DATE	SCALE (A3)
NS6-910	B · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see lines	see lines	Claude / not checked

5 Reused 12 kW drawings

Spindle, bearings, seals, disc, lock (NP1 rev B) · generator Rev C (G6: no fins, ribs, fan blades) · yaw system NY



VIEW A · HALF SECTION (upper half sectioned)



VIEW B · FLANGE FROM REAR

NOTES

1. Blank: open-die forging 42CrMo4, forging ratio ≥ 3, quenched and tempered to Rp0.2 ≥ 650 MPa, Rm 900 to 1100 MPa; KV ≥ 27 J at −40 °C (3 specimens, longitudinal, taken from the flange zone). Certificate EN 10204 3.1.
2. Ultrasonic test of the blank to EN 10228-3 quality class 3; MT of the finished part (fillets, shoulders, thread) to EN ISO 9934, no linear indications.
3. Bearing seats: Ø140 f6 front (cone must slide for preload), Ø140 j6 rear; Ra 0.8; cylindricity 0.006; run-out of both seats and of Ø400 and Ø200 to common axis A-B ≤ 0.01.
4. Flange rear face and front face perpendicular to A-B within 0.02; flatness 0.02. Ø400 seal surface Ra 0.2 to 0.4, plunge ground (no lead), hard chrome 30 µm or wear sleeve.
5. Thread M135×2-6g with undercut DIN 76-D; precision locknut KMT 27 (or equivalent with locking screws).
6. Bore Ø90 H8 for the cooling insert NP1-110 (installed with liquid nitrogen shrink, interference 0.02 to 0.06).
7. Two radial holes Ø6 from the bore to under each bearing cone seat at 90° (bottom dead centre at 0°) for PT100 bearing temperature sensors; deburr, edges R0.5.
8. All fillets and the shoulders rolled or shot peened; no stamping on the part: mark serial number on the flange OD rear chamfer by etching.
9. Protect: phosphate + oil in storage; seats and seal surfaces taped until assembly.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

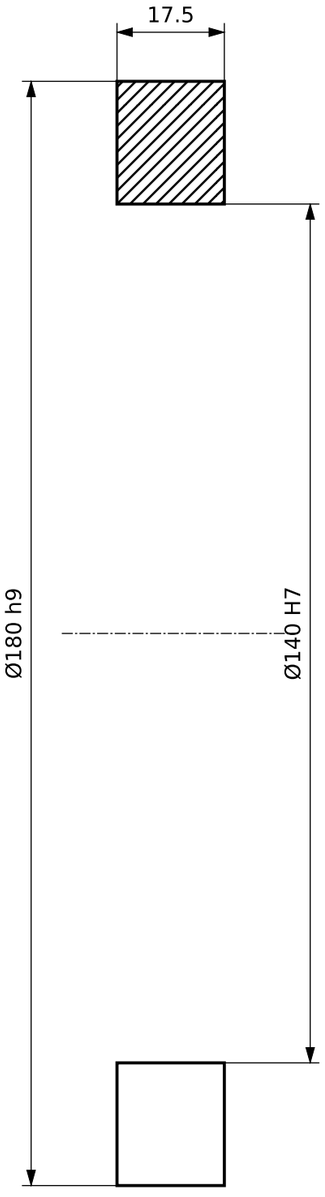
12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

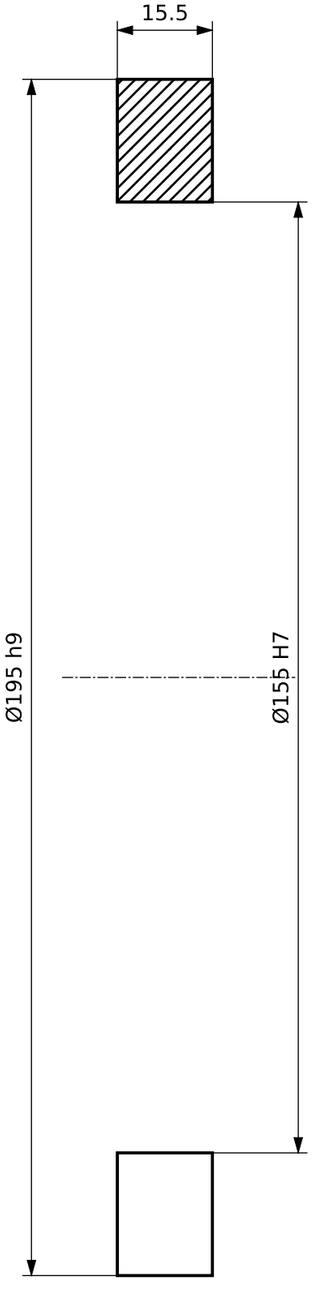
SPINDLE (STATIONARY MAIN SHAFT)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-100	B · 2026-11-03	1:2.86 (A) / 1:4.5 (B)
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
42CrMo4+QT (EN 10083-3), forged	56.8 kg	Claude / not checked

NP1-102 REAR SEAL SLEEVE (Rev B)

NP1-101 SECTION





NOTES

1. Faces parallel within 0.005 (it sets the bearing preload path).
2. NP1-102: same material and hardening as NP1-101; O-ring 155 × 3 in a bore groove (static seal); slides onto the Ø155 shoulder before the rear cone, front face against the cone back face.
3. Rev B: OD 160 → 180 so the front seal cover clears the KMT 27 nut (doc 17).
4. Ø180 surface induction hardened 55 to 60 HRC, depth 1 mm, ground Ra 0.2 to 0.4, no lead.
5. Bore chamfer 1.5×30° on the cone side.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

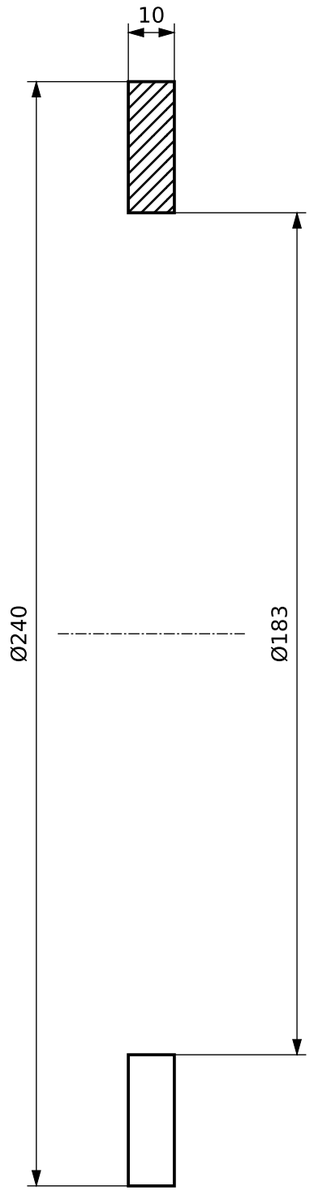
12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

FRONT SPACER + REAR SEAL SLEEVE NP1-102

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-101	B · 2026-11-03	1:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
42CrMo4+QT, surface induction hardened	1.4 kg	Claude / not checked

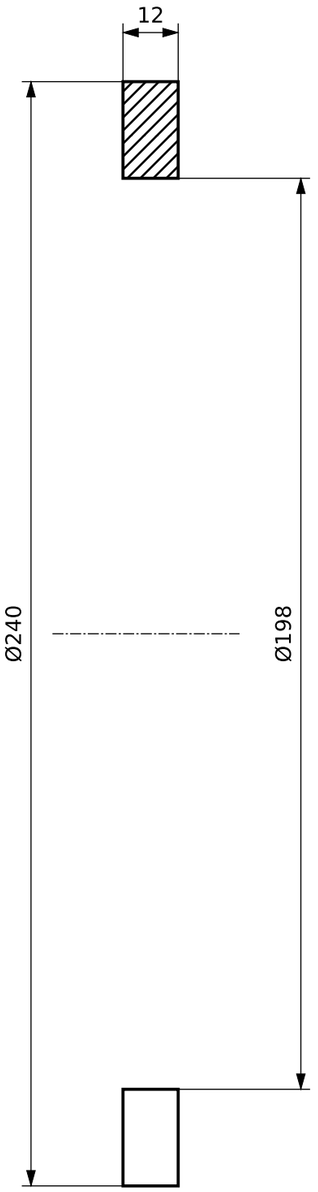
FRONT COVER: lip seal 180×200×10 HNBR + V-ring

6× Ø9 PCD 225; seal bore H8, Ra 1.6



REAR COVER: lip seal 195×220×12 HNBR on NP1-102

6× Ø9 PCD 225; seal bore H8, Ra 1.6



NOTES

1. Rev B (doc 17): both bores are larger than every part they pass. Front cover goes on after the KMT 27 nut and the rolling-torque check; rear cover is bolted to the drum before the drum goes over the fitted rear cone (Ø191 envelope) onto the sleeve NP1-102.
2. Seal lips face the bearing (grease retention); V-ring outside on the front cover runs against the spacer face.
3. Purge: grease exits under the lip; collect with absorbent ring at service.
4. Pilot on the housing bore Ø210, 2 mm deep (not shown), for concentric seals.

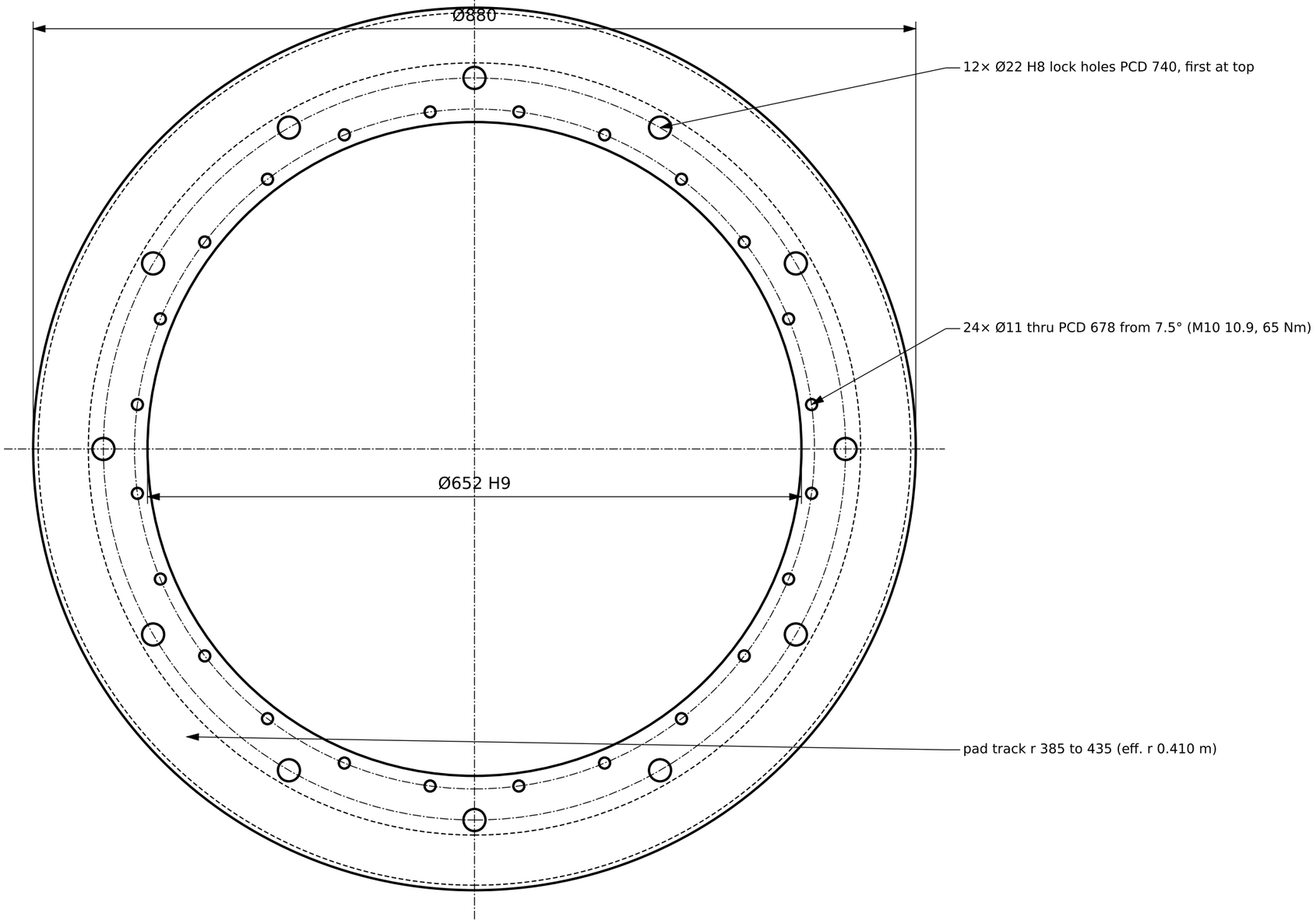
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

MAIN BEARING SEAL COVERS (FRONT / REAR)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-211	B · 2026-11-03	1:1
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355J2	2.8 kg	Claude / not checked

VIEW · Rev B · thickness 16 · slides on from the front over the Ø646 drum shell



NOTES

1. Faces ground parallel 0.03, flatness 0.05, Ra 1.6. Axial run-out mounted on the drum ≤ 0.10.
2. Two spring-applied hydraulically released calipers (bought), at 90° and 270° (horizontal), each ≥ 13.3 kN clamp force on 16 mm disc; brake torque 8.7 kNm at μ 0.4.
3. Rev B (doc 17): bore Ø640 → Ø652 (fits over the drum shell from the front), OD Ø800 → Ø880 (pad track outside the Ø710 flange ring, room for the caliper behind the disc), lock holes moved here from the ring.
4. Corrosion: zinc-flake coating outside the pad track; pad track left bare (it is cleaned by the weekly brake test, doc 11).

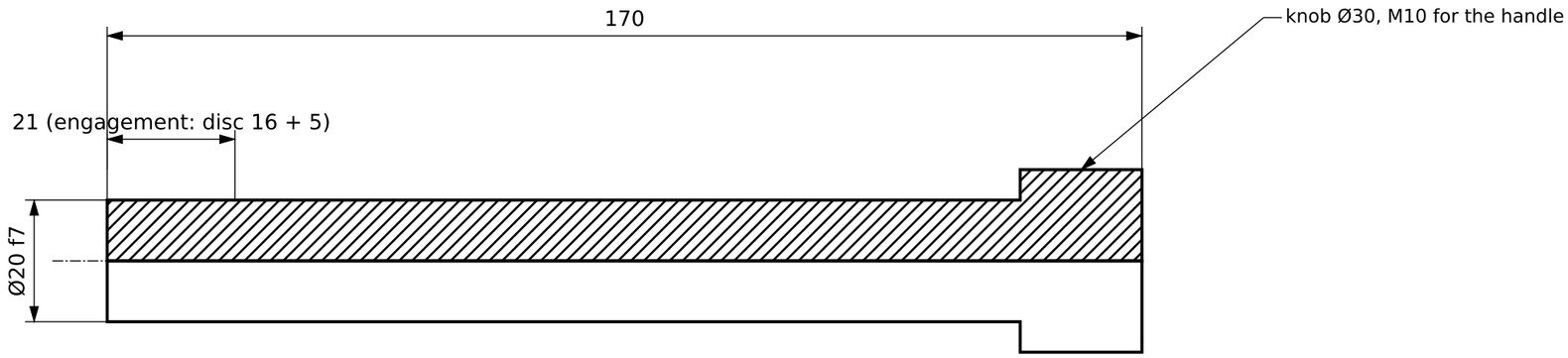
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

BRAKE DISC AND ROTOR LOCK HOLES

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-230	B · 2026-11-03	1:4.8
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
C45E+N (or S355J2+N), 16 mm	34.5 kg	Claude / not checked

LOCK PIN



NOTES

1. Rev B (doc 17): pin Ø20 at the top (12 o'clock) on PCD 740, entering the disc from behind; holder NP1-321 sits on the top edge of the bedplate front plate and also carries the two overspeed sensors. Shear at 8.7 kNm: 23.5 kN, τ = 75 MPa (allowable 300); bearing on the 16 mm disc 73 MPa.
2. Stroke 40: retracted, the nose is 19 mm behind the disc and clear of the Ø710 ring and the rear cover (r ≥ 360).
3. Holder has a proximity switch that reports 'pin engaged' to the safety chain (no rotor release while engaged, doc 11).
4. Insert only at standstill with the brake applied.

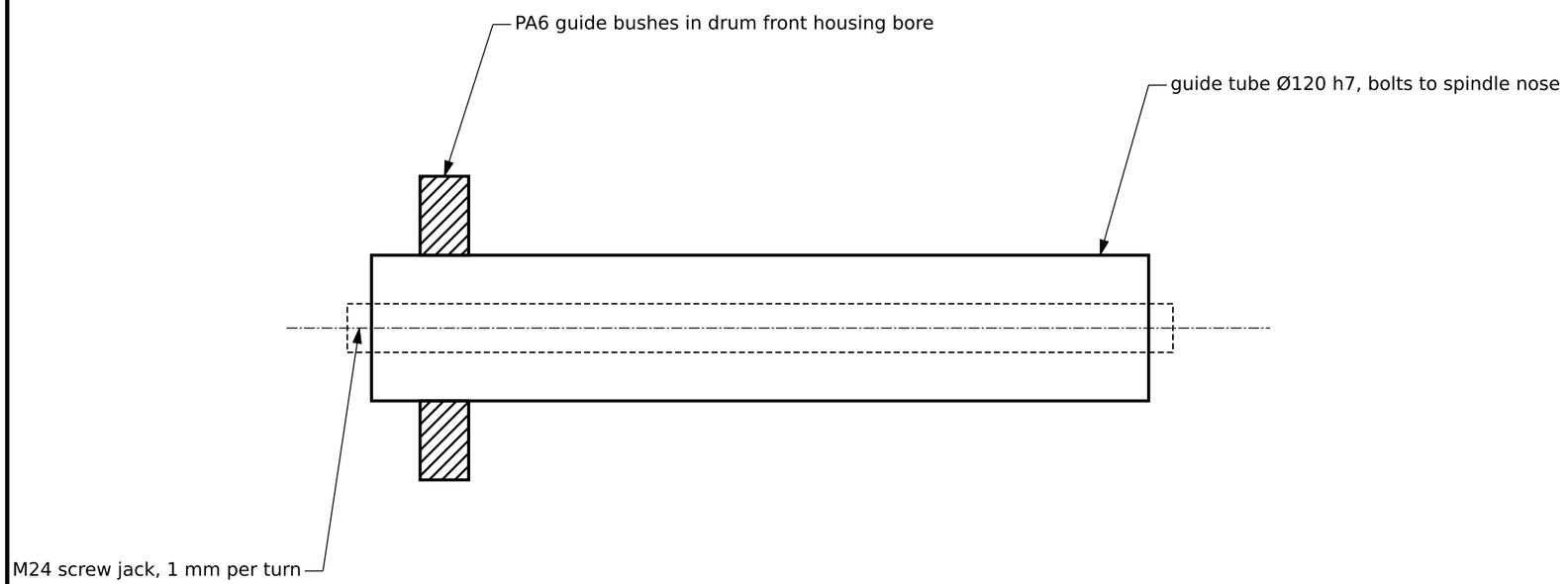
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

ROTOR LOCK PIN AND HOLDER

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-320	B · 2026-11-03	1:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
pin 42CrMo4+QT; holder S355J2	≈ 4 kg	Claude / not checked

GUIDE TUBE ON SPINDLE NOSE



NOTES

1. Radial magnetic pull with the drum off-centre is ≈ 1.9 kN per mm of eccentricity (doc 07) and grows quickly; never let the magnet ring approach the stator by hand.
2. Procedure: spindle vertical, nose up; guide tube bolted to the nose; drum hung from the crane, centred by two PA6 bushes on the tube before the magnets reach the stator; lowered by the screw jack until the rear cup reaches the rear cone.
3. Gap check during lowering: 4 non-magnetic feelers (G10, 1.5 mm) in the gap at 90°.

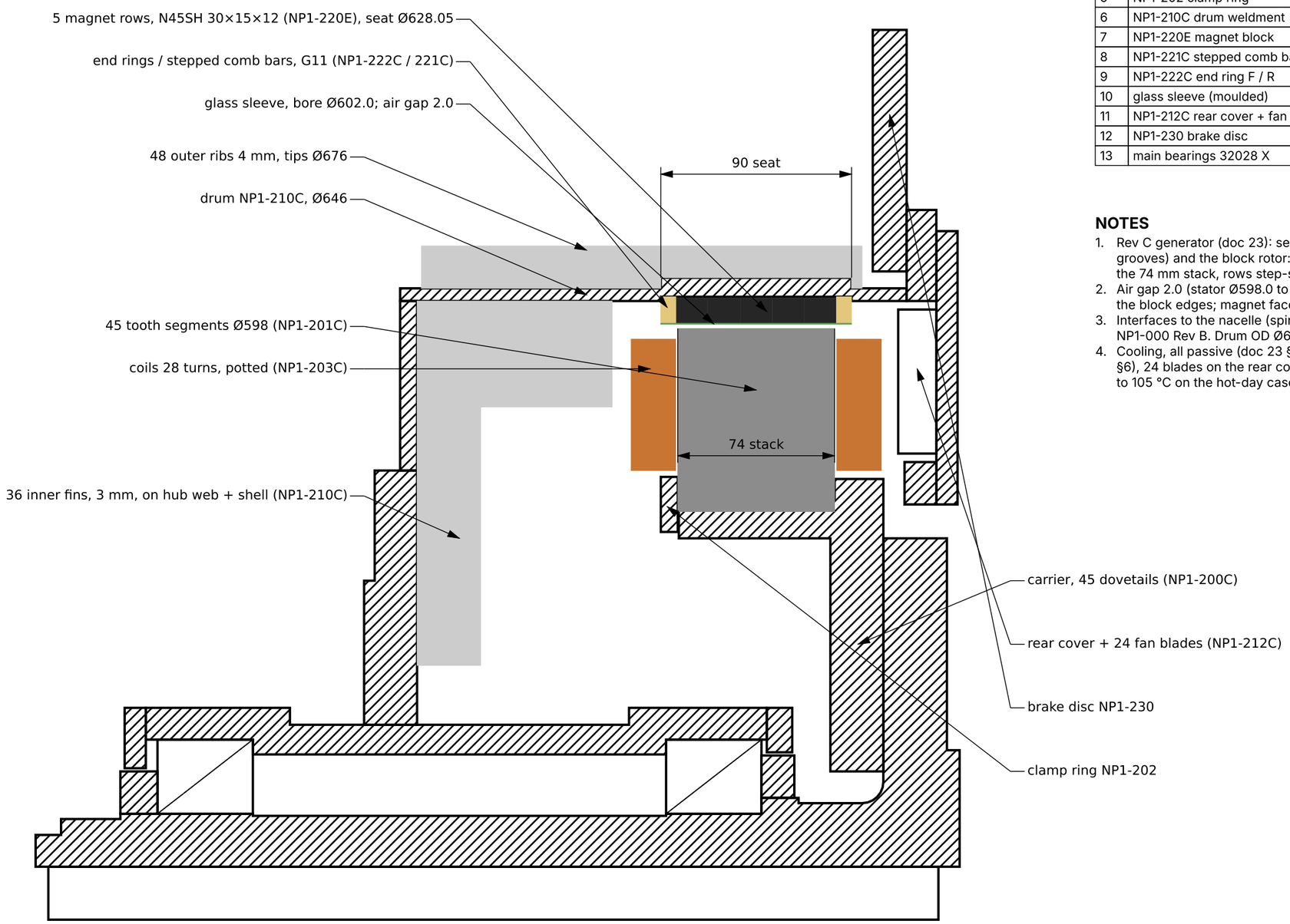
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

DRUM ASSEMBLY GUIDE (MAGNET SAFETY TOOL)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-T01	B · 2026-11-03	1:5
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355 / PA6 bushes	≈ 30 kg	Claude / not checked

GENERATOR SECTION (upper half, 1:2.2)



#	Part	Qty	Material / note
1	NP1-100 spindle (stationary)	1	42CrMo4+QT
2	NP1-200C stator carrier	1	S355J2+N, 45 dovetails
3	NP1-201C tooth segment stack	45+3	M270-35A bonded, 205 sh.
4	NP1-203C coil on segment	45+3	28 t × 4 × Ø1.55, cl. 200
5	NP1-202 clamp ring	1	1.4404
6	NP1-210C drum weldment	1	S355J2+N, seat Ø628.05
7	NP1-220E magnet block	200+10	N45SH 30×15×12
8	NP1-221C stepped comb bar	40+4	G11
9	NP1-222C end ring F / R	1+1	G11
10	glass sleeve (moulded)	1	E-glass/epoxy 0.5–0.9
11	NP1-212C rear cover + fan	1	5083, 24 blades
12	NP1-230 brake disc	1	C45E+N
13	main bearings 32028 X	2	bought

NOTES

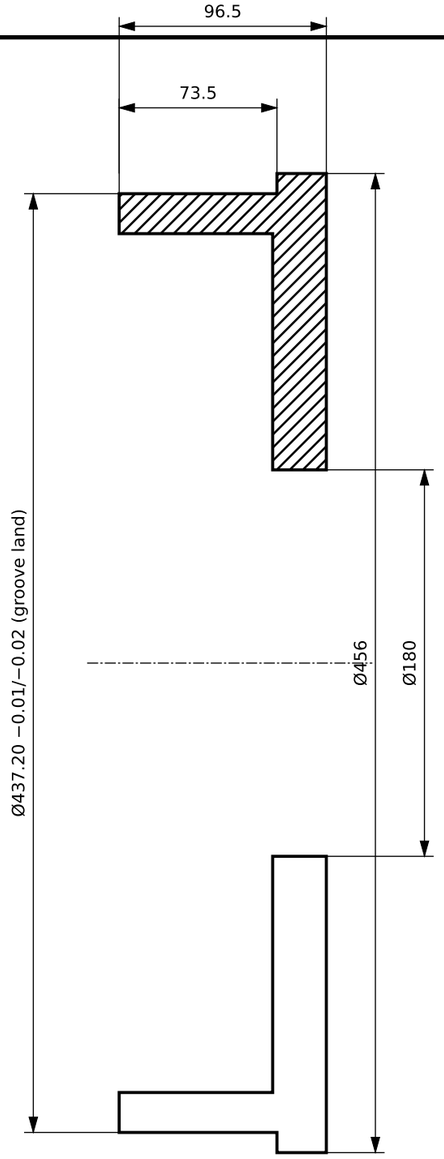
1. Rev C generator (doc 23): segmented stator of doc 18 (45 tooth segments in carrier dovetail grooves) and the block rotor: one N45SH block 30 × 15 × 12 per pole across, five rows along the 74 mm stack, rows step-skewed 0.20° by the stepped comb bars.
2. Air gap 2.0 (stator Ø598.0 to sleeve bore Ø602.0); sleeve 0.52 at the block centre, 0.89 at the block edges; magnet face r 301.52.
3. Interfaces to the nacelle (spindle, bearings, hub flange, disc, bedplate) unchanged from NP1-000 Rev B. Drum OD Ø646 unchanged.
4. Cooling, all passive (doc 23 §4 and §4a): carrier → spindle → finned insert path (doc 14 §6), 24 blades on the rear cover, 36 inner fins and 48 outer ribs on the drum; hot spot 94 to 105 °C on the hot-day cases (limit 130).

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

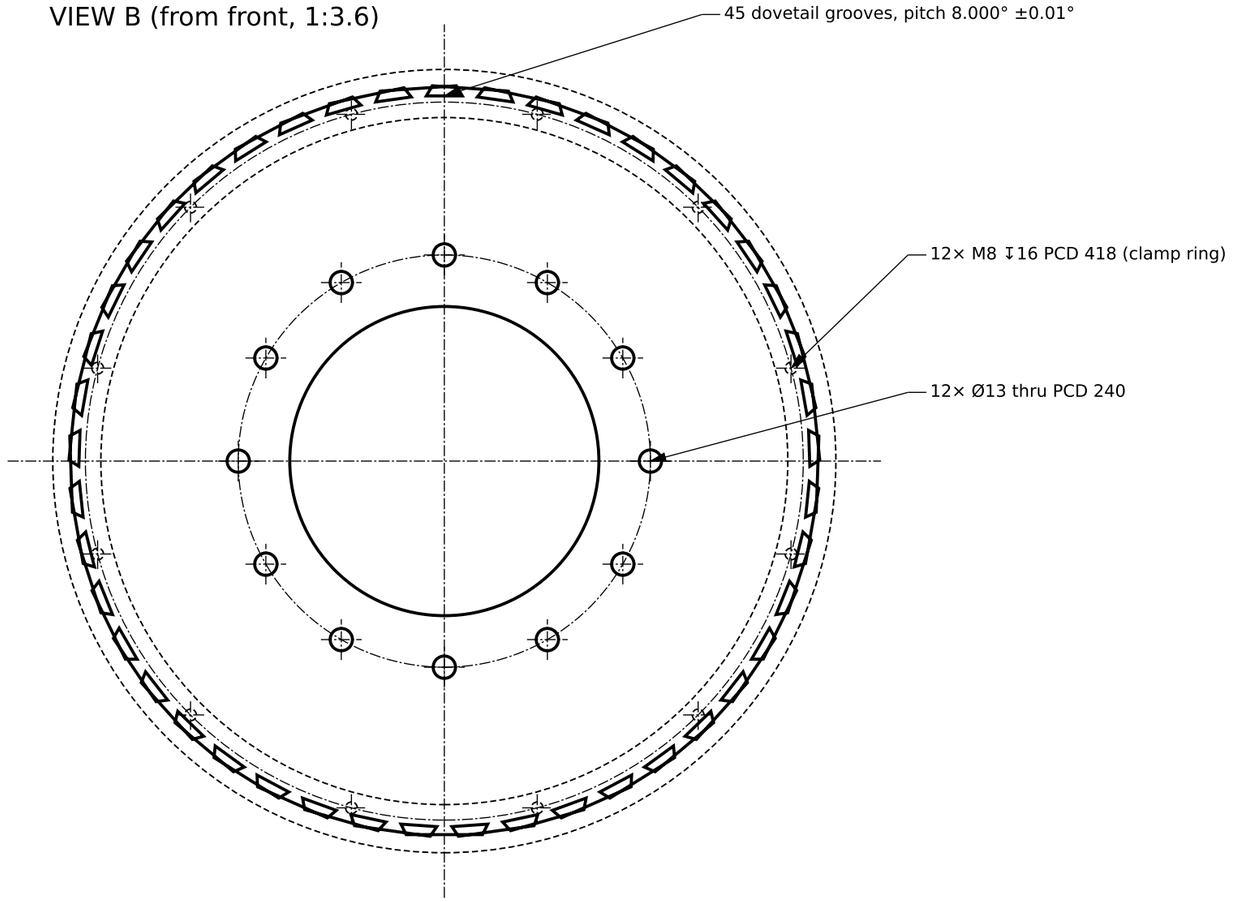
GENERATOR REV C · ASSEMBLY SECTION

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-G00	C · 2026-11-05	1:2.2
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
see parts list	≈ 330 kg drivetrain (no hub)	Claude / not checked

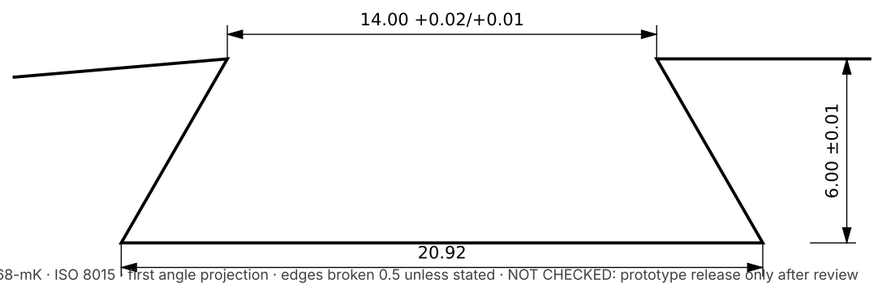


SECTION A-A (1:2.86)

VIEW B (from front, 1:3.6)



DETAIL C: GROOVE (5:1), flanks 60°



Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

NOTES

1. Rev C (doc 18 §5.1): grooves replace the shrink fit and the 3 keys. Grooves open to the front (x 343.5), closed by the rear stack stop (x 417); broached or wire-EDM.
2. Groove angular position ±0.01°, groove seat radius R212.60 ±0.01 to the bolting-face axis; foot clearance 0.01 to 0.03 (slides by hand when clean), fixed by VPI resin, clamped by NP1-202 (12× M8 A4-80, 20 Nm).
3. Run-out of groove lands and rear stop face to the bolting face and Ø180 spigot ≤ 0.02; bolting face flatness 0.02.
4. Bolt to the spindle flange: 12× M12 10.9 + hardened washers, 110 Nm, 2 fitted dowels Ø10 H7 PCD 240; Al-filled paste 0.05 on the full face (heat path, doc 14 §6).
5. Stress relieve 580 to 620 °C, 2 h, before final machining. No paint on grooves and bolting face; elsewhere epoxy primer 60 µm.

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

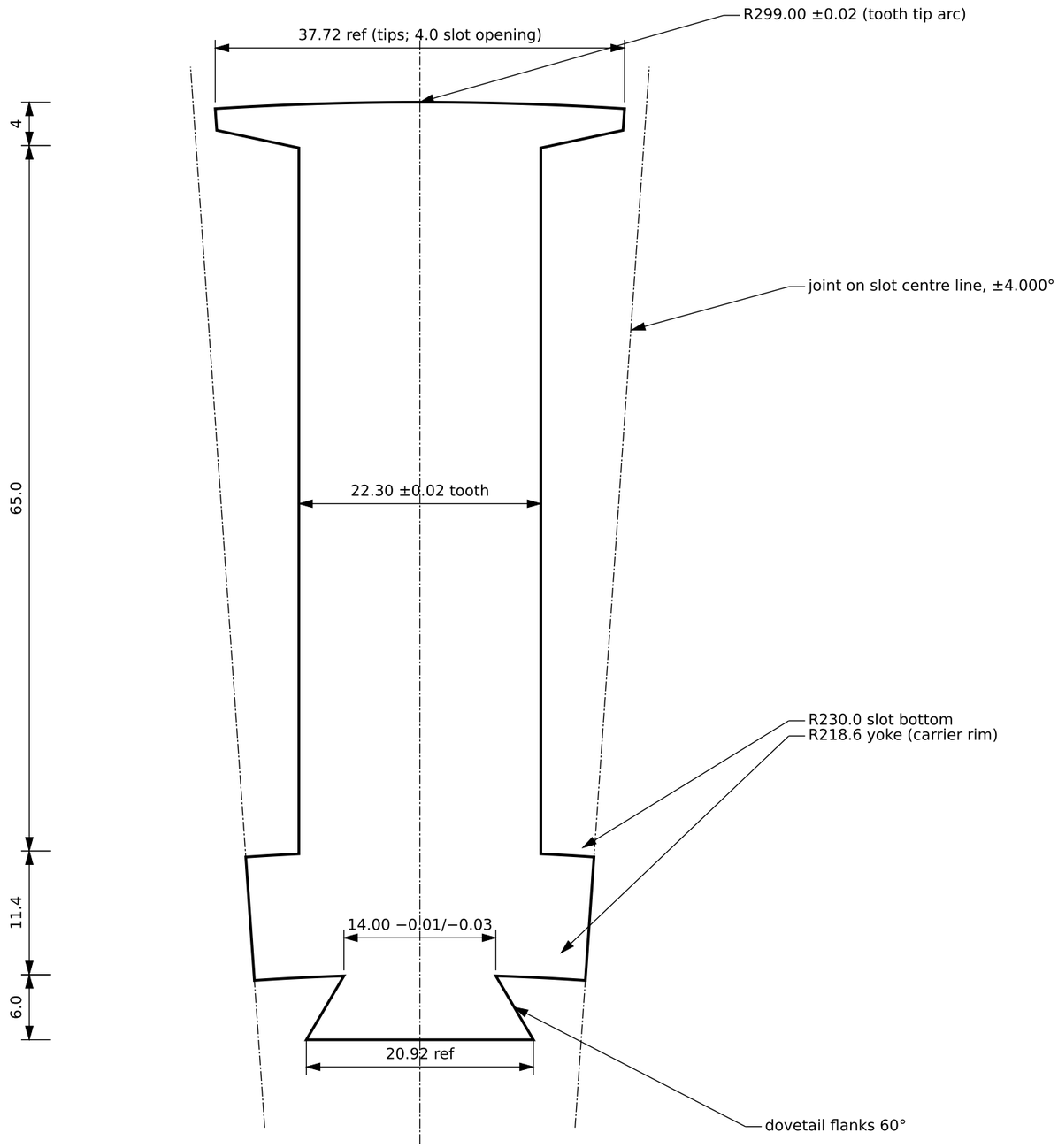
STATOR CARRIER WITH 45 DOVETAIL GROOVES

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-200C	C · 2026-11-05	1:2.86 / 1:3.6 / 5:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N ring forging or plate, stress relieved	40.6 kg	Claude / not checked

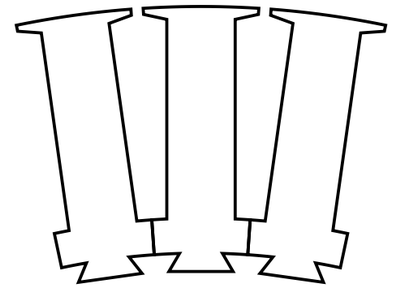
NOTES

1. DXF layer OUTLINE = cutting contour. Laser cut (P1), burr ≤ 0.02, heat zone ≤ 0.1; or progressive die (series).
2. Magnetic contour identical to the doc 14 one-piece lamination; joints on the slot centre lines in the yoke (8.000° pitch).
3. Stack: 205 sheets, 74.0 ±0.1 (sheets counted, height measured per segment), bonded in fixture NP1-T03 at 1.0 to 1.5 MPa, varnish cure per supplier.
4. Checks per stack: tip radius R299.00 ±0.02 on gauge, foot width 14.00 −0.01/−0.03, slot opening edges straight 0.05 over 74; Franklin test ≥ 10 Ω·cm².
5. Do not machine tooth tips after stacking. 45 + 3 spare segments per stator; all identical.

SEGMENT (2:1), radial direction up



3 OF 45 SEGMENTS (1:2)



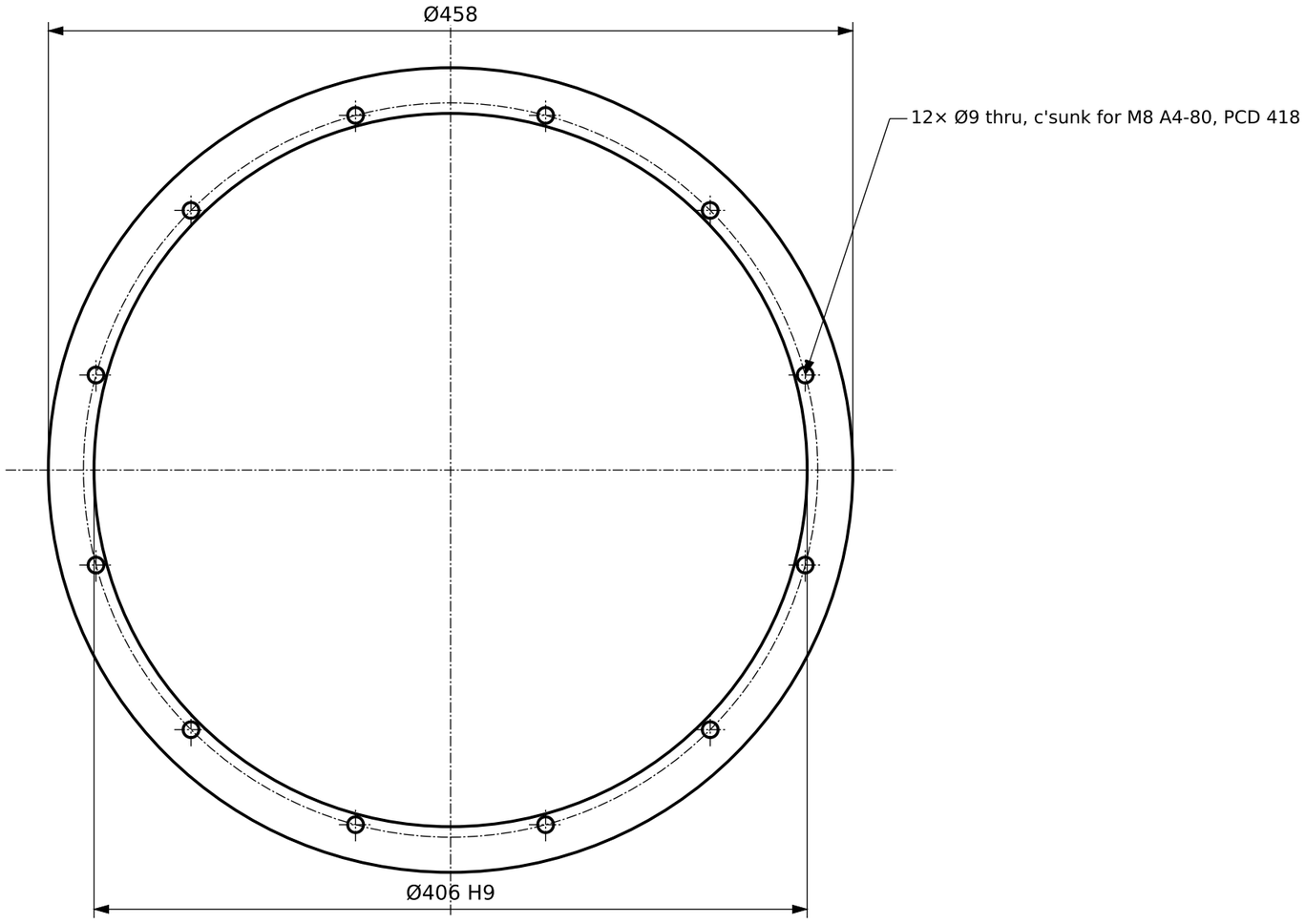
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

STATOR TOOTH SEGMENT LAMINATION (45 + 3 PER STATOR)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-201C	C · 2026-11-05	2:1
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
M270-35A, 0.35 mm, self-bonding varnish	1.07 kg per stack	Claude / not checked

VIEW (thickness 8.0, faces parallel 0.05)



NOTES

1. Laser cut from 8 mm plate, faces machined flat 0.05.
2. Must not touch the coils: inner end-winding radius ≥ 232; ring OD 229.

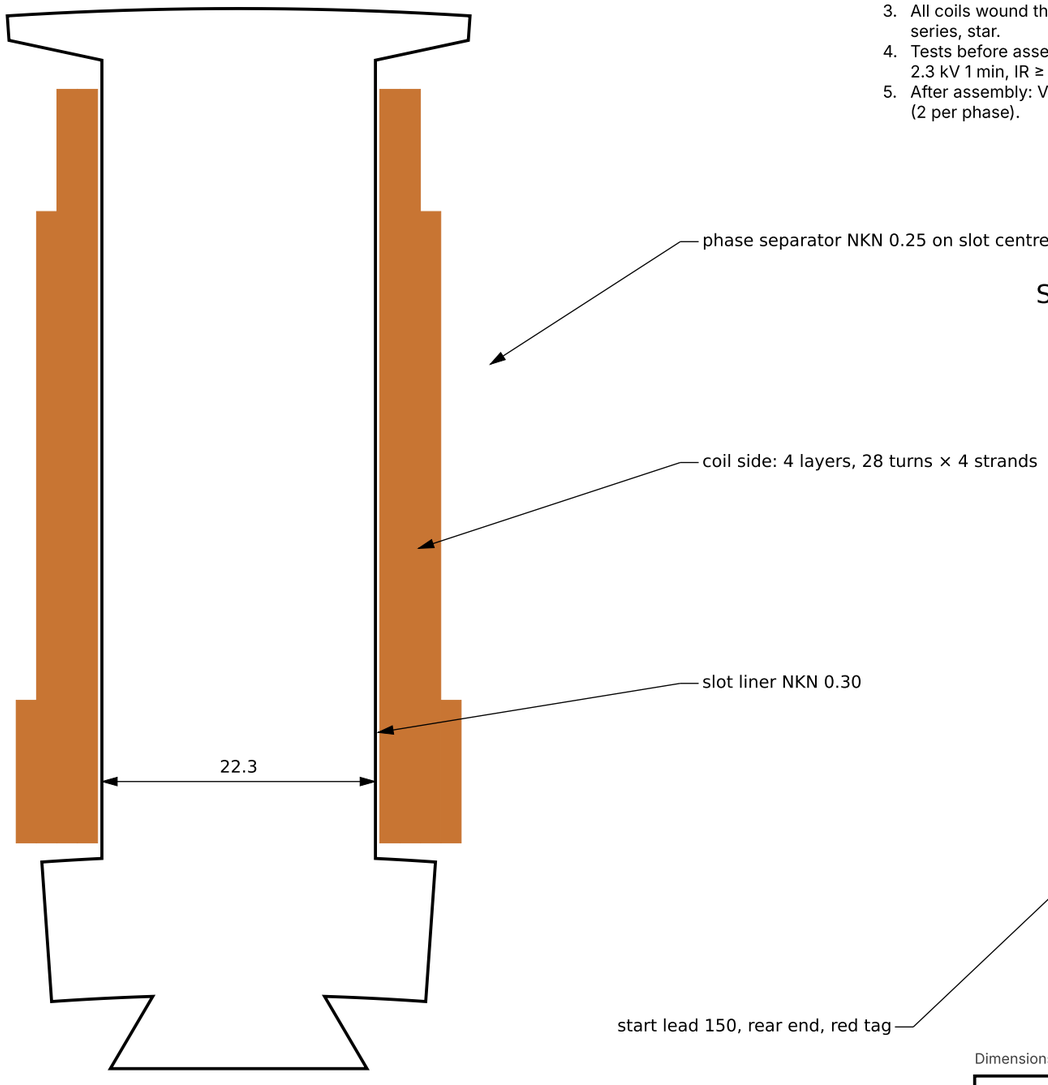
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

STATOR FRONT CLAMP RING

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-202	B · 2026-11-05	1:3.3
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
1.4404 (non-magnetic)	2.2 kg	Claude / not checked

COIL SIDES IN THE SLOT (2:1)

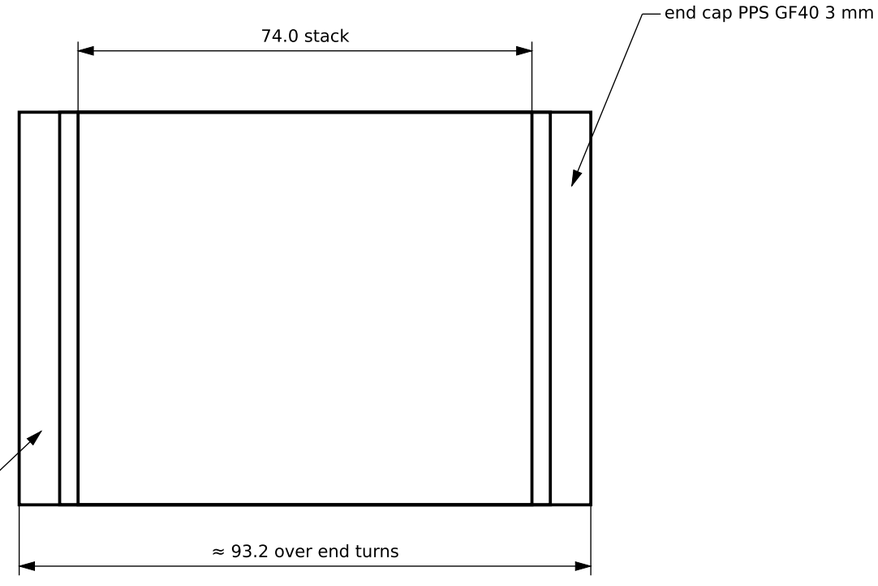


Layer (from tooth flank)	Positions	Used
1	37	37
2	37	37
3	31	31
4	17	7
5	3	0 (reserve)
Total (28 turns × 4 strands)	125	112

NOTES

1. Wind on chuck NP1-T04 by the dovetail, 4 parallel strands Ø1.55 (≤ 1.66 over enamel), 25 to 30 N per strand, layer by layer from the tooth flank, ≤ 60 rpm (doc 18 §5.6).
2. 28 turns per coil (Rev C keeps doc 18): 373 V line at 178 rpm, magnets 100 °C; 389 V at 60 °C. Wire per coil ≈ 6.6 m per strand; R20 per phase 0.217 Ω.
3. All coils wound the same direction; reversal at connection (doc 18 Appendix A): 15 coils per phase in series, star.
4. Tests before assembly (doc 18 §5.7): resistance ±2 % of the lot mean, surge 3.5 kV, hipot to the segment 2.3 kV 1 min, IR ≥ 1 GΩ.
5. After assembly: VPI class H epoxy, end windings potted (doc 18 §5.11, §5.12). PT100 in 6 separator gaps (2 per phase).

SIDE VIEW OF COIL ON TOOTH (1:1, simplified)



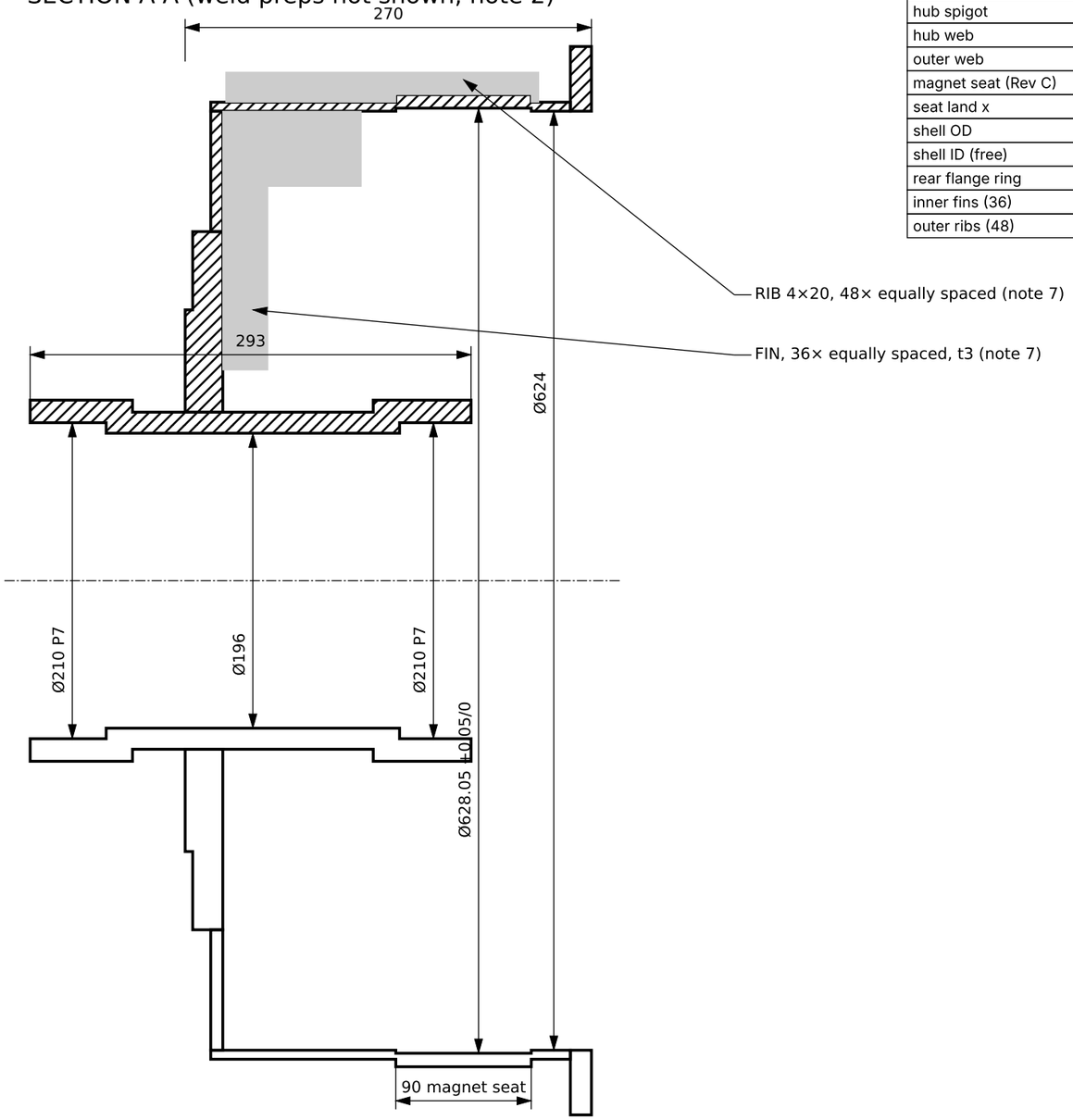
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

COIL ON SEGMENT (WOUND SEGMENT)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-203C	C · 2026-11-05	2:1 / 1:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
Cu enamelled Ø1.55 grade 2, class 200; NKN 0.30; PPS 4540 caps	≈ 0.45 kg Cu per coil	Claude / not checked

SECTION A-A (weld preps not shown, note 2)



Feature	Size	Tolerance / note
cup seats (2)	Ø210 × 45	P7, Ra 0.8, coax. 0.02
cup shoulders	Ø196	square 0.01 to axis
hub spigot	Ø360 × 5	f7; face flat 0.03
hub web	20 thick, Ø464	12× M16 thru PCD 400
outer web	8 thick	full pen. weld to shell
magnet seat (Rev C)	Ø628.05 × 90	+0.05/0, coax. 0.02, Ra 3.2
seat land x	335 to 425	as Rev B
shell OD	Ø646	steel under seat 8.98
shell ID (free)	Ø624	clearance to magnets
rear flange ring	Ø624/Ø710 × 14	M10 PCD 644 + M10 PCD 678
inner fins (36)	t3, x 220-312, r 140-312	L-plate, web + bore, every 10°
outer ribs (48)	4 × 20, x 222-430	tips Ø676, every 7.5°

NOTES

1. Rev C: magnet seat Ø628.05 +0.05/0 (was Ø618.40) for the 12 mm N45SH blocks; steel under the seat 8.98 (FEA: 1.45 T median, 1.7 T at the pole edges, included in the EMF).
2. Welds full penetration at web/housing and web/shell, ISO 5817 B, 100 % MT, UT of the shell seam (ring-rolled shell preferred, doc 18 §4.2). Stress relieve 580 to 620 °C before machining.
3. Machine in one set-up: cup seats Ø210 P7, magnet seat, hub spigot Ø360 f7, rear flange face; coaxiality ≤ 0.02, shoulders square ≤ 0.01.
4. Hub web 12× M16 thru PCD 400; rear flange ring 24× M10 PCD 644 (rear cover) and 24× M10 PCD 678 (disc).
5. Magnet seat: degrease, grit blast Sa 2½ just before bonding; no paint inside. Outside zinc spray + epoxy (C5).
6. Balance finished rotor (drum, magnets, sleeve, disc, cover) to ISO 21940-11 G2.5 at 250 rpm, correction on the rear flange ring.
7. Passive cooling (doc 23 §4a): 36 inner fins t3 S355, L-shaped, fillet-welded both sides to hub web and shell bore, every 10°, 9 mm clear of the end windings; 48 outer ribs 4 × 20 S355 on the shell outside, every 7.5°, intermittent fillet welds 30/60 both sides, ends rounded. Weld fins and ribs before stress relief and machining.
8. Rear flange ring welded to the shell full round, continuous: with the hub web it keeps the 9 mm shell round against the magnet pull (doc 23 §5). Seat roundness 0.05 after stress relief.

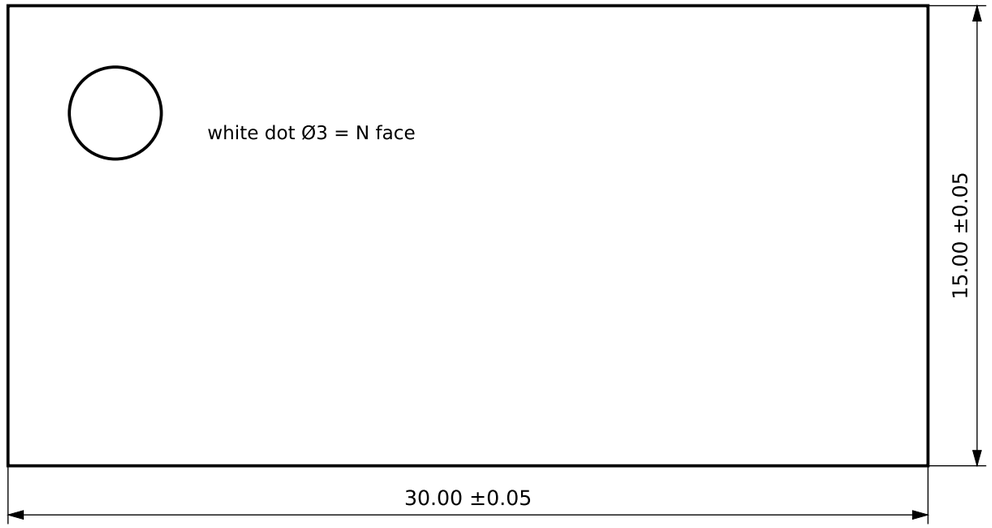
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

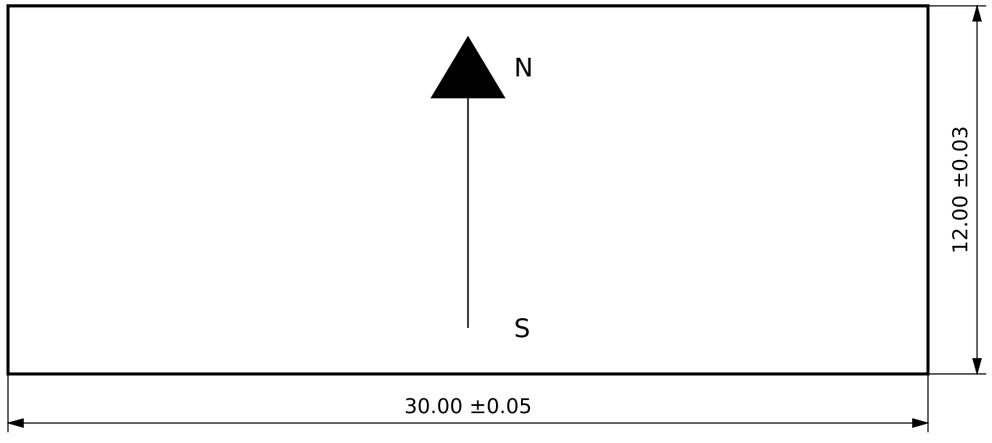
ROTOR DRUM WELDMENT (GENERATOR YOKE + BEARING HOUSING)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-210C	C · 2026-11-05	1:3.8
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355J2+N, welded, stress relieved	91.8 kg + 12.5 kg fins/ribs	Claude / not checked

VIEW A: N FACE (5:1)



VIEW B: END, magnetised through 12 (5:1)



Item	Requirement
Grade	N45SH: Br ≥ 1.32 T, HcJ ≥ 1,592 kA/m (20 kOe), (BH)max ≥ 342 kJ/m³ at 20 °C
Curves	J-H and B-H per batch at 20, 60, 100, 120 °C; αBr, βHcJ stated
Magnetisation	Through the 12 mm thickness, saturated; axis within 2° of the face normal
Tolerances	30 ±0.05, 15 ±0.05, 12 ±0.03; faces flat 0.02, parallel 0.02
Edges	Chamfer 0.2 to 0.3 × 45° all 12 edges; no chips > 0.3
Coating	NiCuNi 10 to 20 µm; 72 h neutral salt spray clean; adhesion OK with the epoxy
Uniformity	Moment of each piece within ±1.5 % of the lot mean; value per serial
Marking	White dot Ø3 on the N face; serial on the packaging position
Quantity	200 + 10 spare per generator; 8 extra for Q1 tests
Packing	Repelling layout in non-magnetic spacers; air freight per IATA PI 953

NOTES

1. One part for N and S poles: N poles face N towards the stator, S poles are the same block flipped (dot hidden against the seat).
2. Solid block (not two 6 mm pieces bonded). If a maker offers only 30 × 15 × 6, the doc 20 bonded pair is the fall-back with identical magnetics.
3. Five blocks per pole along the stack (15 mm pieces = axial segmentation for eddy loss, 4.7 W at rated).
4. Release of N45SH for production is conditional on the 100 °C design basis (doc 19 §16.3, doc 23 §4): bench magnet temperature ≤ 75 °C, temperature watch, no deliberate phase shorting.

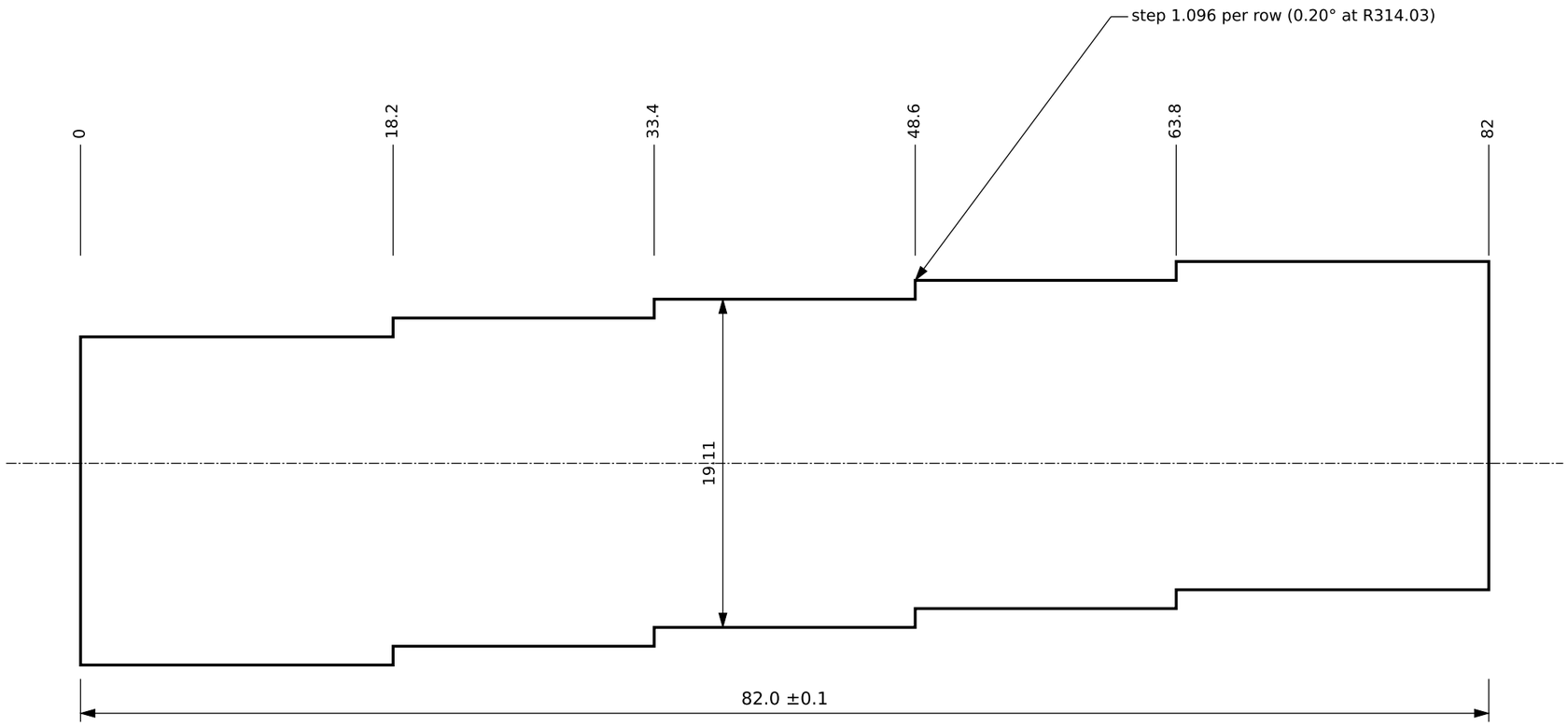
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

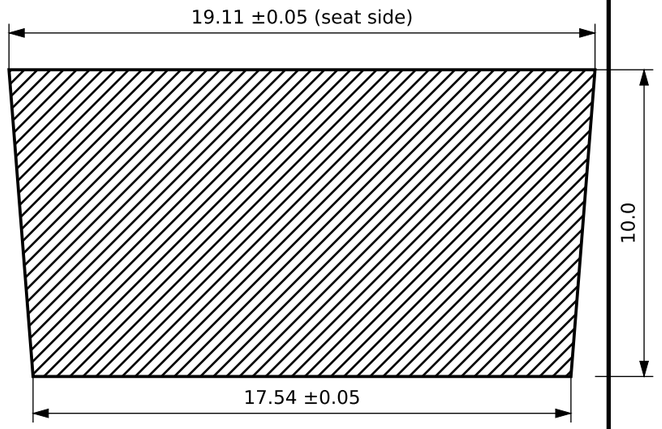
MAGNET BLOCK 30 × 15 × 12 (PURCHASE DRAWING)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-220E	E · 2026-11-05	5:1
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
Sintered NdFeB N45SH, NiCuNi	41.0 g each, 8.21 kg per generator	Claude / not checked

PLAN, seen from the stator (3:1). Steps set the row skew.



SECTION (5:1), flanks follow the block sides



NOTES

1. One bar per pole gap (40 per rotor + 4 spare). Bar sits on the seat (wide face), bonded with the magnet epoxy; ends engage 4 mm pockets in the end rings NP1-222C.
2. Row boundaries (ordinates from the front end): steps of 0.20° each; the bar is identical for every gap. Mark the front end.
3. Side faces are the datum for the blocks: 0.10 glue clearance each side (comb width = gap between neighbouring blocks − 0.20).
4. G11 (class H), ±0.05 on widths; edges deburred, no delamination. Prove on the Q1 panel (doc 18 §10).

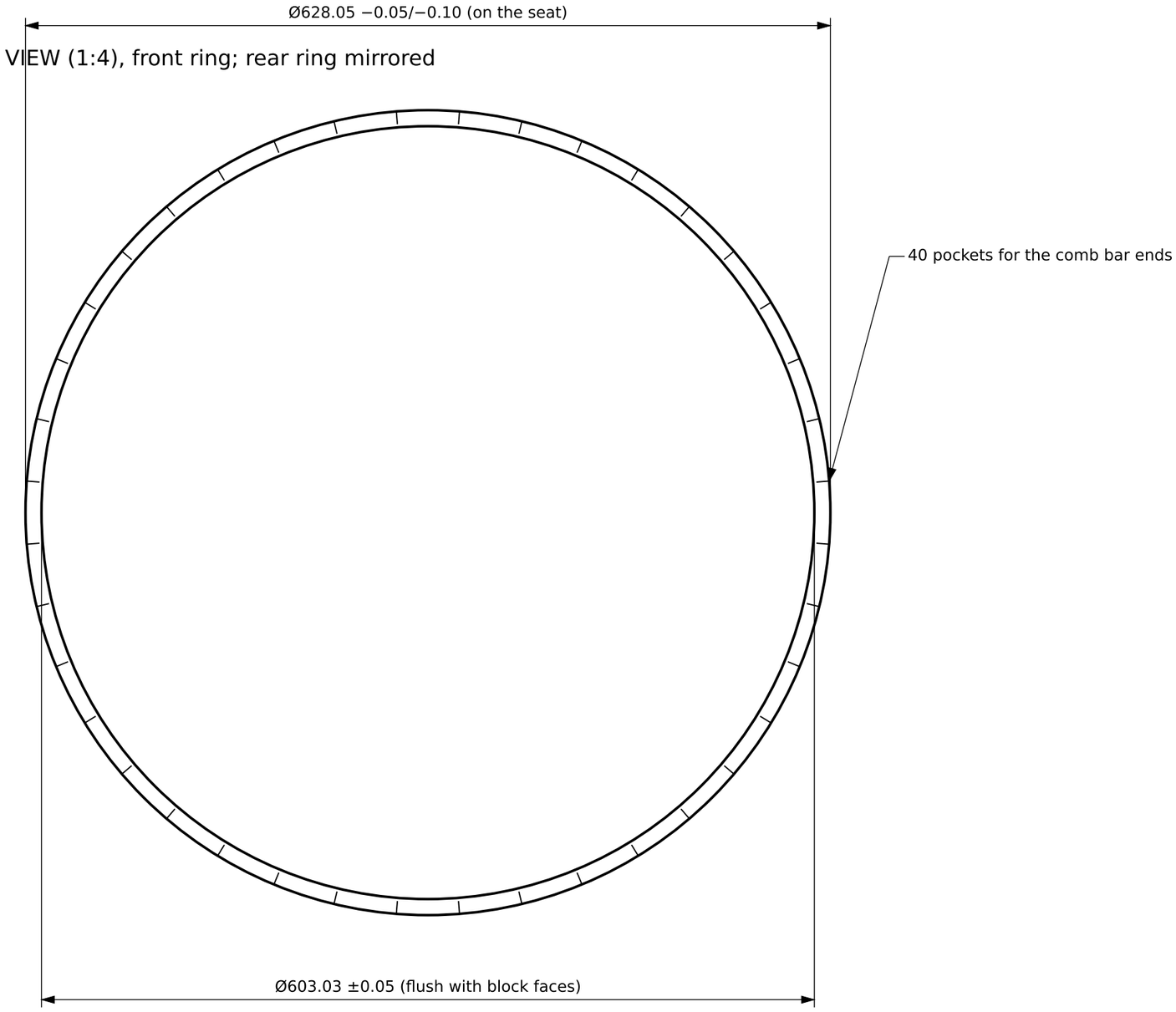
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

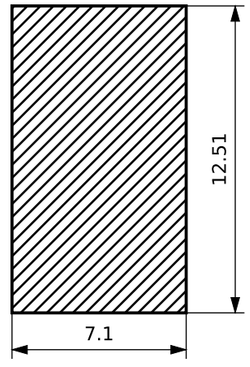
STEPPED POLE COMB BAR (40 + 4 PER ROTOR)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-221C	C · 2026-11-05	3:1 / 5:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
G11 glass-epoxy sheet, CNC machined	≈ 25 g	Claude / not checked

VIEW (1:4), front ring; rear ring mirrored



SECTION (4:1)



NOTES

1. Front ring at x 335 to 342.1, rear ring at x 417.9 to 425 (both 7.1 wide). They close the magnet pack and hold the comb bar ends.
2. Pockets: 19.11 wide (seat side), 10 deep radially, 4 deep axially, on the gap centre lines (4.5° + n × 9°), front ring rotated −0.40°, rear ring +0.40° (row skew).
3. May be made in 4 bonded arcs if no G11 tube of this size is available; joints on pocket centre lines.
4. Bonded to the seat with the magnet epoxy, then covered by the moulded glass sleeve.

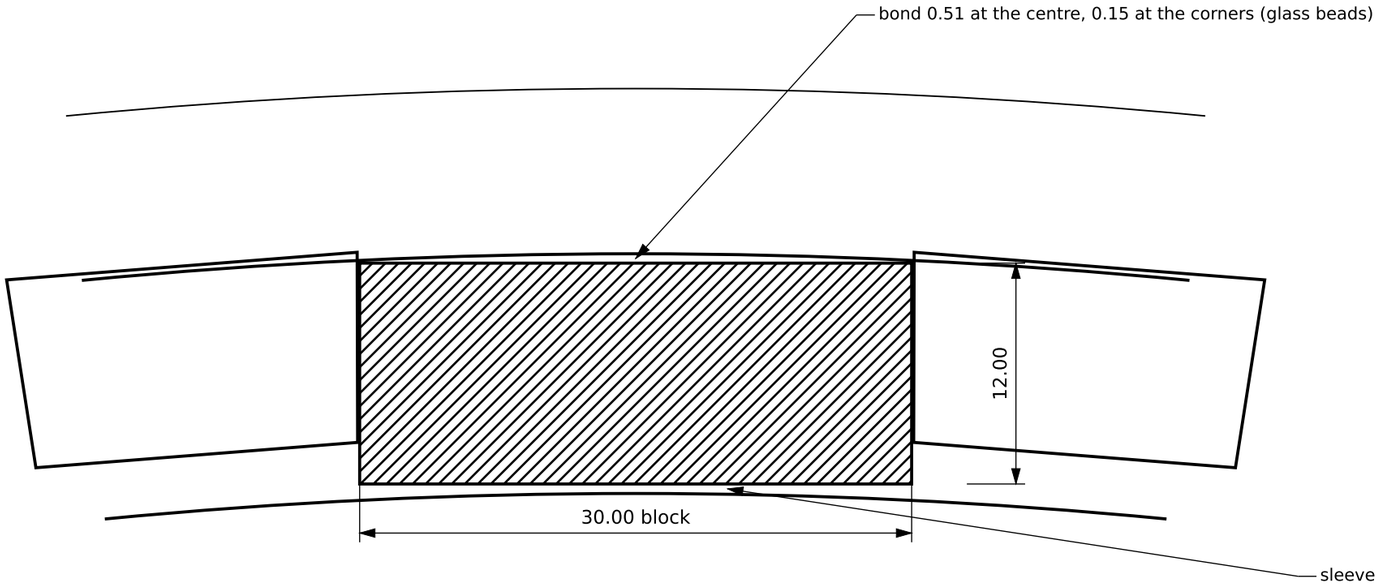
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

MAGNET END RINGS, FRONT AND REAR

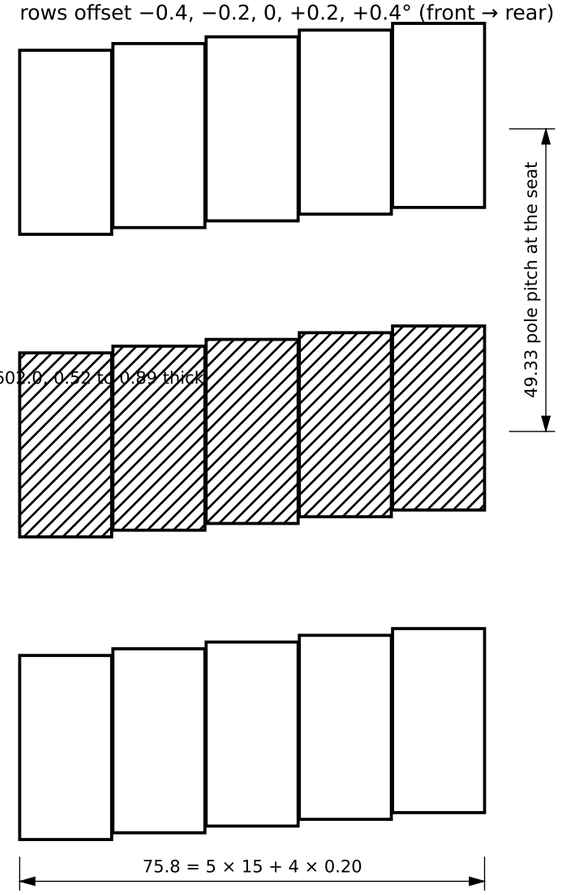
DRAWING NO.	REV / DATE	SCALE (A3)
NP1-222C	C · 2026-11-05	1:4 / 4:1
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
G11 glass-epoxy, machined from tube or segments	≈ 0.12 kg each	Claude / not checked

POLE SECTION (3:1), centre line = pole N1



LAYOUT, 3 POLES UNROLLED (1:1), front row left

rows offset −0.4, −0.2, 0, +0.2, +0.4° (front → rear)



NOTES

1. Bond: two-component toughened epoxy, RT cure, 80 °C post-cure, with 0.15 glass beads (doc 18 §4.2). Blocks placed with jig NP1-T02 against the comb bar faces; all N poles first, then S.
2. Sequence per pole: comb bars and end rings bonded and cured first; blocks row by row from the front; check each row against the template (skew 0.20 ±0.02° per row).
3. Pole map NP1-T10 after cure: every pole centre within ±0.05° of nominal, flux per pole within ±1.5 %.
4. Sleeve: E-glass/epoxy moulded on the Al mandrel NP1-T06 located on both cup seats; bore Ø602.0 ±0.03, thickness 0.52 at the block centres, 0.89 at the edges; covers the end rings.
5. Q1 check (doc 18 §10): sleeve thickness by section 0.45 to 0.95; air gap feeler check 2.0 ±0.1 all round after joining.

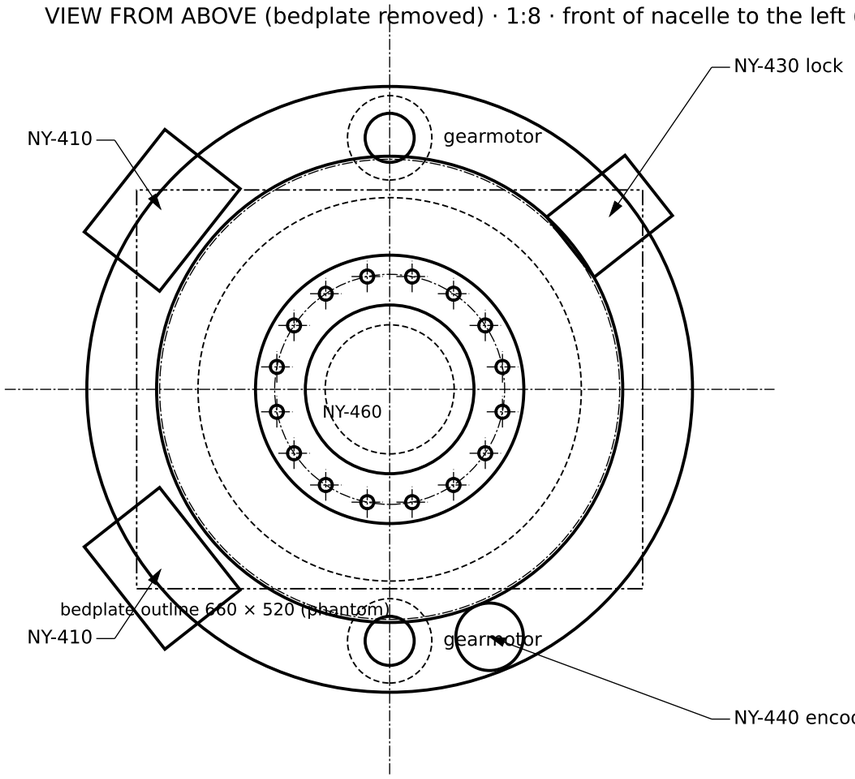
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

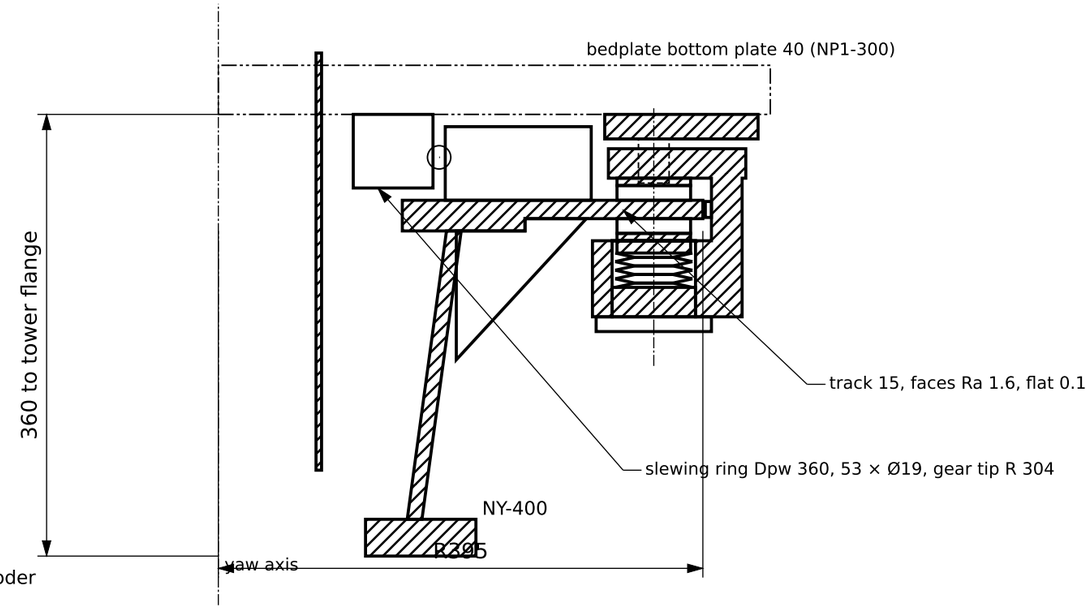
ROTOR MAGNET PACK: POLE SECTION, LAYOUT AND SLEEVE

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-223C	C · 2026-11-05	3:1 / 1:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
assembly	8.2 kg magnets	Claude / not checked

VIEW FROM ABOVE (bedplate removed) · 1:8 · front of nacelle to the left (−X)



SECTION A-A through the pad unit at 141.8° · 1:5



Item	Part	Qty	Make/buy	Mass
1	NY-400 tower-top adapter (S355J2+N weldment)	1	make	107 kg
2	Slewing ring Dpw 360, ext. gear z150 m4, ind. hardened teeth	1	buy	≈ 55 kg
3	Gearmotor 0.18 kW 4-pole, 1:1,680 planetary, 2 Nm motor brake	2	buy	≈ 22 kg
4	NP1-420 drive bracket (doc 16)	2	make	7.7 kg
5	NY-410 friction pad unit	2	make	13 kg
6	NY-430 yaw lock	1	make	6.5 kg
7	NY-440 encoder + cam limit switch unit	1	make/buy	6.9 kg
8	NY-460 cable hanger	1	make	10 kg
9	Bolts: 16 M16 inner, 16 M16 outer (from below), 16 M20 tower	48	buy	
10	Heated 2D ultrasonic + heated cup/vane on the nacelle mast	1+1	buy	

NOTES

1. No hydraulics on the yaw system. Holding: 2 spring-applied motor brakes (fail-safe) through the gear train; damping: 2 passive pad units, 2.4 to 3.6 kNm.
2. Design yaw moment held 8 kNm by ONE pinion (backlash) without friction credit; gearbox static rating ≥ 1,500 Nm output.
3. All bolts 10.9, preload with calibrated torque, M16 90 kN, M20 110 kN. Grease the ring per supplier (−40 °C synthetic).
4. Yaw rate 0.5 °/s at 1,500 rpm. One VFD drives both motors; the safety chain opens the brake supply directly.

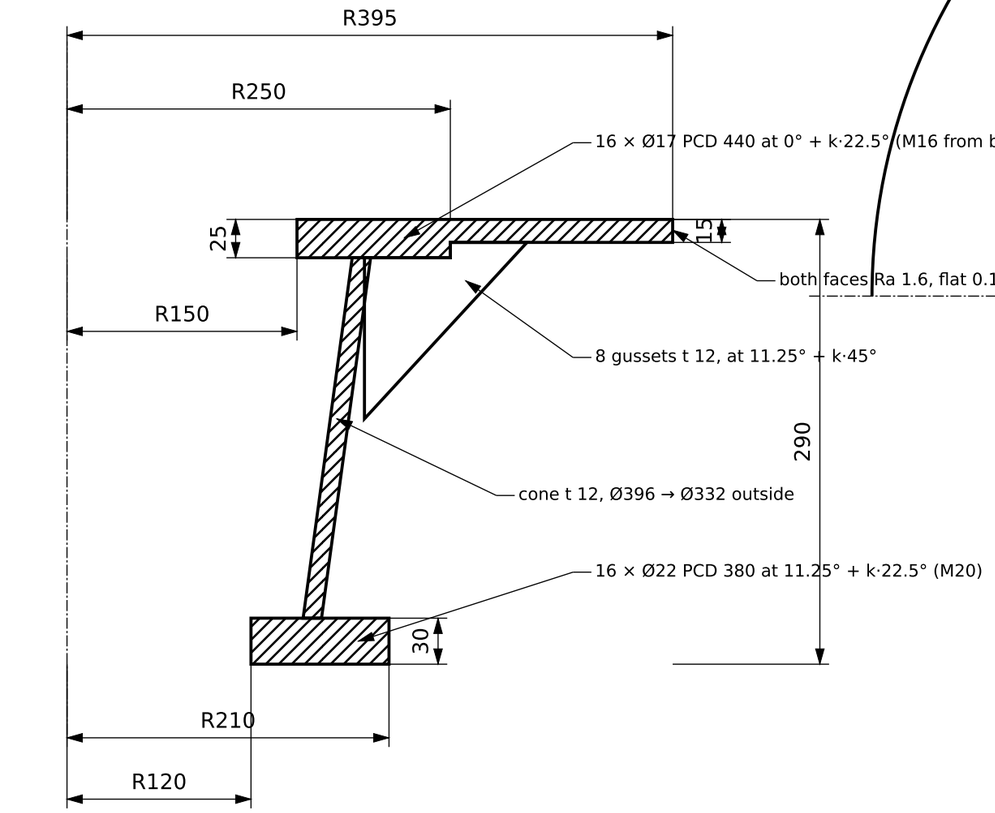
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

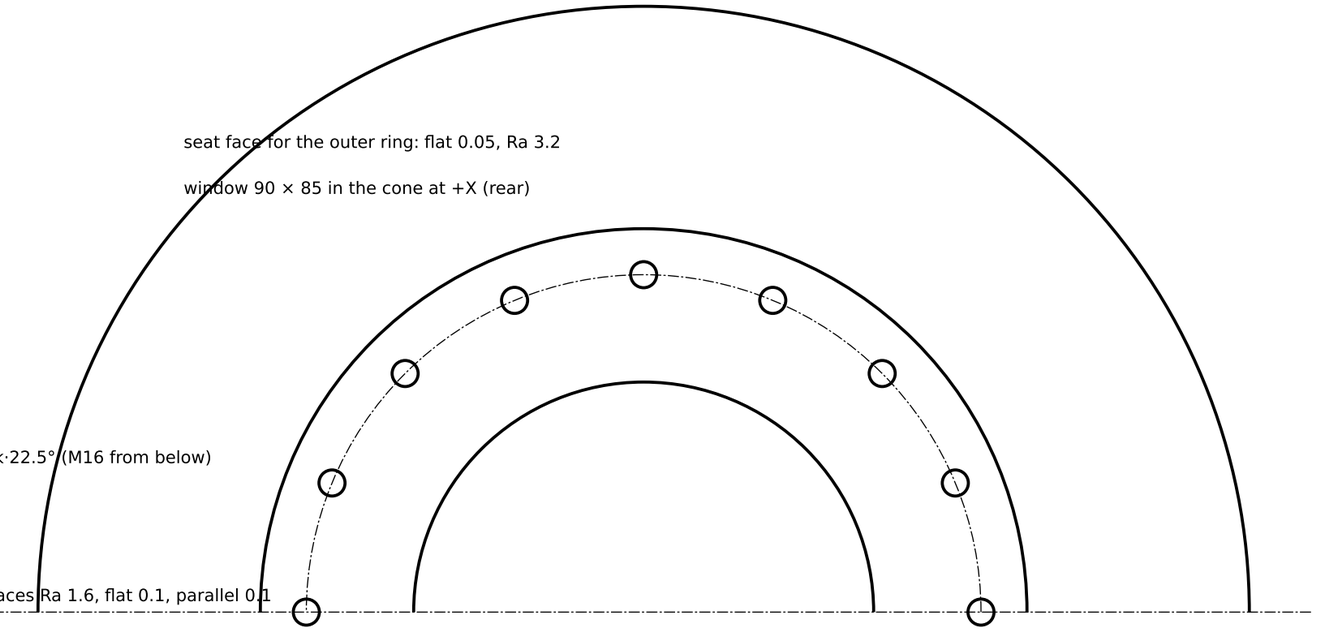
YAW SYSTEM · GENERAL ARRANGEMENT

DRAWING NO.	REV / DATE	SCALE (A3)
NY-000	A · 2026-11-05	1:5 / 1:8
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see parts list	≈ 275 kg (yaw incl. adapter)	Claude / not checked

HALF SECTION (between gussets) · 1:4



PLAN FROM ABOVE · 1:4



NOTES

1. Weld cone to seat plate and to flange with full-penetration welds (ISO 5817 B), gussets with 5 mm double fillets; stress relieve, then machine seat, track faces and flange in one set-up.
2. Seat and track are one machined plate: 25 mm blank, track turned to 15 mm (r 250 to 395). Track faces left bare under the pads; all other surfaces C5-VH coating (ISO 12944, zinc-rich primer).
3. Bolt loads at the SLM tilt moment 34.6 kNm: outer ring 19.1 kN per M16 (preload 90 kN), tower flange 32.0 kN per M20 (preload 110 kN).
4. Replaces the NP1-400 proposal of doc 16 (25 mm plate to Ø840, 143 kg): track moved in to r 395 and thinned to 15 mm because the pads only clamp 5.3 kN.
5. Gussets stay inside r 300 and below the track so the lower pad jaws (r ≥ 318) pass all the way round.

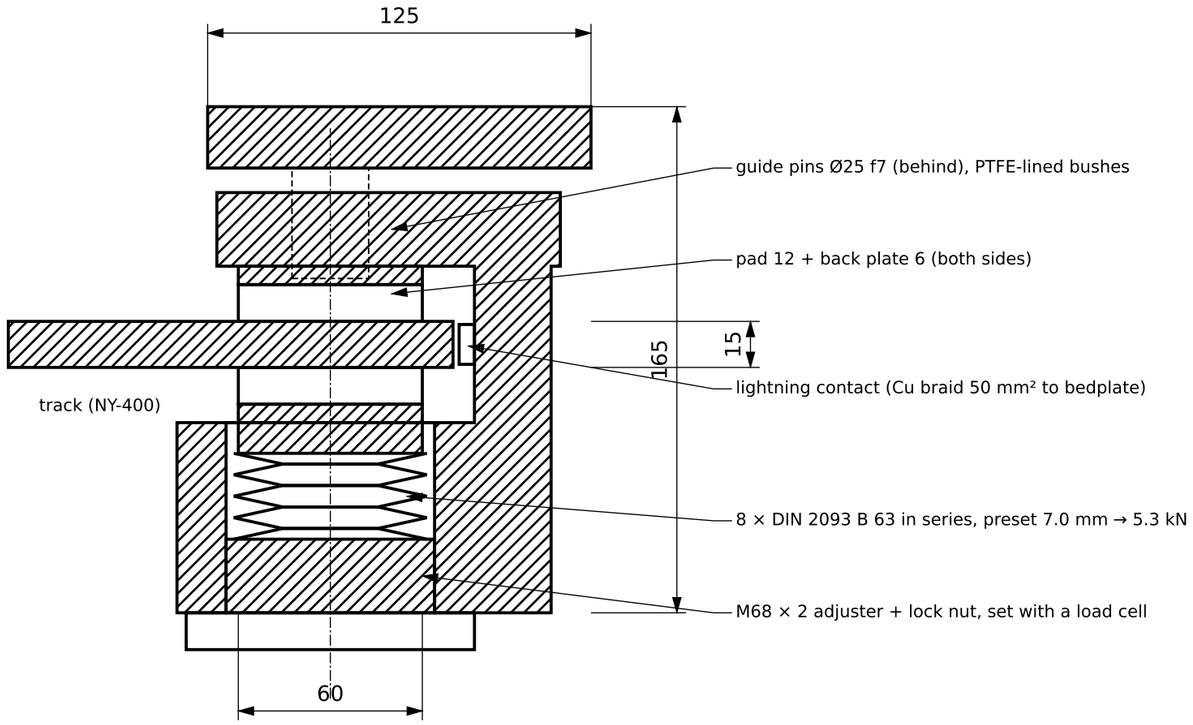
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

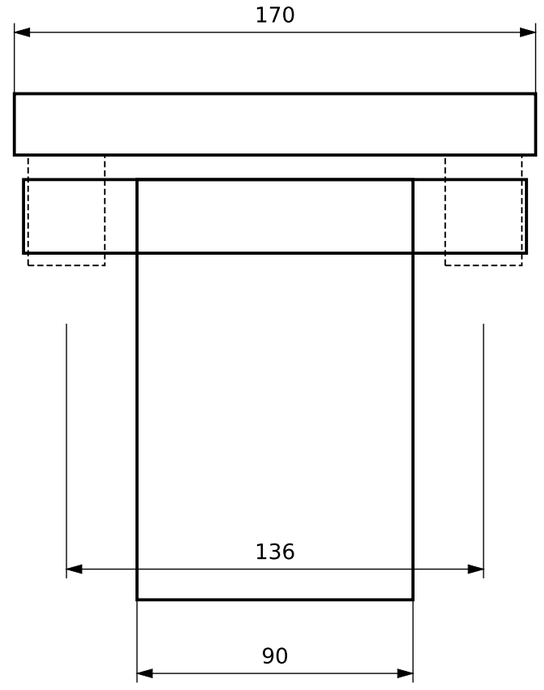
TOWER-TOP ADAPTER WITH FRICTION TRACK

DRAWING NO.	REV / DATE	SCALE (A3)
NY-400	A · 2026-11-05	1:4
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N plate and tube, weldment	107 kg (model)	Claude / not checked

SECTION IN THE RADIAL PLANE · 1:2



VIEW FROM OUTSIDE (radial) · 1:2



Item	Part	Qty	Material / spec
1	Bracket 170 × 125 × 20, 4 × Ø17 to the bedplate	1	S355J2+N
2	C-frame (upper jaw, web, lower jaw), machined	1	S355J2+N
3	Guide pin Ø25 f7 × 55, pressed into item 1	2	1.4462 duplex
4	Pad 90 × 60 × 12 on 6 mm back plate	2	organic industrial pad, μ 0.40 ± 20 %
5	Piston Ø60 × 10	1	1.4462
6	Disc spring DIN 2093 B 63 (63 × 31 × 2.5)	8	51CrV4, zinc-flake
7	Adjuster M68 × 2 + lock nut	1+1	1.4462
8	Lightning contact block + Cu braid 50 mm²	1	CuSn / Cu

NOTES

1. Floating C-frame: the springs push the lower pad up; the reaction pulls the frame down onto the upper pad, so both faces clamp 5.3 kN at any pad wear.
2. Two units give 2.4 to 3.6 kNm (μ 0.32 to 0.48), 3.0 kNm nominal, 1.51 × the 95th percentile operating yaw moment.
3. Pad pressure 0.98 MPa, sliding speed 3.1 mm/s, heat 26 W. Estimated wear 6.6 mm in 20 years of 9 mm usable (to confirm by test).
4. Spring stack at 0.50 h0 per disc: one year of wear lowers the clamp by 8 %. Re-set at the yearly service (load cell between the pads).
5. Pad change without lifting the nacelle: back off the adjuster, swing the frame down on the pins, slide the pads out sideways.

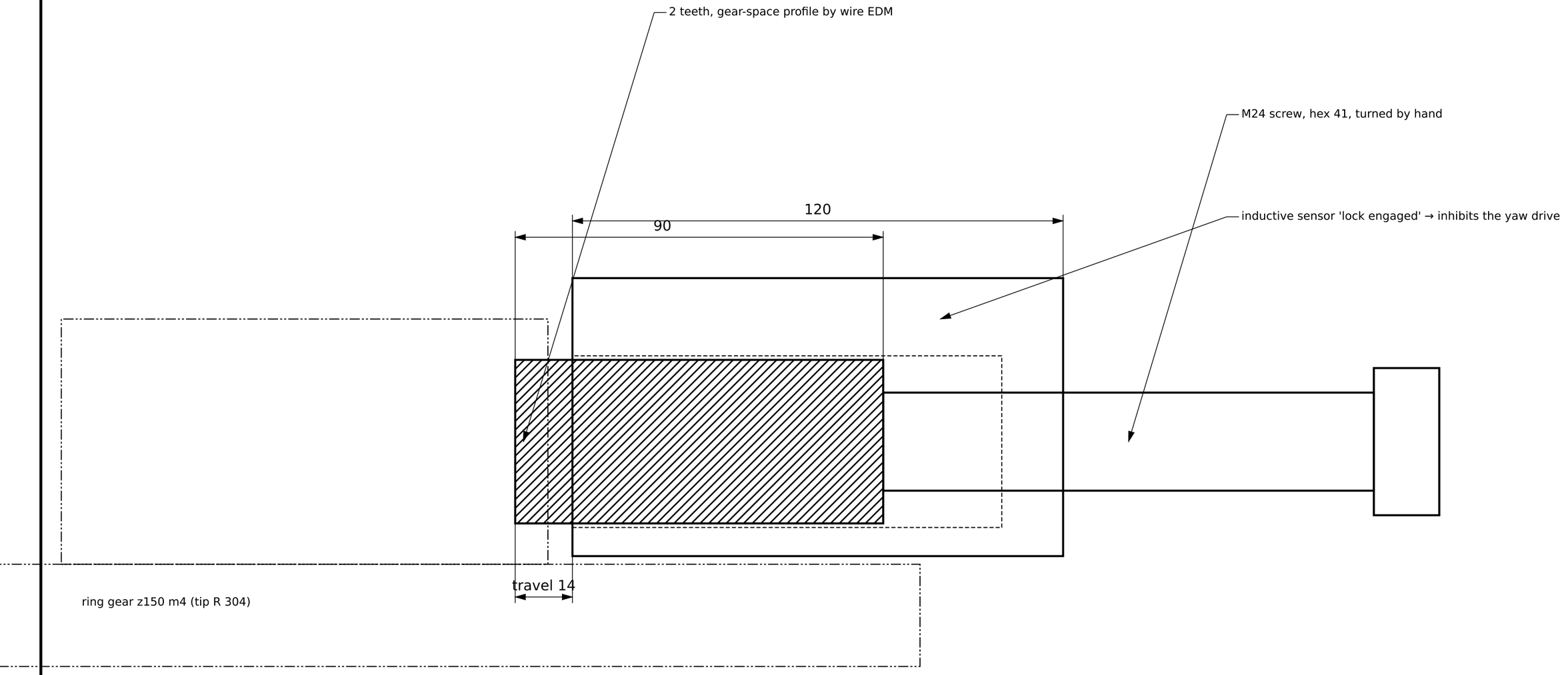
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

FRICTION PAD UNIT (PASSIVE, SPRING-LOADED)

DRAWING NO.	REV / DATE	SCALE (A3)
NY-410	A · 2026-11-05	1:2
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
frame S355J2+N, pins 1.4462	13 kg (model)	Claude / not checked

SECTION IN THE RADIAL PLANE (engaged) · 1:1



NOTES

1. Use: service only (yaw drives locked out). Turn the nacelle until a tooth space is opposite the block (encoder target), then screw the block in.
2. Strength: design yaw moment 8 kNm → 26.7 kN at the gear pitch radius on two teeth; tooth root shear and bending of the block S ≥ 3 (42CrMo4+QT, 900 MPa).
3. The housing bolts to the bedplate underside with 4 × M16 at the rear left corner (38.2°). Grease the slide; boot over the screw.

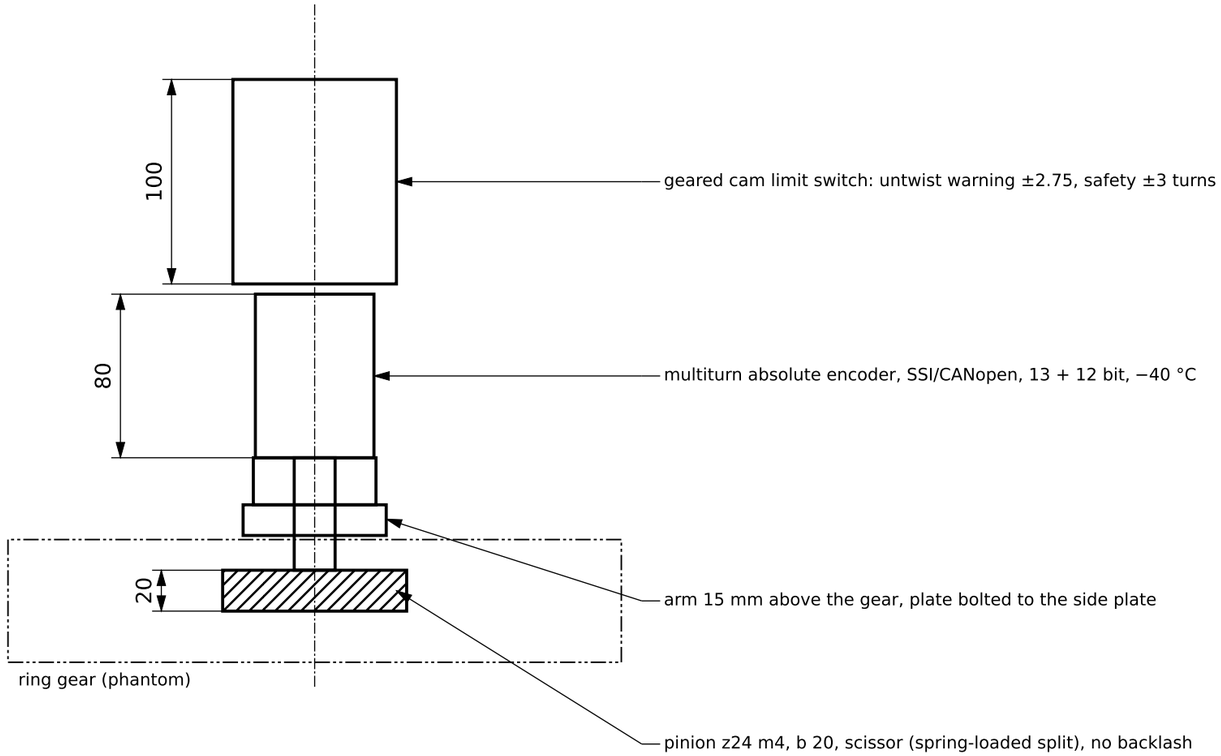
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

YAW LOCK (SERVICE)

DRAWING NO.	REV / DATE	SCALE (A3)
NY-430	A · 2026-11-05	1:1
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
housing S355J2+N, block 42CrMo4+QT	6.5 kg (model)	Claude / not checked

ELEVATION, VIEW TOWARDS THE SIDE PLATE · 1:3



NOTES

1. Ratio ring/pinion 150/24 = 6.25: ±3 nacelle turns = ±18.75 turns at the switch.
2. Centre distance 348 mm at -68° (right side, behind the drive), outside the side plate so the instruments stand beside it.
3. Cam contacts are positively opened, wired into the hardwired safety chain (PL d). The encoder is only for control (yaw angle, twist count, untwist).
4. Plausibility: proximity switch at 0° on a ring tooth marker checks the encoder every revolution.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

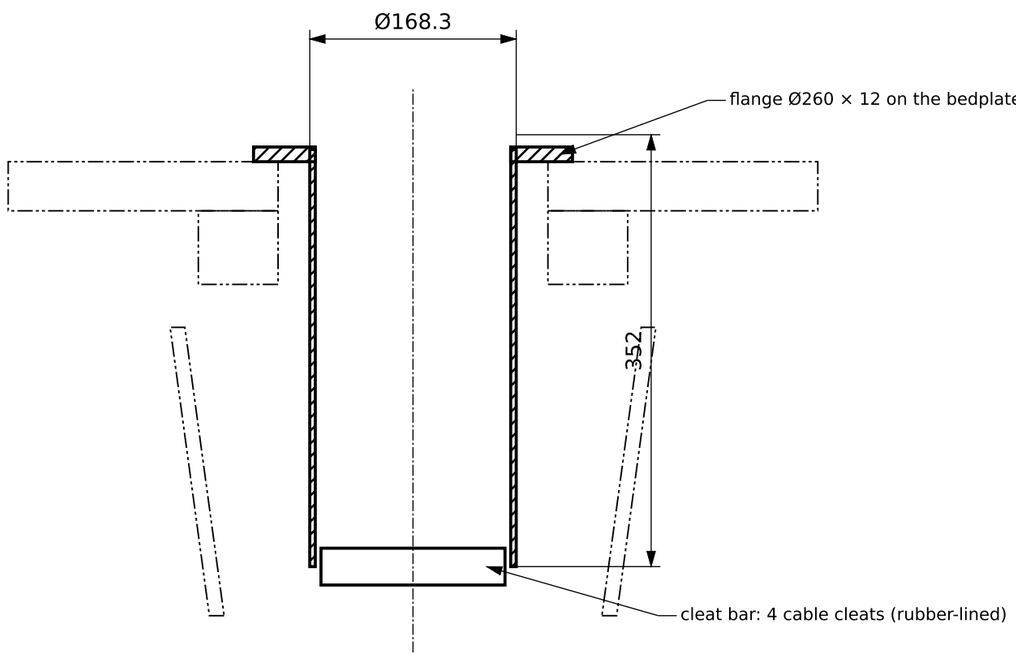
12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

YAW ENCODER AND CAM LIMIT SWITCH UNIT

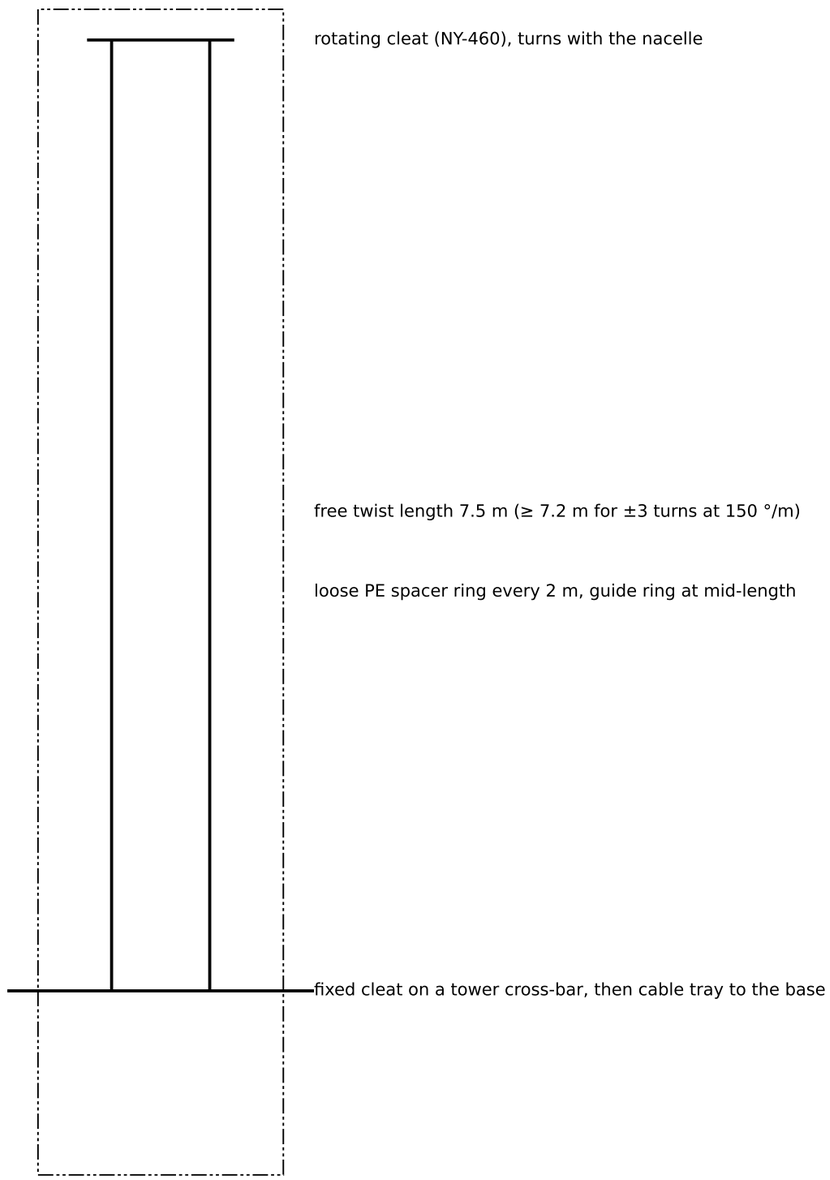
DRAWING NO.	REV / DATE	SCALE (A3)
NY-440	A · 2026-11-05	1:3
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
bracket S355J2+N, pinion 42CrMo4+QT	6.9 kg incl. instruments	Claude / not checked

Cable (all torsion-rated for wind turbine loops, −40 °C)	Ø approx.
Generator 4G6 + screen, 0.6/1 kV (doc 12)	20 mm
48 V auxiliary / backup supply 2 × 10 mm²	14 mm
Safety chain + control signals 12 × 1.0 mm², screened	12 mm
Industrial Ethernet (Cat 6A, torsion) + fibre option	12 mm
Earthing: separate Cu braid via the yaw lightning contact	–

HANGER IN THE YAW BORE · 1:5



TWIST LOOP IN THE TOWER (not to scale)



NOTES

1. Untwist by the controller at ±2.5 turns when the 10 min mean wind is below 5 m/s; safety chain opens at ±3 turns (NY-440 cam switch).
2. No slip ring for power (doc 07). Cables leave the bedplate inside the hanger tube, so they never touch the rotating ring or the adapter.
3. Inspection through the adapter window (NY-400) at every service: sheath wear, cleats, twist count against the encoder.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

CABLE HANGER AND TWIST LOOP

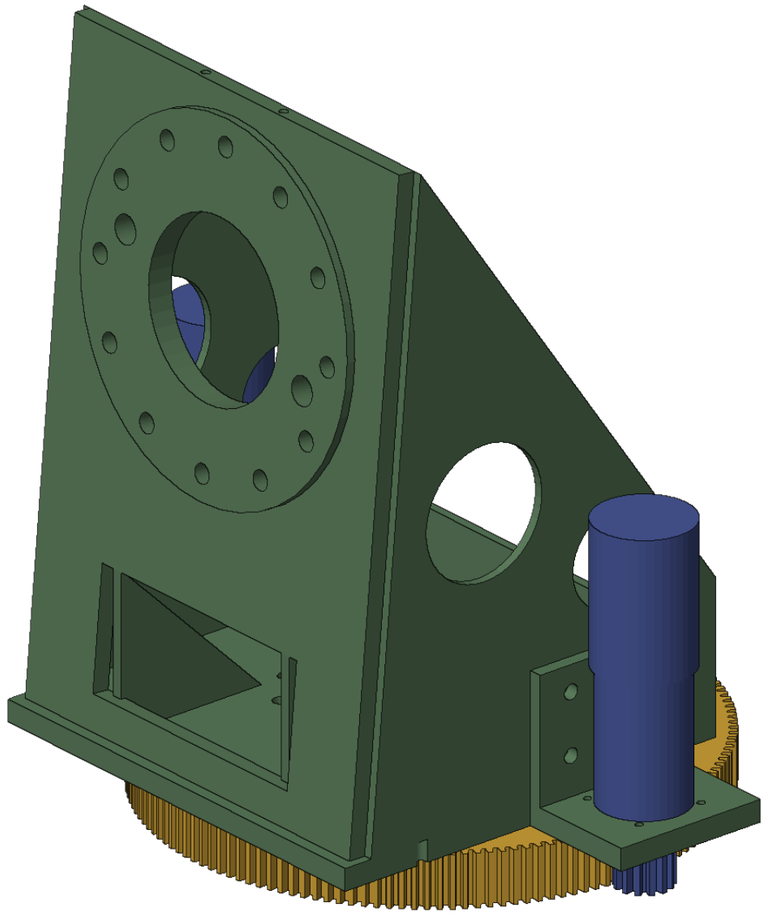
DRAWING NO.	REV / DATE	SCALE (A3)
NY-460	A · 2026-11-05	1:5 / NTS
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
tube S355 Ø168.3 × 4.5, flange S355J2+N	10 kg (model)	Claude / not checked

6 Assembly sequence

Workshop steps 1–11, site steps 12–13

6 Assembly sequence

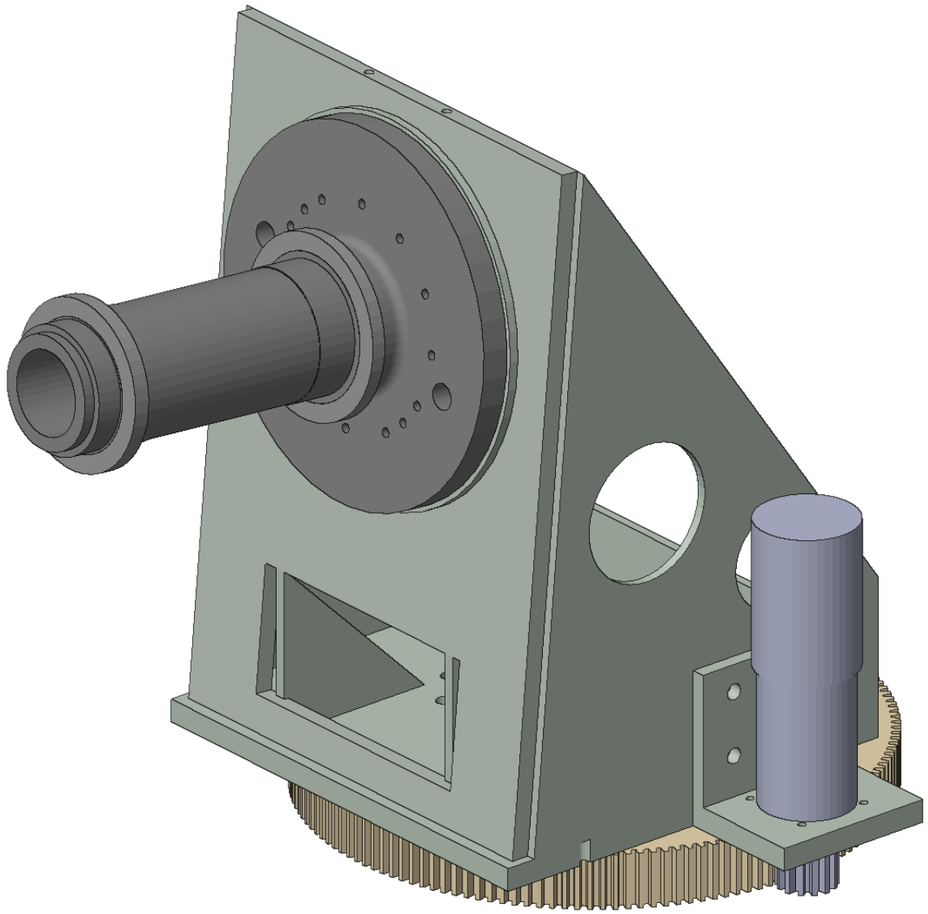
Step 1 Bedplate on the stand; yaw slewing ring, gearmotors and their brackets



- Set the NS6-300 bedplate upside-down on the assembly stand; check the yaw flange flatness (0.05) and the front-face angle (5.00° ± 0.05°) and record them.
- Fit the yaw slewing ring inner ring to the bedplate: 16 × M16 HV, cross pattern, 250 Nm (soft gearing mark towards the lock side).
- Fit the gearmotor brackets NP1-420 (4 × M16, 195 Nm) and the two gearmotors (4 × M10, 49 Nm). Set the backlash with the brackets' shims: 0.15–0.30 mm at the soft point.
- Check: ring turns freely by hand with both motor brakes released; record the turning torque.

Parts (new in colour, earlier steps light): NS6-300, NS6-911-I, NS6-911-O, NS6-912-L, NS6-912-R, NP1-420-L, NP1-420-R. Torques: NS6-900.

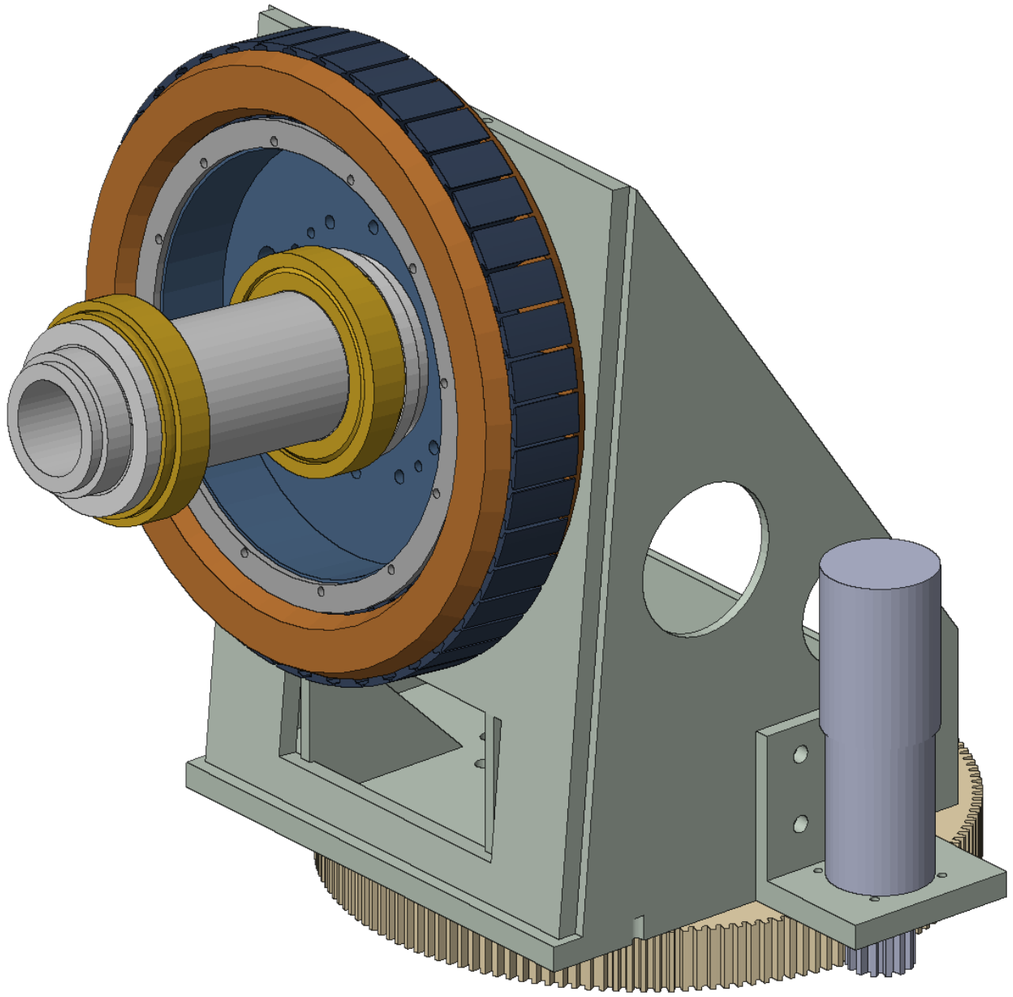
Step 2 Spindle with front spacer and rear seal sleeve bolted to the bedplate front plate



- Clean the spindle flange and the Ø410 pad; fit the Ø200 spigot; insert 12 × M20 10.9 from inside and torque to 540 Nm in three passes.
- Fit the front spacer NP1-101 and rear seal sleeve NP1-102 (heat to 120 °C), cables through the two Ø28 holes before the stator goes on.
- Check: flange face square to the yaw axis (5° tilt) within 0.05/400 with the bedplate level.

Parts (new in colour, earlier steps light): NP1-100, NP1-101, NP1-102. Torques: NS6-900.

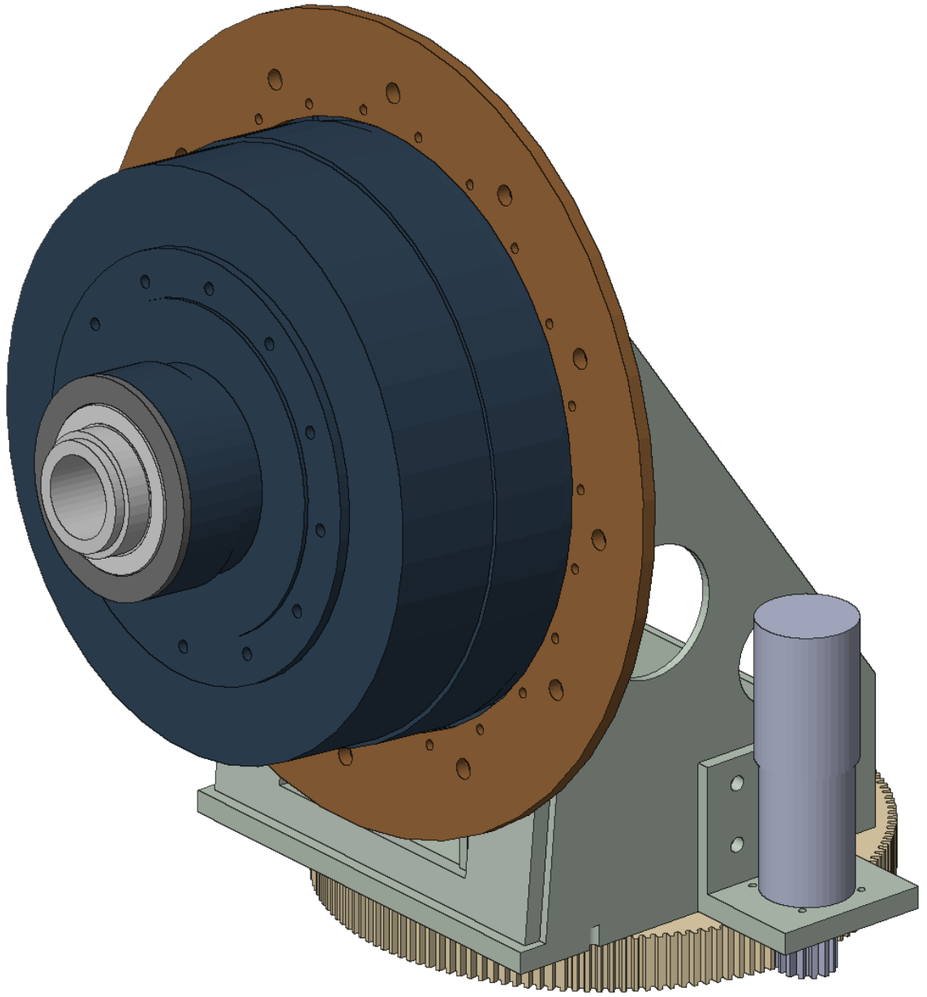
Step 3 Stator: carrier, wound tooth segments, clamp ring; bearing cones on the spindle



- Assemble the stator: 45 wound tooth segments NP1-203C slid into the carrier dovetails NP1-200C, clamp ring NP1-202 (doc 18, Rev C instructions).
- Insulation test 1 kV, winding resistance balance ≤ 2 %, surge test (doc 18 G07). Bolt the stator carrier to the spindle: 12 × M12, 110 Nm.
- Heat the bearing cones (110 °C, induction) and fit them on the spindle seats; pack grease (Mobilith SHC 220, 60 % fill).

Parts (new in colour, earlier steps light): NP1-200C, NP1-201C, NP1-203C, NP1-202, NS6-913-F, NS6-913-R. Torques: NS6-900.

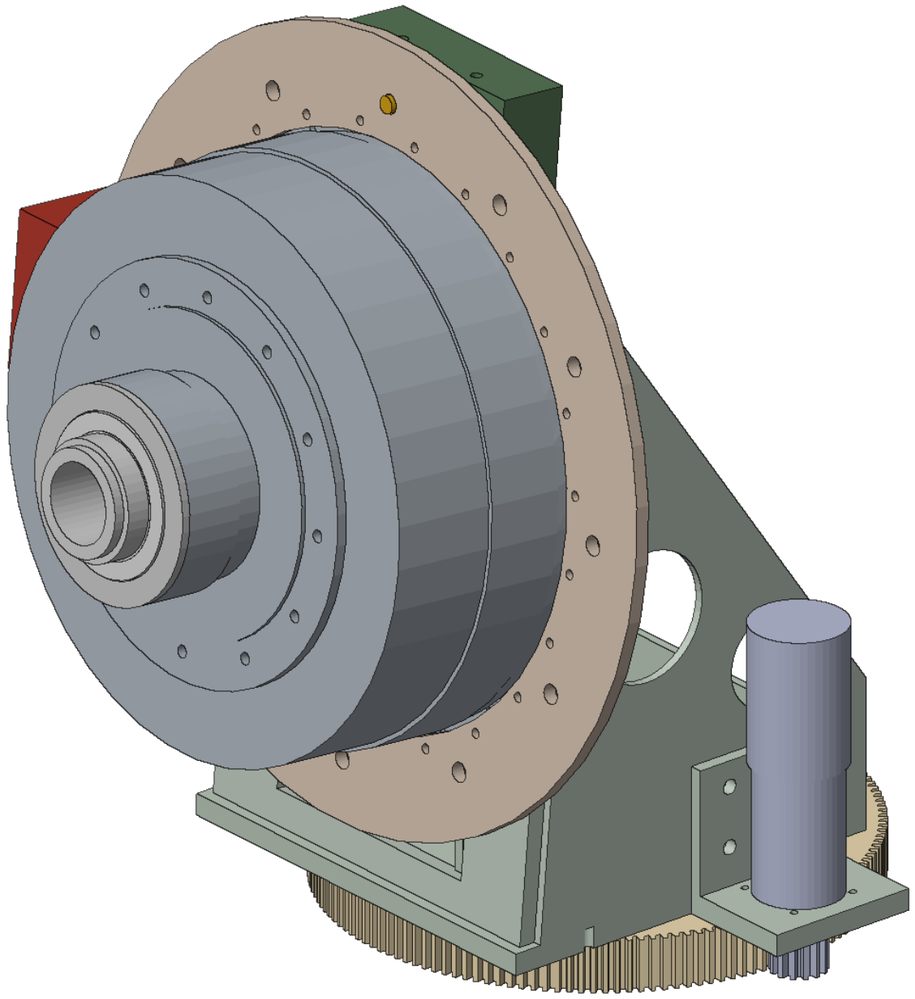
Step 4 Rotor drum with magnets lowered over the stator on the guide tool; disc, seal covers, rear cover



- The drum with the magnets is the dangerous step: use the drum assembly guide NP1-T01 and the magnet safety rules of doc 18 (no steel tools within 1 m, two people).
- Lower the drum with cups and seals onto the bearings on the guide screw; set the bearing preload with the KMT 27 locknut (rolling torque 6–8 Nm).
- Fit the brake disc (12 × M12, 110 Nm), the front and rear seal covers and the plain rear cover NS6-610 (24 × M8, 20 Nm). Remove the guide.
- Check: air gap ≥ 1.4 mm all round (feeler at 8 positions), drum run-out ≤ 0.2 mm, no rubbing when turned by hand; back-EMF test at 60 rpm.

Parts (new in colour, earlier steps light): NP1-210C, NP1-220E, NP1-230, NP1-211-F, NP1-211-R, NS6-610. Torques: NS6-900.

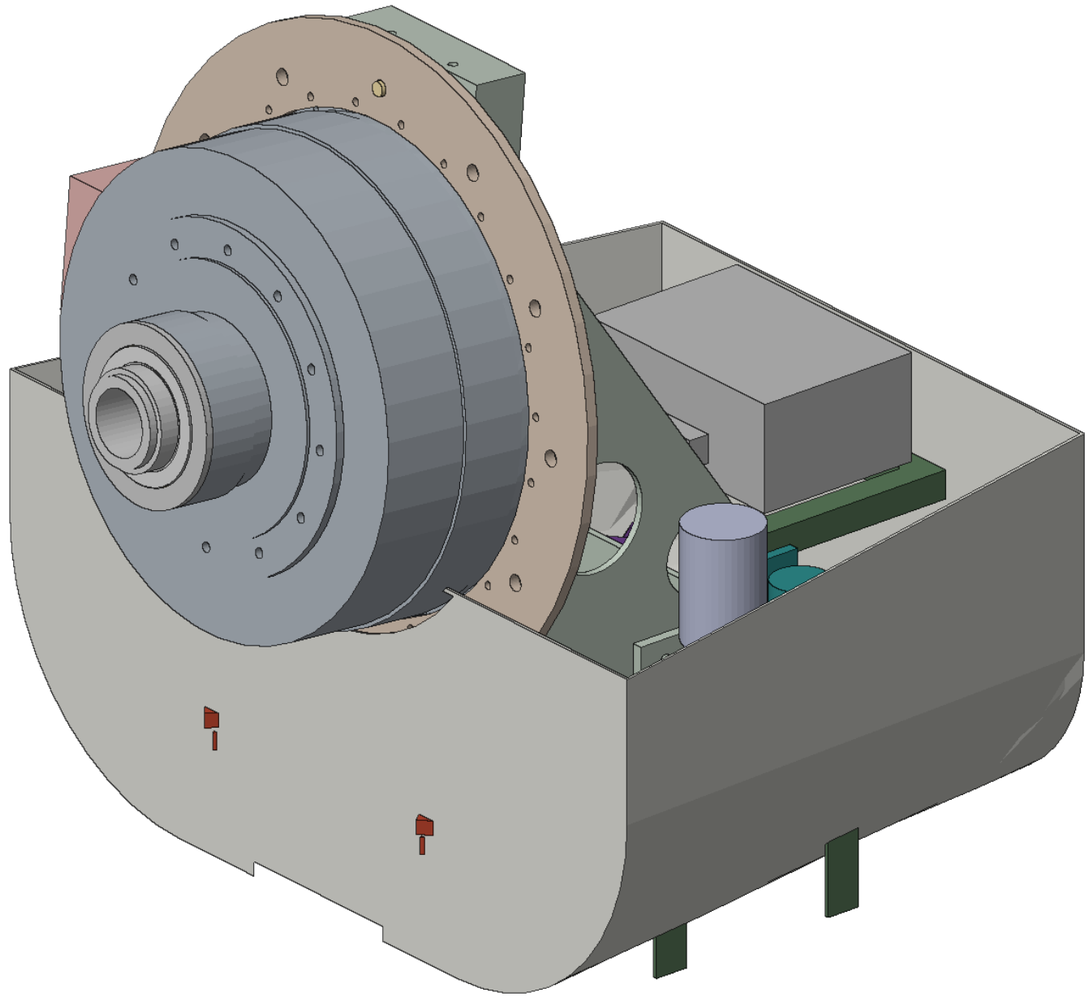
Step 5 Brake caliper on its bracket, lock pin holder with overspeed sensors



- Fit the caliper bracket NP1-310 (left only) and the spring caliper; set the pad gap per the supplier; check the release with the 24 V supply.
- Fit the lock-pin holder with the two overspeed sensors (3 mm from the disc); test the lock pin in all 12 holes.
- Check: brake holding torque test 1.5 kNm (torque arm on the drum) and overspeed sensor pulses (24 per turn).

Parts (new in colour, earlier steps light): NP1-310, NS6-915, NP1-320, NP1-321, NS6-914. Torques: NS6-900.

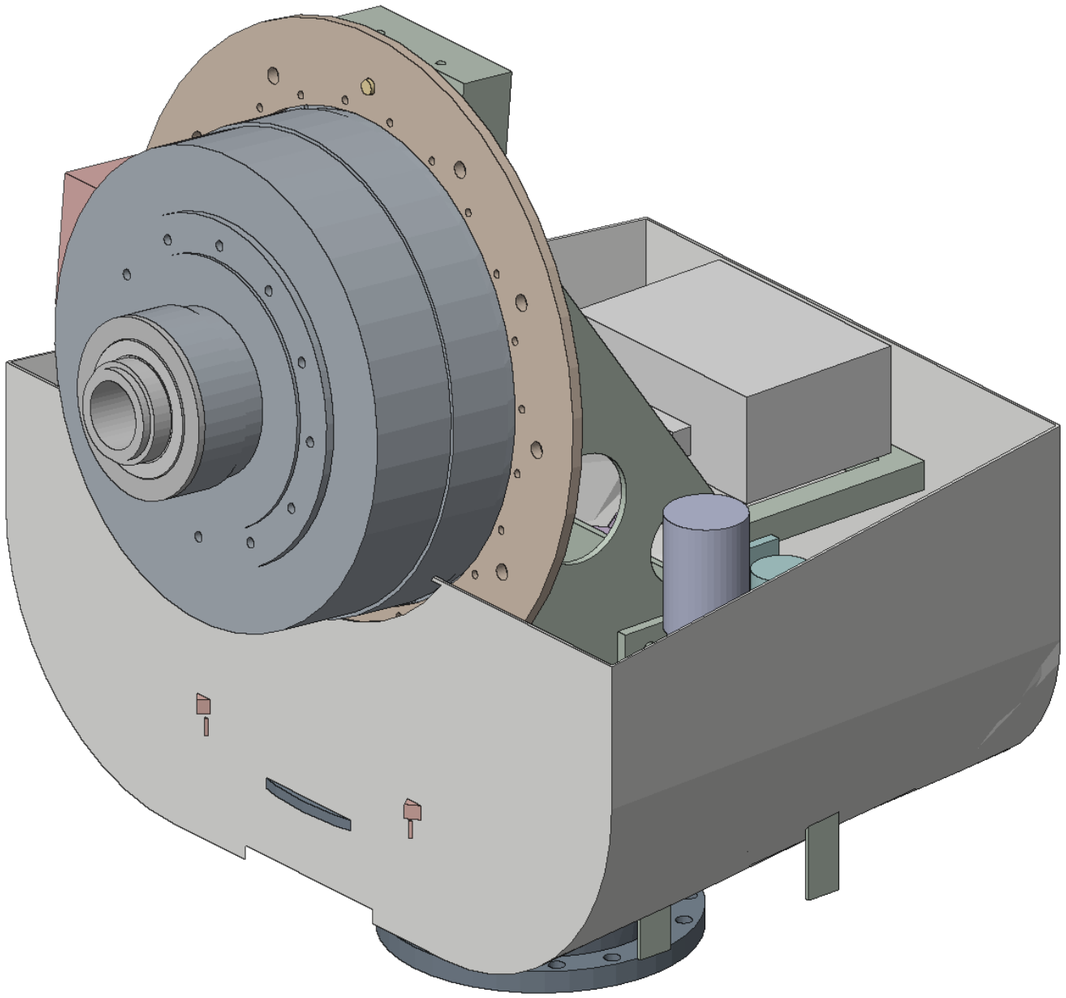
Step 6 Yaw pad units, lock, encoder and cable hanger; cover frame, lower cover tray and junction box



- Fit the two yaw pad units NY-410 (preset spring stack 3.7 mm), the yaw lock NY-430, the encoder unit NY-440 and the cable hanger NY-460.
- Fit the cover arms NS6-710, the lower GRP tray NS6-700-1 on rubber mounts, the rear rail and the junction box NS6-730; wire W1, W4, W5, W6 to the box.
- Check: pad clamping force (spring compression), encoder zero at yaw 0°, cam switch at ±2.5 turns.

Parts (new in colour, earlier steps light): NY-410-L, NY-410-R, NY-430, NY-440, NY-460, NS6-710, NS6-700-1, NS6-730. Torques: NS6-900.

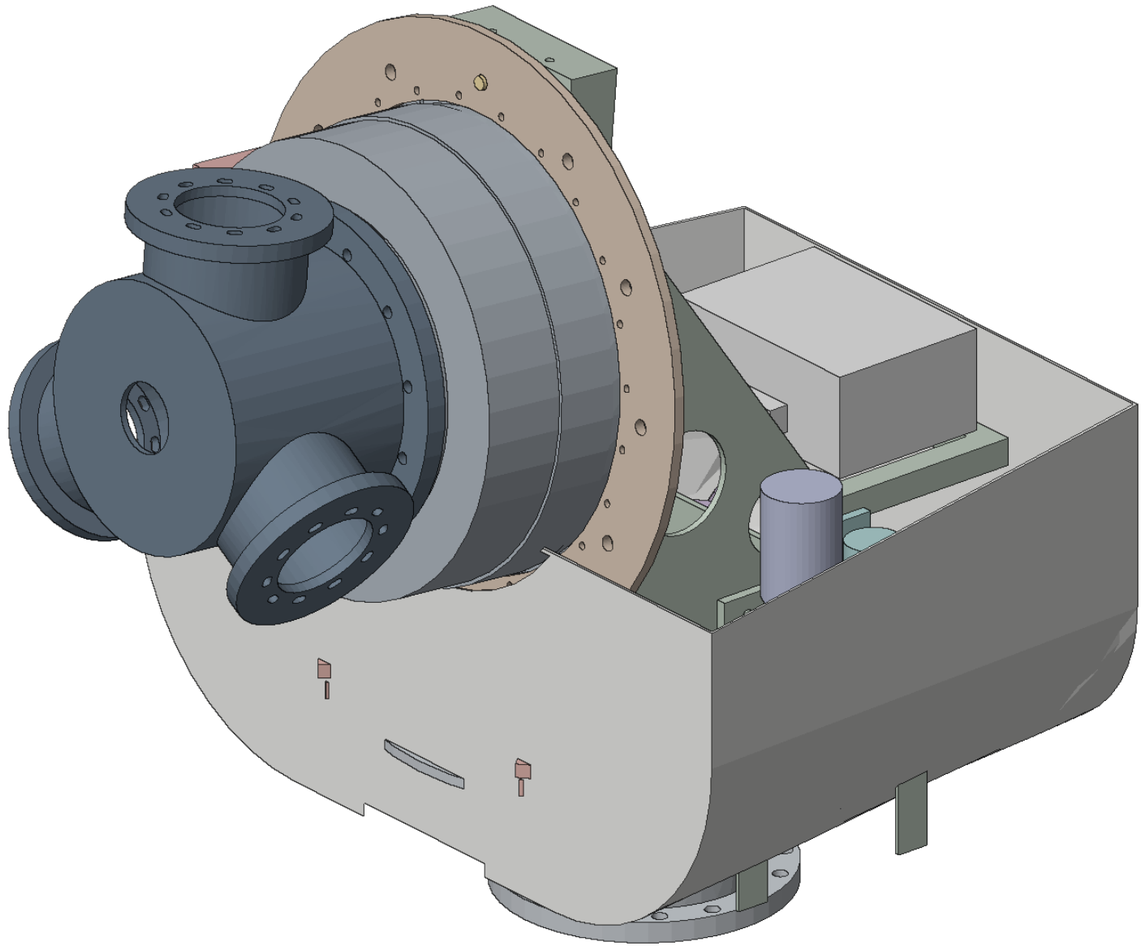
Step 7 Tower-top adapter on the yaw outer ring (nacelle now complete for transport)



- Lower the nacelle onto the tower-top adapter NY-400 (outer ring 16 × M16, 250 Nm). The nacelle is now a transport unit (≈ 790 kg) with lifting points on the bedplate.
- Check: yaw turns 360° with the drives, pads slip at 4–5 kNm (spring balance on a 1 m lever).

Parts (new in colour, earlier steps light): NY-400. Torques: NS6-900.

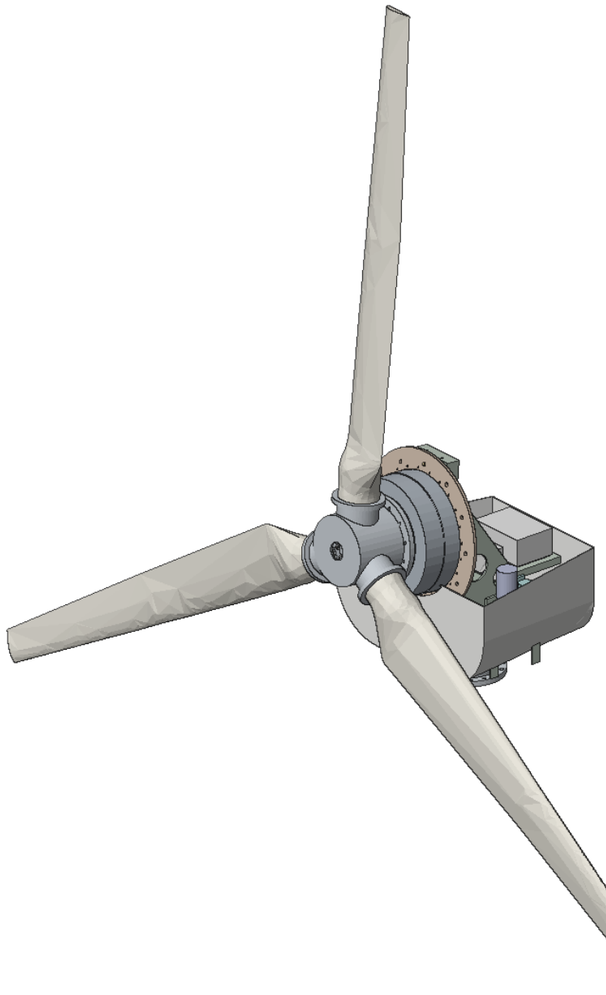
Step 8 Fixed hub on the drum flange (12 × M16 10.9)



- Fit the hub NS6-200 on the drum spigot, 12 × M16 10.9, 250 Nm in 3 passes, cross pattern. Mark arm 1 at the lock-pin azimuth.
- Check: hub face run-out ≤ 0.3 mm at the root flanges.

Parts (new in colour, earlier steps light): NS6-200. Torques: NS6-900.

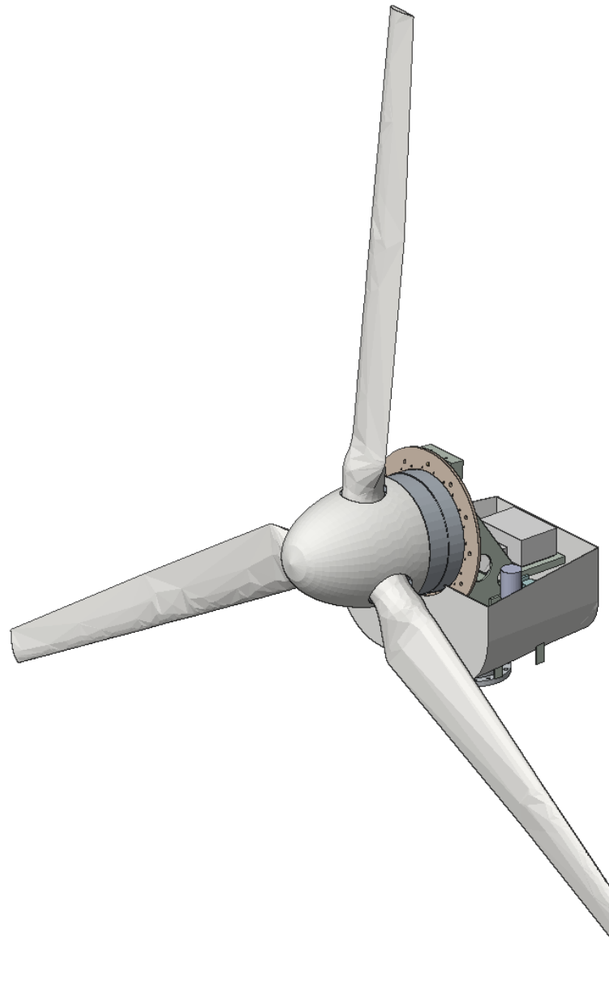
Step 9 Blades with pitch washer plates, set to −2.0° with the pitch gauge



- Screw the M12 studs into the blade inserts (Loctite 270, 40 Nm). Mount B1 on arm 1, B2, B3 by their static-moment record.
- Fit washer plates; set each blade to −2.0° ± 0.1° with the pitch gauge NS6-135 at r = 2 000 mm; torque 95 Nm + 90° in a star pattern; re-check pitch.
- Check: blade tip track within ±15 mm; rotor turns freely.

Parts (new in colour, earlier steps light): NS6-100-1, NS6-100-2, NS6-100-3, NS6-130, NS6-105. Torques: NS6-900.

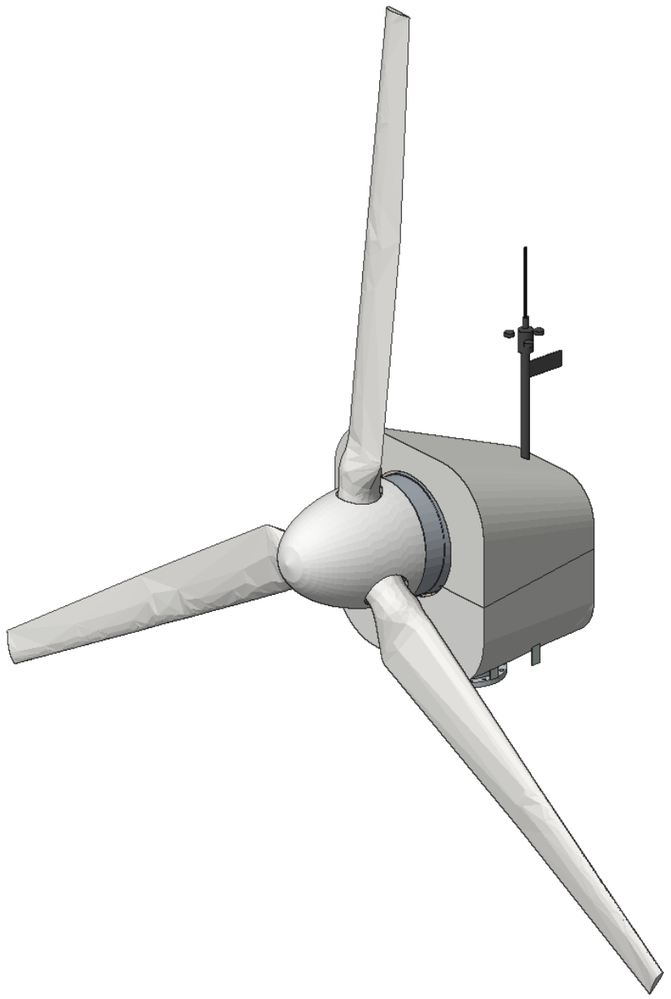
Step 10 Spinner and its brackets



- Fit the three spinner brackets and the spinner (3 × M8 with rubber washers, 12 Nm). Check 5 mm clearance to the drum front web.

Parts (new in colour, earlier steps light): NS6-210, NS6-211. Torques: NS6-900.

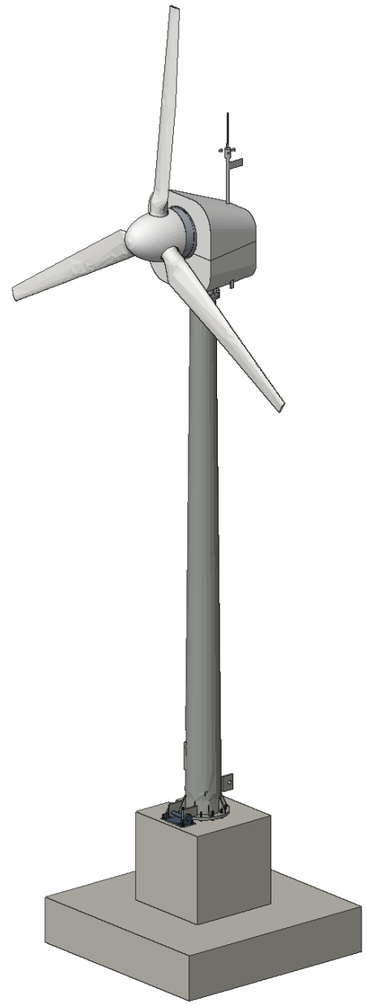
Step 11 Upper cover lid and met mast



- Fit the upper cover lid (14 × M8, EPDM gasket) and the met mast with anemometer, vane and air terminal; bond the air terminal to the bedplate (50 mm² Cu).
- Factory test: spin the rotor with a drive on the hub (doc 15 bench) or leave to site; record insulation, brake, yaw, sensor function.

Parts (new in colour, earlier steps light): NS6-700-2, NS6-720. Torques: NS6-900.

Step 12 On site: footing with anchor cage, hinge brackets, tower lying on the hinge; head bolted to the tower top



- Site: excavate to 1.6 m, blinding, place the bottom mesh, the anchor cage NS6-510 with the template ring (± 3 mm, level ± 1 mm) and the hinge anchors; ducts; earthing ring.
- Pour the footing C30/37 XF4, cure ≥ 7 days. Fit the hinge brackets NS6-410 and the levelling nuts. Lay the tower on trestles with the base lugs on the hinge pin.
- Bolt the head to the tower top flange on the ground (16 × M20 HV, 450 Nm); pull W2 and W5 through the tower; lock the rotor with blade B1 up.

Parts (new in colour, earlier steps light): NS6-500, NS6-510, NS6-410, NS6-400. Torques: NS6-900.

Step 13 Tilt-up with the gin pole and winch, base nuts torqued; cabinet, resistor box, cabling, hand-hole cover



- Fit the gin pole NS6-420 and the rigging; pull the tower up with the winch in wind below 10 m/s; nobody in the fall zone.
- Land the base flange on the levelling nuts, fit the top nuts, grout (30 mm, ≥ 40 MPa), after 7 days torque the anchors to 450 Nm. Remove the gin pole.
- Set the cabinet stand, cabinet and resistor box; connect W3, earthing (≤ 10 Ω), hand-hole cover. Go to the commissioning checklist (§7).

Parts (new in colour, earlier steps light): NS6-420, NS6-620, NS6-600, NS6-800, NS6-401. Torques: NS6-900.

7 Erection, commissioning and inspection

Before power-up

- All torque stripes present; anchor, adapter, hub and blade bolt records signed (NS6-900).
- Tower plumb within 0.3 % (25 mm at the top) in two directions; grout cured; hand-hole cover fitted.
- Earthing ≤ 10 Ω measured; continuity air terminal → bedplate → pad brush → tower → earth ring ≤ 0.2 Ω.
- Insulation: generator phases to earth ≥ 100 MΩ at 1 kV; cables W2/W3 ≥ 100 MΩ.
- Rotor lock pin out; yaw lock retracted; cover lid closed; nobody inside 21 m.

Functional tests (wind below 8 m/s)

- Safety chain: each trip input (overspeed sensors × 2, E-stop, 24 V loss, vibration, converter fault) closes K1 and drops the brake together; record the stop time.
- Brake only (K1 forced open) and resistor only (brake held off) stops from 100 rpm: both stop or crawl as doc 31.
- Overspeed trip at 189 rpm checked by lowering the set point to 60 rpm (test mode).
- Yaw: ±360° with the cable cam stop at ±2.5 turns; yaw-out 90° in ≤ 3 min on the aux battery; exclusion zone 60–77 rpm set in the controller.
- Rotor balance: run at 120 rpm, measure the nacelle 1P vibration; add trial mass to reach ≤ 0.2 kg·m (doc 09 R6.6).
- Tower frequency: tap test or run-up; f1 must be ≥ 3.1 Hz (record; if lower, widen the exclusion zone).
- Power curve and stall controller: 48 h supervised operation with storm ramp and yaw-out tested on the next windy day.

Inspection plan

Interval	Inspection
50 h	Re-torque blade studs, hub bolts, adapter bolts, anchors; re-check pitch and track
3 months	Visual blades (LE erosion, cracks at the root), brake pad wear, cover fixings
Yearly	All structural bolts 10 % sample at 90 % torque; pad units spring gap; slewing ring grease (2 shots); yaw gear grease; bearing grease 30 g; earthing; safety chain test; brake torque test
5 years	Lower the tower: full blade inspection, bearing play, magnet-sleeve inspection, cable loop wear
After a storm > 50 m/s or an ice event	Visual check, yaw-out log, brake slip log (doc 32 §3)

Lowering the tower for service

- Stop, yaw to the hinge direction, lock the rotor with B1 up, lock the yaw; rig the gin pole and winch; slacken the base nuts on the hinge side last.
- Lower to the trestle (head 1.0 m above ground); nacelle and rotor are at working height.

Records to keep with the turbine

- Blade set static moments and masses; magnet batch certificates; weld and UT records (hub, tower, bedplate); torque records; commissioning test results; f1 measured.

8 Open items before the prototype release

- FE checks: hub arm-to-shell weld (doc 30 §11); NS6-300 ring seat stiffness and flatness (doc 32 §1).
- Stall torque measurement on the prototype (docs 28/29): the main design risk of S6.
- Site investigation: soil class (plate-load test on volcanic ash), frost depth, earthing resistivity, local obstacle survey at 8 m.
- Supplier drawings replace the envelopes: slewing ring, gearmotors, bearings, caliper (NS6-910); then rerun the clash check.
- Tower frequency on site (≥ 3.1 Hz); if lower, the exclusion zone in the controller is widened.
- Blade: first-article static test to SLM H × 1.35, insert pull-out tests, mould acceptance against the plug templates.
- Icelandic DSO: confirm 6 kW single-phase connection (doc 12).
- Certification route (IEC 61400-2 type test) and the energy loss at 8 m on low-wind sites: offer the 8 m tower for coast, highland and open land; keep an 18 m option for sheltered sites.
- All drawings NOT CHECKED: an independent check of every sheet before the prototype parts are ordered.